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# Advanced Dielectric Materials for Triboelectric Nanogenerators: Principles, Methods, and Applications

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**Triboelectric nanogenerator (TENG) manifests distinct advantages such as multiple structural selectivity, diverse selection of materials, environmental adaptability, low cost, and remarkable conversion efficiency, which becomes a promising technology for micro-nano energy harvesting and self-powered sensing. Tribo-dielectric materials are the fundamental and core components for high-performance TENGs. In particular, the charge generation, dissipation, storage, migration of the dielectrics, and dynamic equilibrium behaviors determine the overall performance. Herein, a comprehensive summary is presented to elucidate the dielectric charge transport mechanism and tribo-dielectric material modification principle toward high-performance TENGs. The contact electrification and charge transport mechanism of dielectric materials is started first, followed by introducing the basic principle and dielectric materials of TENGs. Subsequently, modification mechanisms and strategies for high-performance tribo-dielectric materials are highlighted regarding physical/chemical, surface/bulk, dielectric coupling, and structure optimization. Furthermore, representative applications of dielectric materials based TENGs as power sources, self-powered sensors are demonstrated. The existing challenges and promising potential opportunities for advanced tribo-dielectric materials are outlined, guiding the design, fabrication, and applications of tribo-dielectric materials.**

## 1. Introduction

Sustainable development desires to address the energy issues through eco-friendly, renewable, and economic strategies.[1–5] Energy harvesting technology that scavenges sustainable electrical energy from wasted mechanical energy around the environment has been recognized as one of the most promising solutions, and diverse energy harvesting devices have been proposed for harvesting vibration energy, wind energy, water energy, thermal energy, and radiant energy, etc.[6–12] Specifically, piezoelectric nanogenerator (PENG),[13,14] triboelectric nanogenerator (TENG),[15,16] and magnetoelastic generator (MEG)[17,18] featuring moving parts have been invented for mechanical energy harvesting. Thermoelectric nanogenerator (TEG),[19,20] pyroelectric nanogenerator (PyNG),[21,22] evaporation-driven hydrovoltaic device (EHD)[23,24] that operate on the Seebeck effect, pyroelectric effect, and hydrovoltaic were also developed. Furthermore, hybrid energy harvesters based on the combination of TENG, PENG, EMG, PyNG, etc., are reported to enhance energy harvesting efficiency, addressing the limitations of individual nanogenerators in adapting to diverse energy sources and meeting the power demands of the load.[25]

As an emerging branch, TENG invented in 2012 by Wang et al. that operates by the coupling of contact-electrification effect and electrostatic induction, demonstrates the advantages of simple structure, diverse materials, portability, and

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multifunctional, which have garnered tremendous attention around the world.[26–29] Although the energy harvesting density cannot compete with conventional EMGs, TENGs deliver superior efficiency, low-frequency decentralized energy generation.[30–32] Meanwhile, the rapid development of the Internet of Things (IoT), digital twins, and artificial intelligence (AI) needs various distributed sensors for information acquisition and perception.[33–36] Conventional batteries or capacitors utilized for powering sensors are inevitably limited in terms of charging, maintenance, and environmental pollution.[37,38] To this end, the self-powered sensing realized by energy harvesters such as TENGs has been recognized as the pervasive energy solution in the future.[39,40] Moreover, TENG itself could be employed as a self-powered sensor by virtue of the relationship between voltage/current response to mechanical agitation, such as the flourishing self-powered chemical, physical sensing, healthcare, etc.[41–45]

Taking the advantages mentioned above, researchers have witnessed the explosive development of TENGs from fundamental mechanisms to versatile applications, wherein the dielectric materials are critical to high-performance TENGs for contact electrification and electrostatic induction processes.[46–48] As the basic principle, triboelectricity or contact electrification (CE) presents in a wide variety of dielectric materials (solid–solid, solid–liquid, solid–gas, etc.), which belong to a highly complex physical and chemical process.[49–54] Generally, the elemental composition, physicochemical structure, dielectric parameter, and trap properties of dielectric materials play a crucial role in the CE and charge transport behavior (charge generation, migration, dissipation, and storage, etc.), which essentially determines the output performance of TENGs.[55–57] Meanwhile, the interface electric field generated by triboelectric charges will also induce dipole, interface polarization of the dielectrics, thereby affecting the electrostatic induction process.[58,59] Besides, dielectric materials exhibit various tunable qualities, such as hydrophobicity, flexibility, stretchability, antibacterial capabilities, biocompatibility, etc., which can fulfill the application requirements of various scenarios for TENGs.

In recent years, dielectric materials have made significant progress for high-performance TENGs, leading to the emergence of novel materials, theories, modification techniques, and applications. Triboelectric series, the primary database for numerous dielectric materials ranked according to their capabilities of gaining or losing electrons, have been proposed for tribo-dielectric pair selection.[60–62] However, the limited materials in the triboelectric series list cannot meet the needs of ultrahigh performance and diverse functionality of TENGs. Therefore, additional modification or original design of tribo-dielectric materials is essential to boost the output performance and endow diversified functional applications.[63–65] Although some existing reviews referred to high-performance materials for TENGs,[66–69] the focus mainly concentrated on materials synthesis and applications, while basis CE and charge transport mechanisms, modification methodological principle, and optimization strategies are still lacking. Generally, the dielectric parameters such as charge trapping, polarization, dielectric constant ($k$), dielectric loss, dielectric strength, etc. could be regulated by various physical and chemical methods.[70–73] Additional strategies, including dielectric coupling, dielectric structure optimization, also endow controllable charge transport characteristics.[74–77] Therefore, the design and fabrication of novel tribo-dielectric materials with improved CE performance inspired by the intrinsic charge transport mechanism will effectively promote the development of TENGs. A more profound, comprehensive summary to elucidate the internal principle of dielectric charge transport mechanism and tribo-dielectric material modification is highly desired.

Herein, we offer an overview of recent advances in tribo-dielectric materials for TENGs in terms of principle, method, and application, as shown in **Figure 1**. We started with the CE mechanism and charge transport behavior of dielectric materials. Subsequently, the operation principle and the basic solid, liquid, and gas dielectric materials for various types of TENGs were summarized. Accordingly, we highlighted the modification principle and method for advanced tribo-dielectric materials regarding physical/chemical, surface/bulk, dielectric coupling, and structure optimization. Furthermore, we demonstrated the fascinating applications of advanced dielectric materials based TENGs in micropower sources, high-voltage power sources, self-powered sensors, self-powered healthcare, and blue energy. Finally, we highlighted the scientific challenges and promising perspectives with respect to the basic mechanisms, strategies of advanced dielectric materials design, modification, and application link, which can serve as a guideline for developing high-performance TENGs.

## 2. CE Mechanisms of Dielectric Materials for TENG

CE has been known as a universal phenomenon for ≈2600 years and exists in all kinds of matter worldwide, such as solid–solid, solid–liquid, and solid–gas materials, as illustrated in **Figure 2a**.[99] Despite being the fundamental basis of triboelectricity, CE possesses a complicated physical process and various basic principles, including the unified physical model, interfacial charge exchange/tunneling mechanism, remain challenging. The situation has significantly shifted due to the development of TENGs as the charges generated by CE can be collected and transferred to the external circuit. The increase in output performance relies on effective CE and the development of electrostatic induction/breakdown efficiency, which are essential components of dielectric utilization and modification. These issues will be discussed more in the following chapters. In this section, we summarized the physical principles of CE and introduced the operation mechanism of various TENGs, providing basic guidance for rational utilization and efficient modification of dielectric materials.

### 2.1. CE of Solid–Solid Dielectrics for AC-TENG

CE commonly exists in solid–solid dielectric materials, whose mechanism could be described by the surface state model. The electron transfer-dominated process during CE has been proven, essentially determined by the work function, energy level, and band structure of the materials.[100,101] As shown in Figure 2b, the contact of two dielectric materials with different valence and conduction band structures will transfer electrons from the dielectric material with the higher valence band to the lower one, realizing the electron state equilibrium. After separation, the net positive and negative charge will deposited on the dielectric surface.


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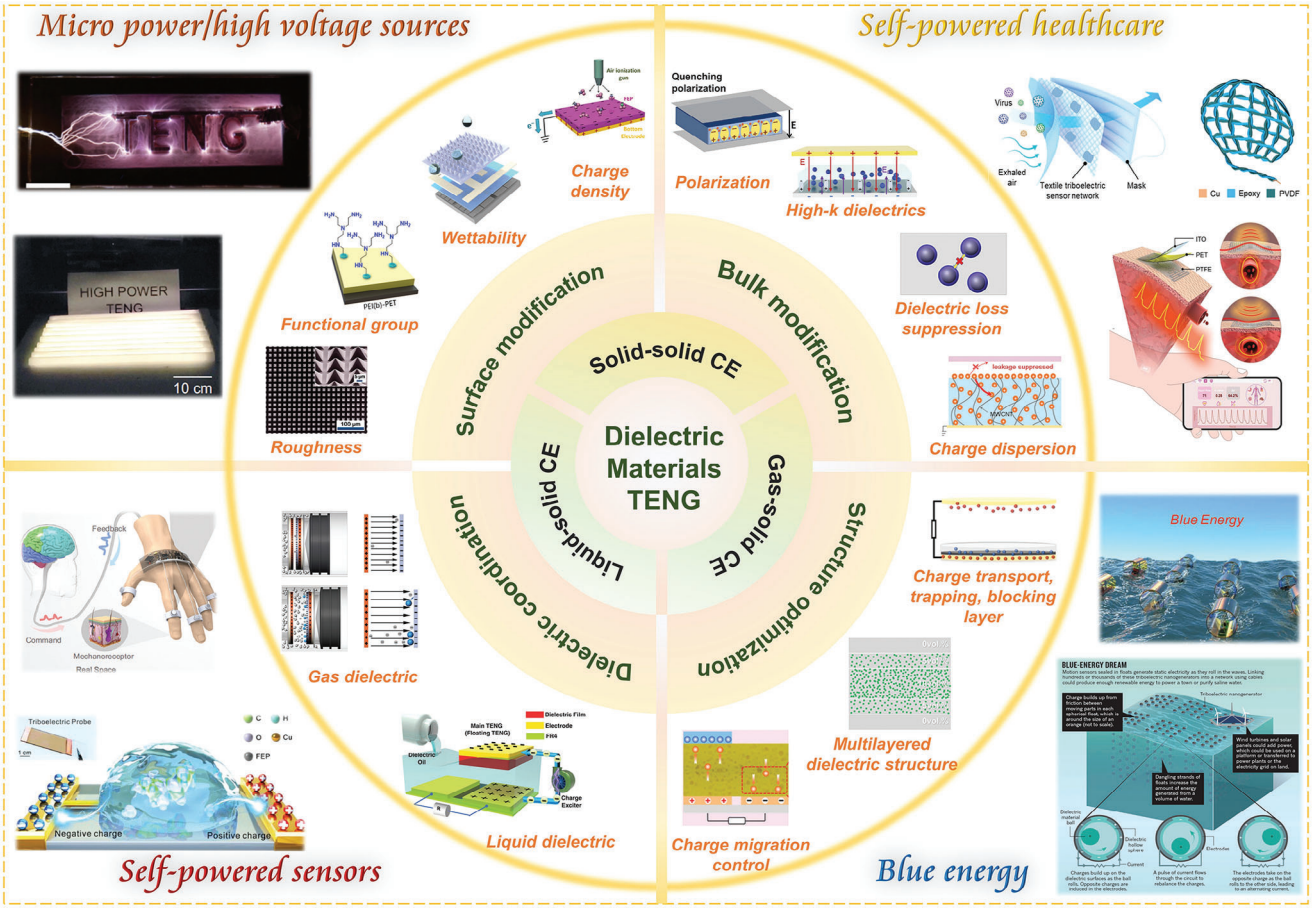


**Figure 1.** Advanced dielectric materials based TENG for diverse applications. Surface modification-roughness, Reproduced with permission ref. [78]. Copyright 2012, American Chemistry Society. Surface modification-functional group, Reproduced with permission ref. [79]. Copyright 2017, American Chemistry Society. Surface modification-wettability, Reproduced with permission ref. [80]. Copyright 2019, Elsevier. Surface modification-charge density, Reproduced with permission ref. [81]. Copyright 2014, Wiley-VCH. Bulk modification-polarization, Reproduced with permission ref. [82]. Copyright 2023, Wiley-VCH. Bulk modification high -*k* dielectrics, Reproduced with permission ref. [83]. Copyright 2022, Wiley-VCH. Bulk modification-dielectric loss suppression. Reproduced with permission ref. [84]. Copyright 2022, American Chemistry Society. Bulk modification-Charge dispersion. Reproduced with permission ref. [85]. Copyright 2023, Wiley-VCH. Structure optimization-charge migration control. Reproduced with permission ref. [86]. Copyright 2023, Wiley-VCH. Structure optimization – multilayered dielectric structure. Reproduced with permission ref. [87]. Copyright 2020, Wiley-VCH. Structure optimization -charge transport, trapping, blocking layer. Reproduced with permission ref. [88]. Copyright 2022, Elsevier. Dielectric coordination-gas dielectric, Reproduced with permission ref. [89]. Copyright 2019, Wiley-VCH. Dielectric coordination-liquid dielectric, Reproduced with permission ref. [90]. Copyright 2022, Wiley-VCH. Micro-power sources/high voltage sources, Reproduced with permission ref. [91]. Copyright 2021, Wiley-VCH. Reproduced with permission ref. [92]. Copyright 2021, the authors. Self-power healthcare, Reproduced with permission from ref. [93]. Copyright 2022, Wiley-VCH. Reprinted with permission from ref. [94]. Copyright 2022, Wiley-VCH. Self-power sensors, Reproduced with permission ref. [95]. Copyright 2022, the authors. Reproduced with permission ref. [96]. Copyright 2023, Wiley-VCH. Blue energy, Reproduced with permission ref. [97]. Copyright 2020, Wiley-VCH. Reproduced with permission ref. [98]. Copyright 2017, Springer Nature.

The electron acts as the charge carrier was also affirmed via the thermionic and photoelectron emission effect.[102–105] Relevant results demonstrated that the generated charge decayed faster at higher temperatures, indicating the existence of CE electrons on the dielectric surface. Additionally, the electron cloud model was proposed, where the molecular structure of dielectric materials could be explained by chain alignment and orbitals.[103,106,107] As shown in Figure 2c, the overlap in electron clouds or wave functions results in an equilibrium distance (bond length) between two atoms. When two atoms from different dielectric materials come into close proximity and contact by external pressure or force, electron cloud overlapping occurs to form an asymmetric double-well potential. Electrons can transition from a higher energy level on one surface to a lower energy level on the other, leading to the overall development of net charge upon separation.[107] This was also supported by quantum mechanical analysis based on density functional theory (DFT), where the delocalization of electron wave functions was found upon stress or strains.[108,109]

Solid dielectric-based alternating current TENGs (AC-TENGs) are the most common devices for collecting CE charges, mainly divided into four operating modes: Contact-separation (CS) mode, Single electrode (SE) mode, Sliding mode, and Freestanding (FS) mode. CS-TENG can be classified into two main categories: dielectric-to-dielectric and conductor-to-dielectric (Figure 2d(i)). Dielectric-to-dielectric CS-TENG is


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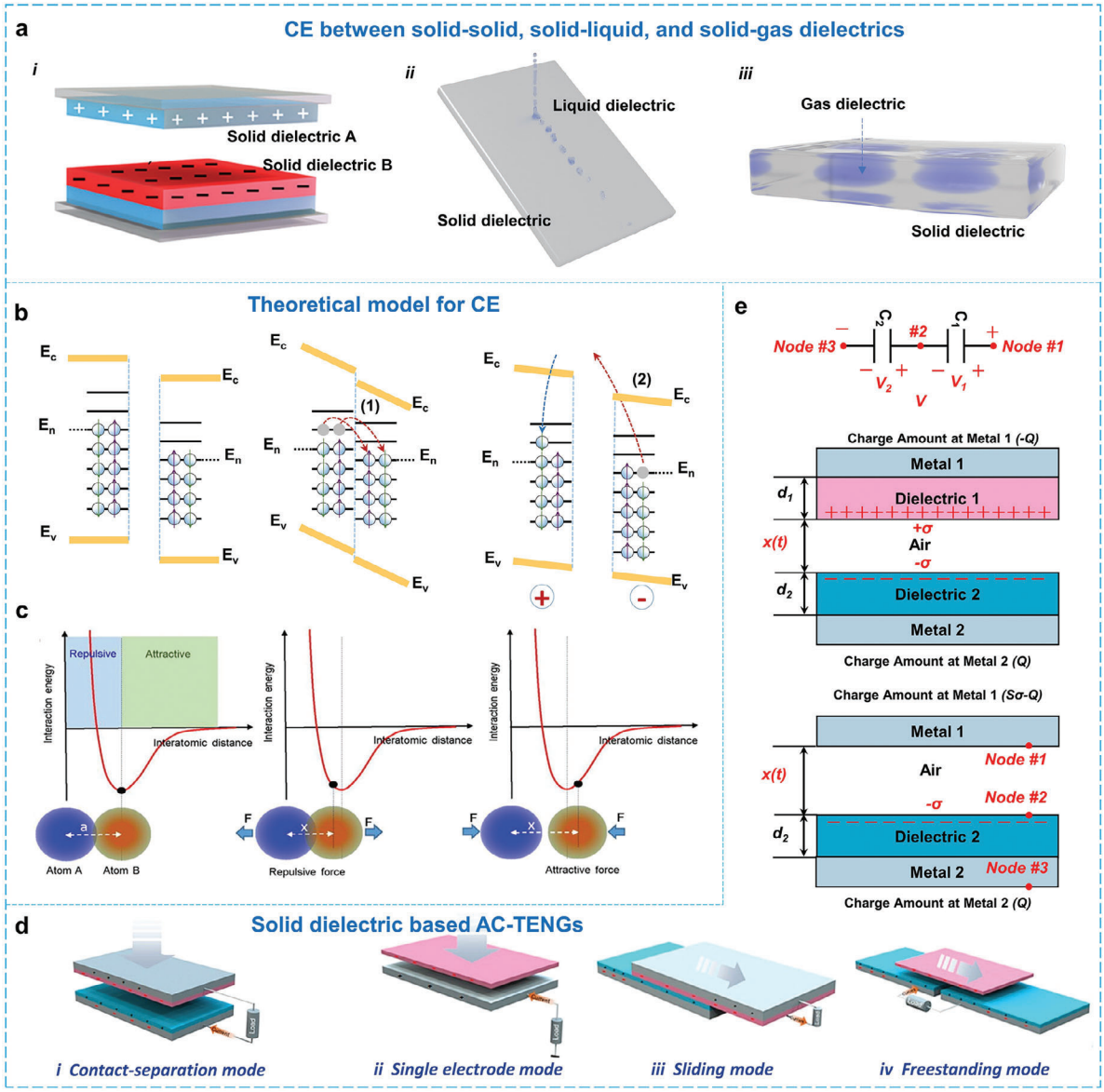


**Figure 2.** CE mechanism of solid–solid dielectric and AC-TENGs. a) Schematic diagram of CE of solid–solid, solid–liquid, solid–gas cases. b) Energy band diagrams for the solid–solid CE. c) Interatomic interaction potential of two atoms at the equilibrium, repulsive, and attractive position. d) Solid dielectric based AC-TENGs. e) Theoretical models for CS-TENG. a(i) Reproduced with permission ref. [106]. Copyright 2018, Elsevier. b,c) Reproduced with permission ref. [105]. Copyright 2019, Elsevier. d) Reproduced with permission ref. [111]. Copyright 2021, American Chemical Society. e) Reproduced with permission ref. [110]. Copyright 2015, Elsevier.

usually made by layering two dielectric materials with distinct electron-capturing properties and attaching metal electrodes to opposite back surfaces. As to the conductor-to-dielectric type, metal serves as both electrode and triboelectric material, coupling with the other tribo-negative or tribo-positive dielectric. The corresponding working mechanisms of such two types are similar. Generally, AC-TENG inherently exhibits capacitive behavior and the intrinsic capacitance $C$ can be represented as

$$C = \frac{\varepsilon_0 S}{d_0 + x(t)} \tag{1}$$

where $\epsilon_0$ represents the permittivity of vacuum; $S$ represents the dielectric area; $x$ represents the separating distance; $d_0$ represents the effective dielectric thickness[110]

$$d_0 = \sum_{i=0}^{n} \frac{d_i}{\epsilon_{ri}} \tag{2}$$

where $d_i$, $\epsilon_{ri}$ represent the thickness and relative permittivity of each dielectric layer, including the air gap.

Researchers have extensively studied the physical models of CS-TENG, establishing the equivalent circuit based on first-order lumped parameters and the theoretical $V$–$Q$–$x$ relationship


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through the capacitive model. The potential difference $V$ between two electrodes of any TENG can be divided into two components. The first component arises from tribo-charges, which contribute to the voltage as $V_{oc}(x)$. The transferred charge $Q$ between electrodes also contributes to a portion of the potential difference. Subsequently, the $V$–$Q$–$x$ relationship as the governing equation of CS-TENGs can be expressed by the following equations

$$V = -\frac{Q}{S\varepsilon_0}(d_0 + x(t)) + \frac{\sigma x(t)}{\varepsilon_0} \tag{3}$$

$$V_{oc} = \frac{\sigma x(t)}{\varepsilon_0} \tag{4}$$

$$Q_{sc} = \frac{S\sigma x(t)}{d_0 + x(t)} \tag{5}$$

Single-electrode TENGs employ a different structure by omitting one of the electrodes in CS-TENGs, with the remaining electrode generally grounded to an external circuit.[110,111] This approach offers more flexibility in device design, eliminating the structural constraints associated with dual electrodes. However, the maximum $\eta_{CT}$ can only reach 50% with the charge transfer efficiency ($\eta_{CT}$) defined as the ratio between $Q_{sc}$ and the amount of tribo-charge.[74] Sliding-TENGs generate electricity by coupling sliding triboelectrification with in-plane charge separation, which can be manufactured into sophisticated structures such as turntable, cylindrical, and gridding structures.[112] The following expression can be used to derive the intrinsic capacitance $C$ using the parallel plate capacitor model

$$C = \frac{\varepsilon_0 w(l - x(t))}{d_0} \tag{6}$$

where dielectrics have geometric sizes of $l$ in the longitudinal direction and $w$ in the lateral direction. Furthermore, the distribution of charges can be used to assess the potential difference between the two electrodes

$$V = -\frac{d_0}{w\varepsilon_0(l - x(t))}Q + \frac{\sigma x d_0}{\varepsilon_0(l - x(t))} \tag{7}$$

$$V_{OC} = \frac{\sigma x d_0}{\varepsilon_0(l - x(t))} \tag{8}$$

A freestanding TENG is typically made up of a freely sliding/contact layer of triboelectric dielectric as well as two fixed electrodes that can be attached to another triboelectric material. Taking dielectric-based freestanding TENG as an example, the following expression can be used to derive the intrinsic capacitance $C$ using the parallel plate capacitor model

$$C = \frac{1}{\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}} = \frac{\varepsilon_0 S}{d_0 + g} \tag{9}$$

where $g$ represents the interval between the two electrodes. Furthermore, he $V$–$Q$–$x$ relationship can be expressed by the following equations

$$V = -\frac{d_0 + g}{\varepsilon_0 S}Q + \frac{2\sigma x(t)}{\varepsilon_0} \tag{10}$$

$$V_{OC} = \frac{2\sigma x(t)}{\varepsilon_0} \tag{11}$$

$$Q_{SC} = \frac{2\sigma S x(t)}{d_0 + g} \tag{12}$$

Overall, it is evident that the charge density ($\sigma$) plays a crucial role in TENG's output performance. According to the $V$–$Q$–$x$ relationship given in Equations (3)–(5), the output voltage and transferred charge are determined by $\sigma$, which is influenced by the relative dielectric constant ($k$) and thickness of the dielectric materials. To enhance the reliability of TENG throughout extended operation and prevent breakdown under high field strength, it is more effective to increase the relative dielectric constant rather than decrease the thickness. Besides, charge generation and dissipation occur simultaneously during CE. The dielectric material with poor charge storage capacity leads to a significantly lower steady-state surface charge density than the one initially produced by CE. Thus, the dielectric parameter and trap properties of the tribo-dielectric material essentially determine its charge storage performance. The performance figure-of-merit ($FOM_P$) comprising structure $FOM_S$ as well as material $FOM_M$ ($\sigma^2$) is also proposed to evaluate the TENG performance

$$FOM_p = FOM_s \cdot \sigma^2 = 2\varepsilon_0 \frac{E_m}{Ax_{max}} \tag{13}$$

### 2.2. CE of Fluid–Solid Dielectrics for AC-TENG

#### 2.2.1. Liquid–Solid Dielectric Case

Interface interaction between a liquid and solid material is another fundamental CE phenomenon. As illustrated in **Figure 3a**, the adsorption of surface charges and formation of an electrical double layer (EDL) occurs as the water droplet slides on the dielectric material. Then, most surface charges are neutralized at the back side, for instance, by OH-desorption. On the other hand, some surface charges move to the solid–air interface.[113] The overlapped electron cloud can be employed to elucidate the electron transfer mechanism as well, where the electrons are transferred from water molecules to polytetrafluoroethylene (PTFE).[114] For liquid-solid CE, the formation of EDL at solid–liquid interfaces is significant. The traditional EDL model comprises two parallel layers, labeled as the stern layer composed of ions adsorbed to the dielectric surface and the diffuse layer solution ions are attracted to the first layer via Coulomb force (as shown in Figure 3b).[115,116] Wang's group further investigated the source of the initial surface charge on the dielectric material. As illustrated in Figure 3c, the involvement of electrons during liquid–solid CE was verified via the thermionic emission effect. The results showed that most dielectric surface charges


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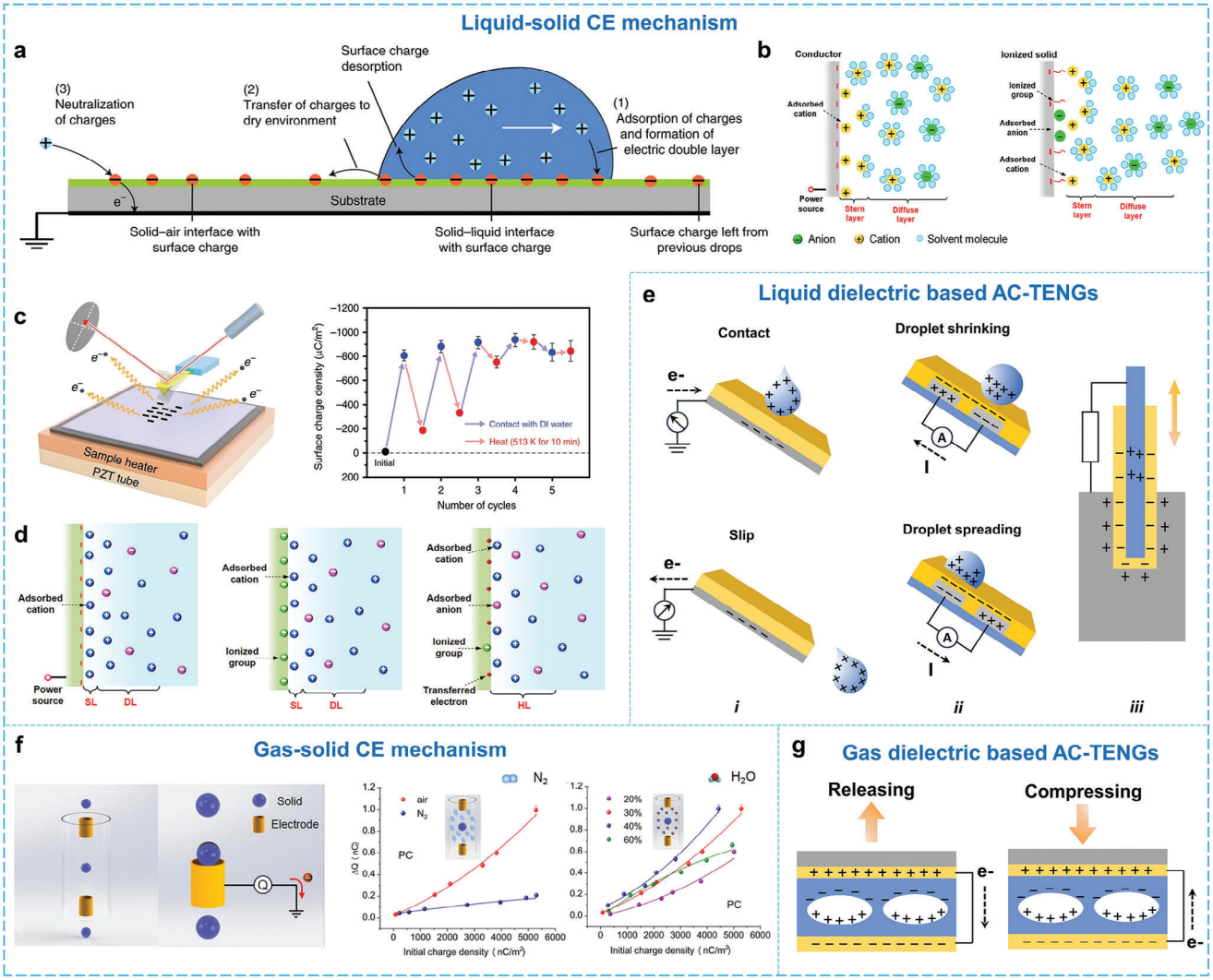


**Figure 3.** CE mechanism of fluid-solid dielectric and AC-TENGs. a) Schematic diagram of the solid–liquid CE process. b) Traditional EDL model on the electrode and ionized solid dielectric surface. c) The decay of the liquid–solid CE charge at different substrate temperatures. The charge density on a solid surface in the charging and heating (at 513 K for 10 min) cycles. d) EDL charge distribution at a liquid–solid interface. e) Liquid dielectric-based AC-TENGs. f) Schematic diagram of the solid–gas CE test platform and the detected CE signal. g) Gas dielectric based AC-TENGs. a) Reproduced with permission ref. [113]. Copyright 2022, the authors. b) Reproduced with permission ref. [52]. Copyright 2021, American Chemical Society. c) Reproduced with permission ref. [117]. Copyright 2020, the authors. d) Reproduced with permission ref. [118]. Copyright 2020, American Chemical Society. f) Reproduced with permission ref. [121]. Copyright 2023, the authors.

dissipated at temperatures higher than 433K.[117] The remaining surface "sticky" charges belong to the ions. Further, the "two-step" EDL model was proposed, as shown in Figure 3d. The first stage involves the collision of liquid molecules with a dielectric membrane due to thermal motion or pressure-induced CE, which results in electron cloud overlap and electron transmission. Subsequently, the free ions in the liquid are attracted to the charged dielectric layer to create the diffuse layer.[118]

The liquid–solid CE demonstrates the feasibility of employing the dynamic properties of fluid for interacting with the solid interface to generate electricity, where two typical droplet TENGs and fluid/wave TENGs are proposed.[111] The liquid offers a greater contact area, a continuous and soft contact with the solid dielectric interface, which allows TENG to maintain stronger wear resistance and a longer triboelectric effect. Droplets contain rich kinetic and electrostatic energy converted from the friction with air during descent, etc. Droplet-TENG mainly operates in the single-electrode mode (Figure 3e(i)) and freestanding mode (Figure 3e(ii)). As to the single-electrode mode, macroscopic electrical signals can be observed as the water droplet periodically contacts, slides, and falls off the dielectric film owing to the liquid–solid CE and electrostatic induction processes. For independent-layer mode TENG, as the droplets alternately enter and leave above the electrode, the positive charge carried by the droplets alternately shields the negative charge on the surface of the dielectric, causing free electrons on the electrode to flow through external loads.[119] Moreover, continuous liquids or waves also enable energy harvesting from various sources, and


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a typical fluid/wave TENG is shown in Figure 3e(iii). Specifically, the negative surface charge was sustained at the solid dielectrics. Immersing solid dielectric into the liquid will induce positive ions at the interface to screen the negative charges. The subsequent separating process will cause an unbalanced potential, which drives free electrons to flow through the external circuit until the new balance is created. The opposite process happens as the solid dielectric immerses again. The periodic movements of fluid/wave will generate AC triboelectric signals. The energy harvesting of water, oil, or organic fluid/waves has been explored.

#### 2.2.2. Gas–Solid Dielectric Case

CE between gas–solid dielectric materials also exists, and Lee et al. reported the first gas–solid TENG by filling different gases into the porous elastomer.[120] Sun et al. initially investigated the CE mechanism through a noncontact TENG and found that the transferred charge increases with surface area, movement distance, and the initial charges of the dielectric material (Figure 3f).[121] Meanwhile, electron transfer was speculated to be the primary mechanism, which could be enhanced during polar gas conditions. Owing to the lack of fixed volumes and unpredictable movement, CE between gas–solid dielectric materials generates relatively low output performance. Additional studies on the gas–solid interface CE mechanism considering gas compression are anticipated to provide a more thorough understanding of the physical mechanism.

The gas–solid TENG operates by the contact between gas and solid dielectrics, particularly the porous elastomer illustrated in Figure 3g.[119] The pressure fluctuations were produced when an elastomer undergoes compression deformation, and strong contact electrification occurs between the gas and solid dielectrics with the charge/ion transfer process happening simultaneously. The release of applied pressure will cause instantaneous asymmetric charge distribution, delivering the electric potential difference between the upper and lower electrodes. Subsequently, a continuous electric signal will flow through the external circuit to bring a new balanced potential. The other elastomer compression will induce an opposite current flow due to the reduced electric potential difference. Therefore, the repeated compression and release process generates AC output. Generally, the gas–solid triboelectric performance could be regulated by the gas composition, pore size, and the applied pressure.

### 2.3. DC-TENG

The direct-current TENG (DC-TENG) is another significant triboelectric energy harvester based on the principle of CE and electrostatic breakdown effect. Typical foundational models of DC-TENG include the sliding structure and CS structure, as shown in **Figure 4a**. A sliding-mode DC-TENG typically comprises a friction electrode for CE with a triboelectric layer and a charge-collecting electrode (CCE) for collecting the electrostatic breakdown charges.[122] Specifically, the friction electrode will contact the triboelectric layer to generate surface charges, as shown in Figure 4b. Then, a strong electric field was built between the CCE and triboelectric layer, causing air gap breakdown and generating a DC in the external circuit. The surface charge density of a DC-TENG is limited by the following equation

$$\sigma_{\mathrm{DC-TENG}} = k \times \min(\sigma_{\mathrm{triboelectrification}}, \sigma_{\mathrm{c,\ triboelectrification\ breakdown}}) \quad (14)$$

where $k$ represents the electrode structure factor, $\sigma_{\mathrm{triboelectrification}}$ is the CE surface charge density, $\sigma_{\mathrm{c,\ electrostatic\ breakdown}}$ is the collected charge density by the electrostatic breakdown. The $\sigma_{\mathrm{triboelectrification}}$ largely determines the occurrence of electrostatic breakdown, and the optimization of $k$ and $\sigma_{\mathrm{c,\ electrostatic\ breakdown}}$ is crucial for the enhancement of output performance. In addition, it should be noted that CCE cannot collect all the triboelectrification charges and the ratio of effective charges involved in the breakdown process to triboelectrification charges is consistently less than 1.

A theoretical model was suggested to explain the electrostatic breakdown process of DC-TENGs.[123] The model identifies three discharge zones around a sliding-mode DC-TENG, labeled as 1st domain, 2nd domain, and 3rd domain (Figure 4c). The area of electrostatic breakdown between CCE and triboelectric layer is 1st domain, which was discussed above. In this case, the charge flow process can be reckoned as the cascaded-capacitor-breakdown model, with the equivalent circuit comprising a capacitor $C_{\mathrm{CCE\text{-}TL}}$ (facing surfaces of CCE and triboelectric layer before breakdown), a Zener diode $D_{\mathrm{Z,CCE\text{-}TL}}$ as the judgment for breakdown, and an equivalent resistor $R_{\mathrm{CCE\text{-}TL}}$ (when the breakdown takes place, as shown in Figure 4d (red part). Then, the area of electrostatic breakdown between FE and triboelectric layer is defined as 2nd domain, exhibiting the same electric characteristics as 1st domain (the electrostatic breakdown between conductor and dielectric layer). Finally, 3rd domain is the area of electrostatic breakdown between CCE and FE (conductor and conductor), which can be easily observed in the presence of high resistance or capacitive loads accompanied by bright arcs. Combining the above processes, the equivalent circuit model of sliding-mode DC-TENG (Figure 4d) and the comprehensive "cask model" was obtained.

CS mode DC-TENGs with different working principles have also been investigated for a broader range of application scenarios. As demonstrated in Figure 4e, overcharged area discharges can be obtained by employing asymmetrical contact pairs (different in contact area).[124] The asymmetrical structure, proper soft/rigid contact pairs, and enough soft layer thickness are necessary premises for the configuration of such TENGs. Additionally, another typical configuration of CS mode DC-TENGs with three electrodes is shown in Figure 4f. Charges move via the external circuit from the bottom to the top electrode first and then return via the ionized air channel created by electrostatic breakdown, generating a continuous pulsed DC output by repeatedly separating and contacting.[125]

## 3. Basic Tribo-Dielectric Materials and Charge Transport Mechanism

Dielectric materials and the dynamic behavior of surface/bulk charges are important research topics of TENGs, which are also


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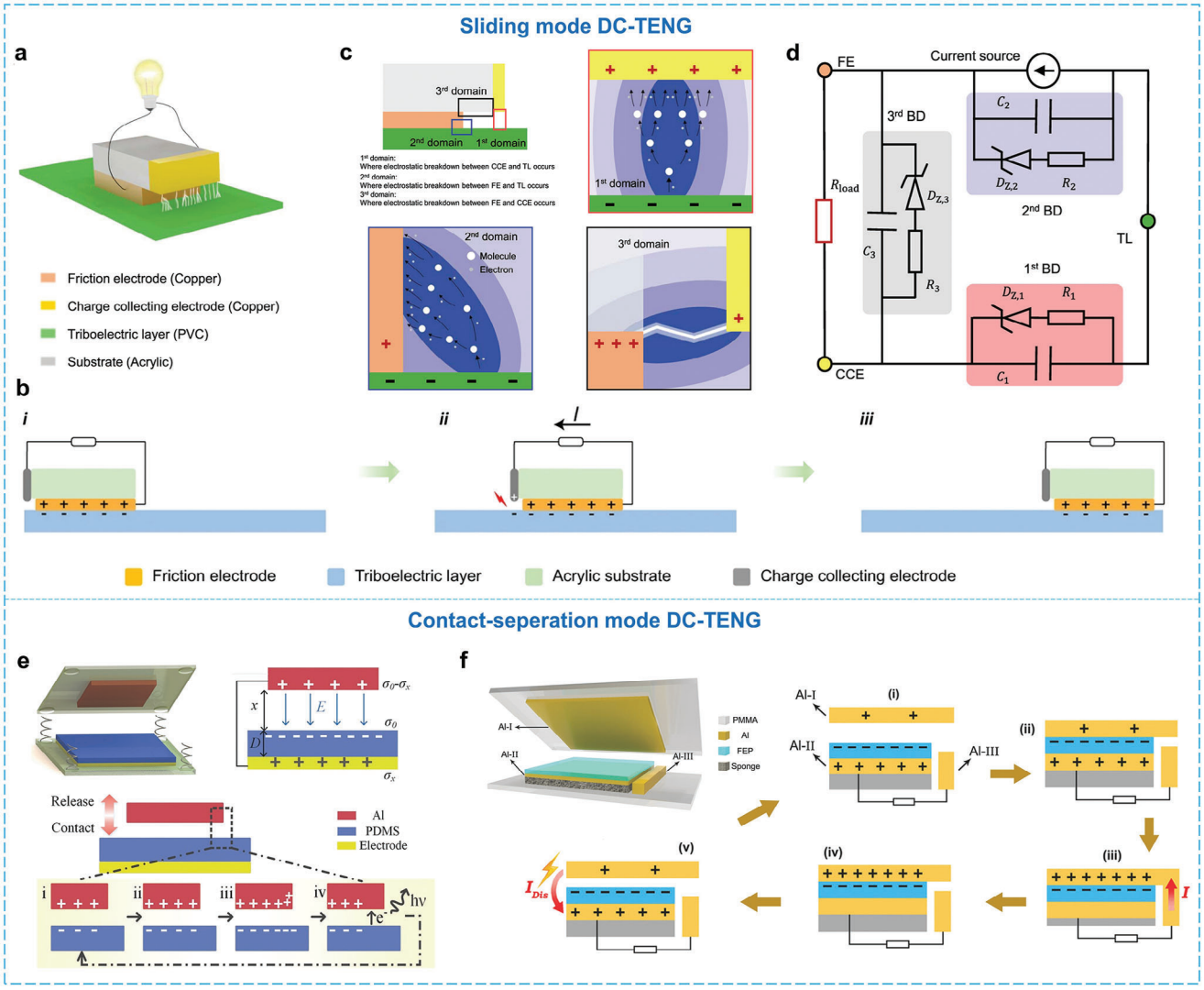


**Figure 4.** Mechanism of DC-TENG. a) Schematic of the sliding mode DC-TENG. b) Working principle of the sliding mode DC-TENG. c) Discharge domains of the sliding mode DC-TENG. d) Equivalent circuit of the sliding mode DC-TENG. e) The CS mode DC-TENG with asymmetrical contact pairs. f) The CS mode DC-TENG with three electrodes. a) Reproduced with permission ref. [123]. Copyright 2023, the authors. b) Reproduced with permission ref. [122]. Copyright 2021, the authors. c,d) Reproduced with permission ref. [123]. Copyright 2023, the authors. e) Reproduced with permission ref. [124]. Copyright 2016, WILEY-VCH. f) Reproduced with permission ref. [125]. Copyright 2018, WILEY-VCH.

crucial elements determining the output performance. In this section, we first introduced the typical solid, liquid, and gas dielectric materials adopted for TENG to cover basic material selection principles. Subsequently, we summarized the charge transport mechanism of dielectric materials to facilitate the reasonable selection of materials and maximum utilization of generated tribo-charges.

## 3.1. Common Dielectric Materials for TENG

### *3.1.1. Solid Dielectric Materials*

Solid dielectric materials acting as triboelectric negative/positive or triboelectric active layers are crucial for high-performance TENGs. The triboelectric series describes and ranks the material regarding their trend to gain or lose electrons during CE, which has been recognized as a helpful tool for TENG material selection. Basically, the position of dielectric materials in the triboelectric series determines the charge transfer efficiency. The first triboelectric series was established in the 18th century, then various scholars gradually updated and expanded it.[128,129] In 2019, Zou et al. measured and quantified the triboelectric charge density (TECD) of various dielectric polymers with respect to liquid metal.[126] The normalized TECD-based triboelectric series (the TECD of PTFE with a thickness of 20 mil (≈0.5 mm) was considered as the reference) was obtained, as shown in **Figure 5a**. It can be found that the fluoroelastomer rubber, polyvinyl chloride (PVC), PTFE possess the highest TECD, whereas the opposite polarity TECD is found in oil-resistant


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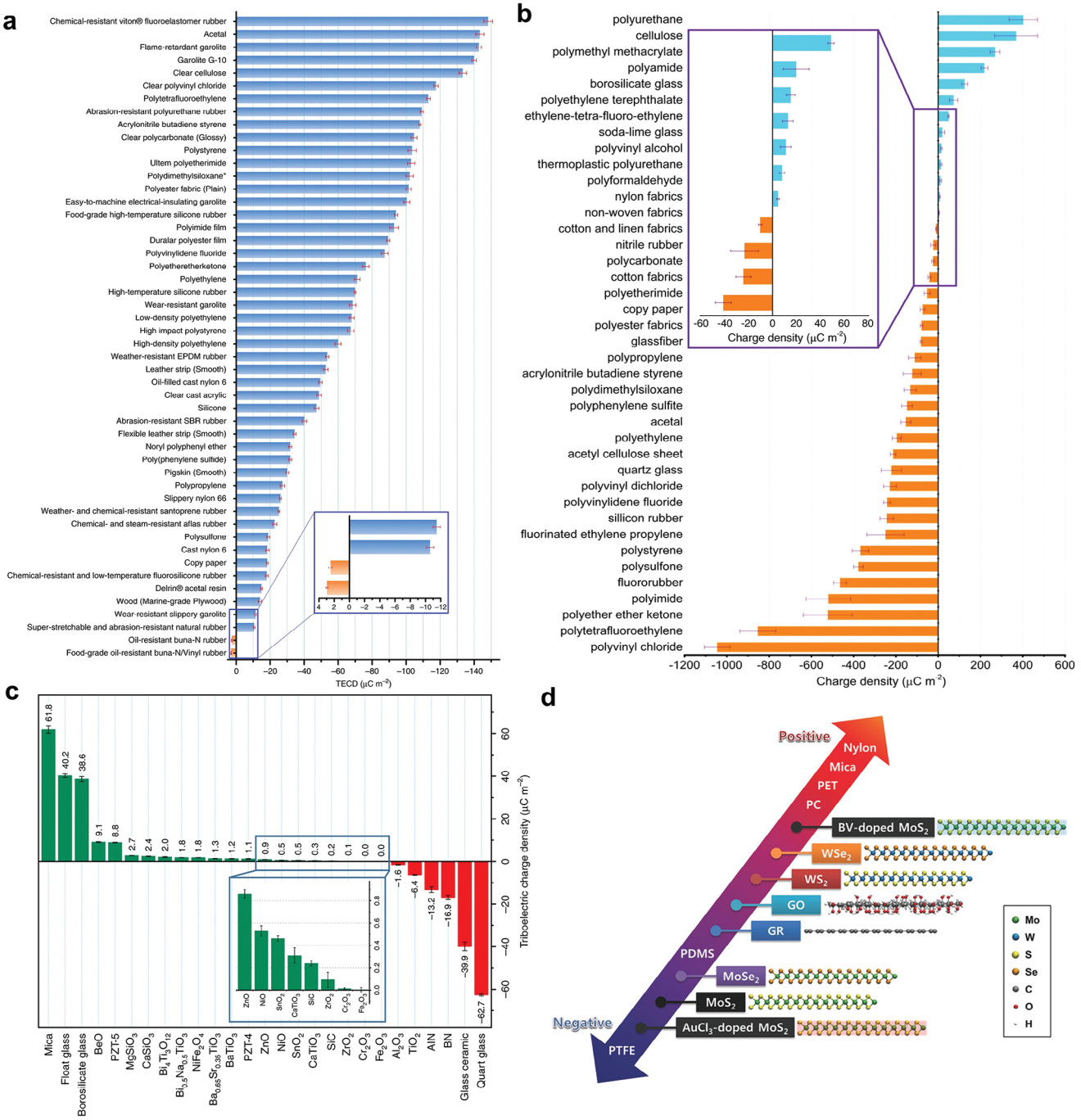


**Figure 5.** Triboelectric series of solid dielectric materials. a) The quantified triboelectric series of various dielectrics. b) Triboelectric charge density of various materials in vacuum conditions. c) Triboelectric charge density of various inorganic nonmetallic materials. d) Triboelectric series of 2D materials. a) Reproduced with permission ref. [126]. Copyright 2019, the authors. b) Reproduced with permission ref. [127]. Copyright 2022, the authors. c) Reproduced with permission ref. [126]. Copyright 2019, the authors. d) Reproduced with permission ref. [129]. Copyright 2018, Wiley-VCH.

buna-N rubber and food-grade oil-resistant buna-N/vinyl rubber. This work established the basis for selecting dielectric solid materials for TENG.

Considering the impact of air breakdown effect, the standardized measurement of intrinsic triboelectric charge density for dielectric materials was further conducted at the vacuum condition, and the updated triboelectric series was given in Figure 5b. It was found that PVC and PTFE tend to obtain electrons, whereas polyurethane (PU) and cellulose are prone to losing electrons.[127] Furthermore, we can better comprehend the discrepancy between the actual and maximized output performance of dielectric material due to the air breakdown effect by adopting the


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proposed universally standardized technique for determining the maximum TECD. Besides, the TECD of inorganic nonmetallic materials was also quantified to obtain the triboelectric series, as shown in Figure 5c.[126] Most inorganic dielectric materials possess positive TECD, indicating their electron-donating property. Besides, it was found that the triboelectric performance is highly related to the work functions, while the dielectric constant demonstrates little influence on the CE performance. A higher output is provided by a larger work function difference between two dielectric materials because there are more discrete surface states for electrons. In addition, the triboelectric series of 2D materials was also reported by Seol et al. According to Figure 5d, all the 2D materials located at the negative side of the triboelectric series, and $MoS_2$ exhibited the most negative electron-accepting ability.[129] This work guides the utilization of 2D materials for functional TENG.

Regarding dielectric materials for DC-TENG, the select rule of dielectric materials was also proposed by considering stability, charge utilization rate, friction coefficient, polarization, and surface charge density.[122] According to the working mechanism of DC-TENG discussed in Section 2.3., the low friction coefficient ($\mu$), low polarization of the triboelectric layer, high surface charge density ($\sigma_{SCD}$) is desired for the high output performance. Specifically, the threshold value of $\mu$ and polarization was set to ≈0.4, and ≈0.6 μC m$^{-2}$ at 10 kV m$^{-1}$. As shown in **Figure 6a**, polydimethylsiloxane (PDMS), Nitrile rubber, and poly(styrene) (PS) is applicable to DC-TENG owing to the relatively large friction coefficient. The utilization of PTFE, fluorinated ethylene propylene (FEP), poly(vinylidene fluoride) (PVDF), PVC offers larger $\sigma_{SCD}$ (>0.15 mC m$^{-2}$), whereas the polyamide (PA), polyethylene terephthalate (PET), polypropylene (PP), polyethylene (PE), and polycarbonate (PC) presents inferior $\sigma_{SCD}$ (Figure 6b). Besides, the relatively strong polarization effect of PVDF and polyetherimide (PEI) brings low DC output (Figure 6c). The relationship between $\sigma_{SCD}$ and the effective charge density of DC-TENG ($\sigma_{DC}$) given in Figure 6d indicates that PVC, PFE, PTFE are the ideal dielectric materials for high-performance DC-TENG, followed by polyimide (PI), polyetheretherketone (PEEK), and polyphenylenesulphide (PPS). Balancing the trade-off between the properties of dielectric material and the high-output principle of devices is particularly crucial.

### 3.1.2. Liquid Dielectric Materials

The triboelectric performance of various liquid dielectric materials has been widely studied. Regarding the physical properties, Pan et al. pointed out that liquid dielectrics with higher polarity and larger dielectric constant present superior output.[130] The role of viscosity and surface tension of liquid dielectric on CE performance was also considered. The dynamic motion of impinging droplets on the solid interface (e.g., deposition, spreading, and rebounding, etc.) is strongly correlated with characteristics of liquid dielectric, including droplet density ($\rho_1$), surface tension ($\gamma_1$), and viscosity, and the diverse motion states exhibit significant differences in CE.[131] First, it was found that the droplet with low surface tension easily breaks and brings residual liquid to the surface. When the droplet begins to spread upon contact with the solid surface without splashing, the inertia mainly determines the initial droplet spreading during descent, and the spreading radius ($R$) increases exponentially over time ($t$) (Equation (15)).[132] After a few milliseconds, the spreading phase is suggested to be influenced mainly by viscosity, showing slower dynamics and following the empirical formula Equation (16)

$$R \propto t^{0.5} \tag{15}$$

$$R \propto t^{0.1} \tag{16}$$

The impinging droplet will travel across the surface until it reaches its maximum radius $R_{max}$, which ultimately detaches from the surface. The ratio of the ultimate droplet radius to the beginning one ($R_0$) is defined as the maximum spreading factor $\beta$, which can be utilized to depict the actual contact area of liquid–solid contact angle, enabling the forecasting of contact angle behavior for different droplets on a solid surface

$$\beta = \frac{R_{max}}{R_0} \tag{17}$$

Further, the droplet remaining on the solid surface can adversely affect the CE performance. To this end, an estimation formula for the largest size of a droplet that can stick to the solid surface ($D_{max}$) was provided, which is ascertained from balancing gravitational and capillary force

$$D_{max}\gamma_l(\cos\theta_r - \cos\theta_a) = \rho_l V_l g \sin\theta \tag{18}$$

where $V_l$ is the droplet volume; $\theta_r$, $\theta_a$, and $\theta$ are receding contact angle, advancing contact angle, and surface inclination angle; $g$ is gravitational acceleration. If the droplet has a maximum size in the spherical cap shape, the following relationship also exists

$$V = \frac{\pi D_{max}^3}{24}\left(2\csc^3\theta_s - 3\cot\theta_s\csc^2\theta_s + \cot^3\theta_s\right) \tag{19}$$

where $\theta_s$ is the static contact angle of such droplets. Besides, the contact time ($t_c$) of a droplet with the dielectric surface is crucial in liquid–solid charge exchange, as the prolonged contact time might hinder the effectiveness of charge transfer. The expression is as follows, where $K_c$ is a constant coefficient (dependent on the solid surface properties).

$$t_c = K_c\sqrt{\frac{\rho_l R^3}{\gamma_l}} \tag{20}$$

The unified CE performance evaluation of liquid dielectric materials remains challenging, considering that both ion transfer and electron transfer exist in liquid–solid CE and will inevitably influence the formation of EDL as well as the triboelectric performance. More recently, Yoo et al. established the first liquid triboelectric series, as shown in **Figure 7a**.[133] The induced charge ($Q_{IN}$) of sorbitol F solution ranks the highest, followed by the glycerol, phosphate-buffered saline and DI water. The role of chemical group, ions valence, ions concentration on the triboelectric performance was systematically clarified. According to Figure 7b, the hydroxyl group in liquid facilitates the output performance, while the hydrocarbon group hinders the triboelectric effect. The


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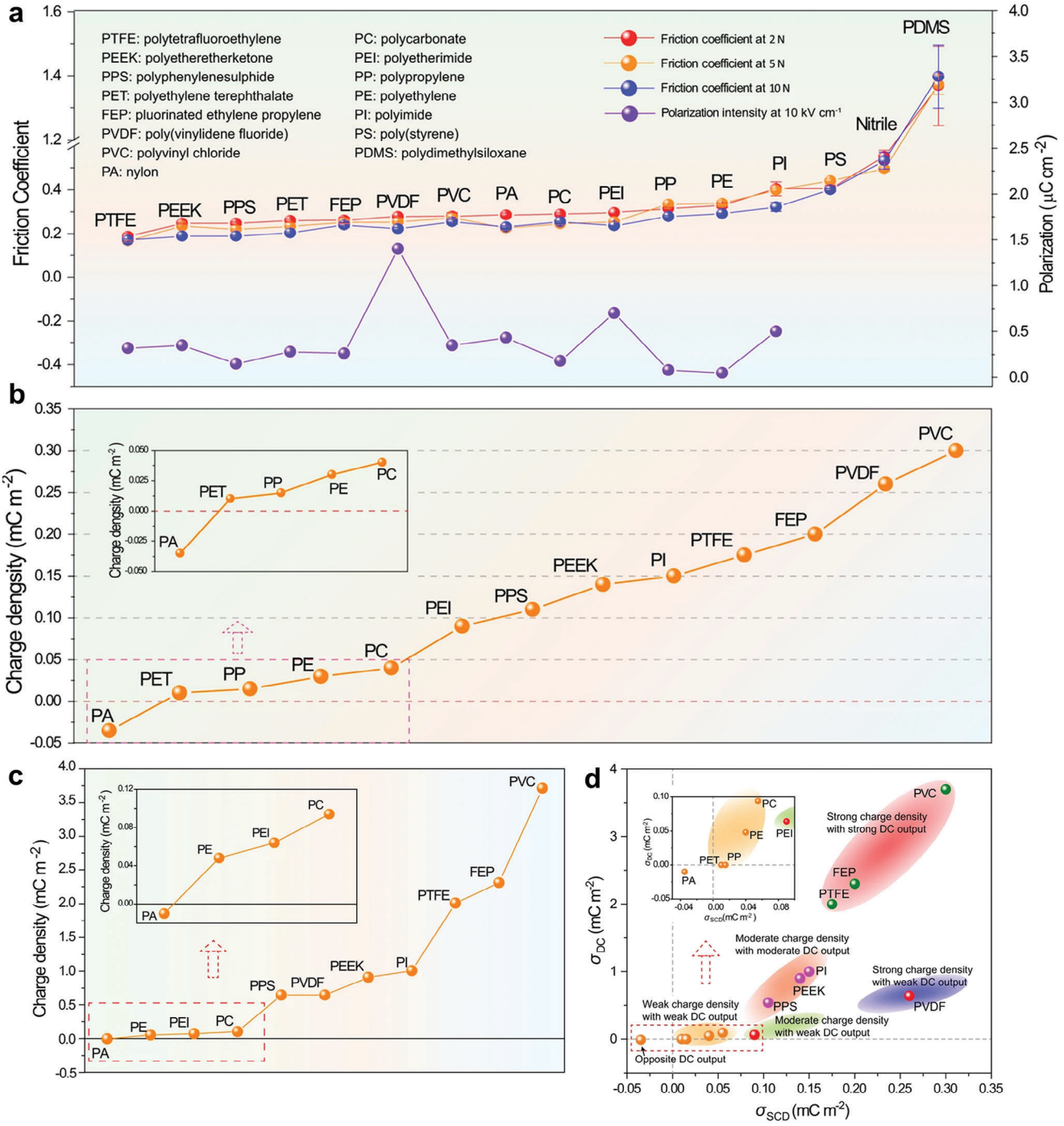


**Figure 6.** Triboelectric dielectric materials for DC-TENG. a) The friction coefficient and polarization of various tribo-dielectrics. b) The surface charge density of different tribo-dielectrics. c) Effective charge density of microstructure-designed DC-TENG. d) Relationship between surface charge density and effective charge density of various tribo-dielectrics. Reproduced with permission ref. [122]. Copyright 2021, the authors.

enhancement effect is ascribed to the hydrogen bonds that favor electron transfer at the liquid–solid interface.[134] Organic dielectrics demonstrate inferior triboelectric performance owing to the van der Waals force based interface interaction. In addition, it was reported that the existence of benzene groups brings a suppression effect on surface charge.

As to oil dielectrics, Li et al. investigated the CE performance of alkanes, cycloalkanes, and aromatic hydrocarbons. The saturated triboelectric output of butyl benzene is the highest, followed by the cyclooctane, *n*-hexadecane, and *n*-dodecanoic (Figure 7c). Subsequently, the frontier molecular orbital and molecular electrostatic potential distribution of oil molecules were analyzed


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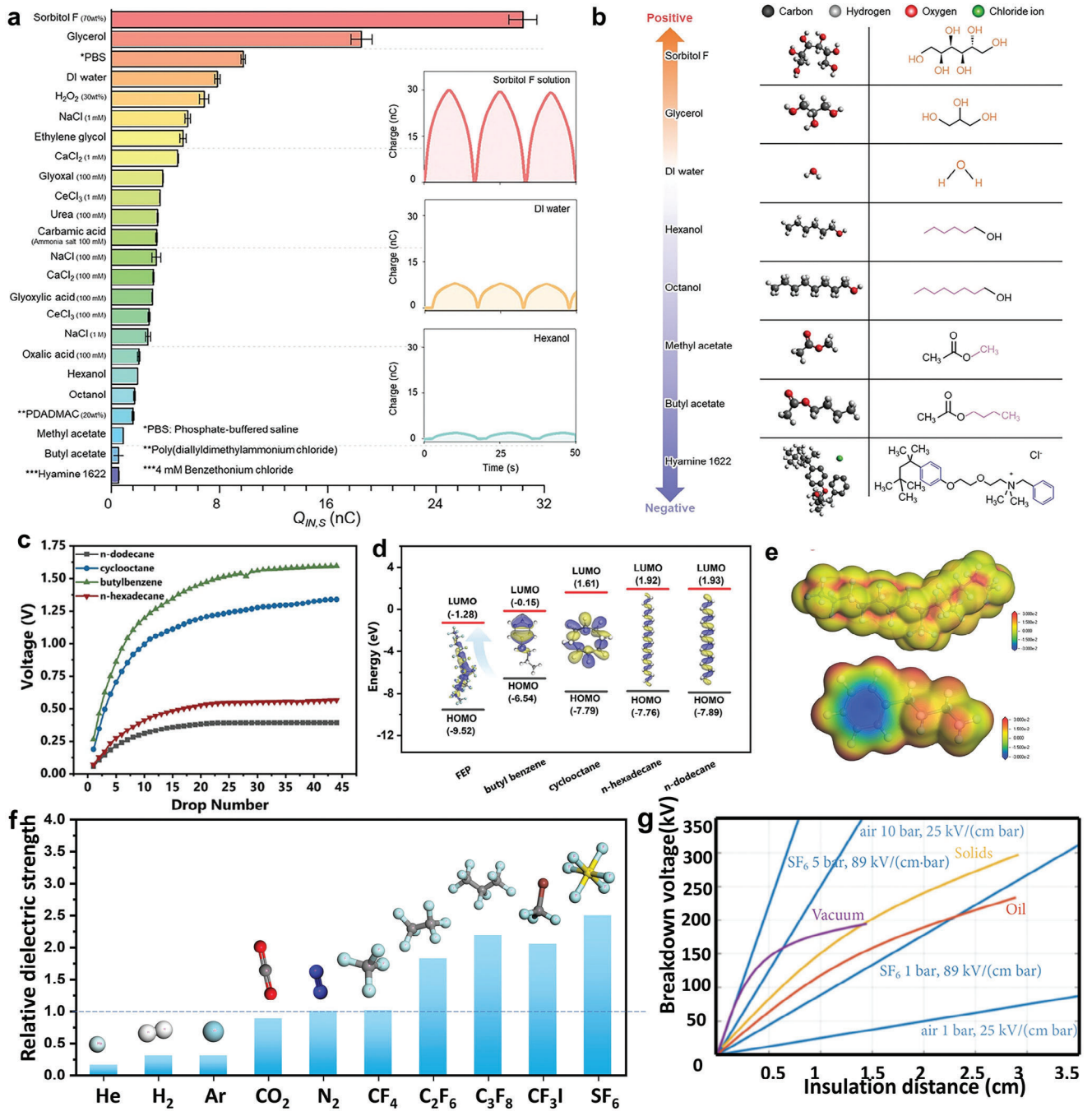


**Figure 7.** Liquid and gas dielectrics for TENGs. a,b) The liquid triboelectric series. c) The triboelectric output voltage of butylbenzene, cyclooctane, *n*-hexadecane, and *n*-dodecane. d,e) The frontier molecular orbital and molecular electrostatic potential distribution of various oil molecules. f) The dielectric strength of various gas dielectrics. g) The dielectric strength of various gases at different pressures, and insulation distance conditions. a,b) Reproduced with permission ref. [133]. Copyright 2023, the authors. c–e) Reproduced with permission ref. [135]. Copyright 2023, the authors. g) Reproduced with permission ref. [139]. Copyright 2023, the authors.

based on the DFT. It was reported that the smaller energy gap between the lowest unoccupied molecular orbital (LUMO) of solid dielectric and the highest occupied molecular orbital (HOMO) of liquid dielectric is beneficial for charge transfer (Figure 7d). Besides, molecules with concentrated electron orbital wave function and pronounced electrostatic potential (ESP) demonstrate higher CE performance (Figure 7e).[135] This work indicates that using quantum chemical calculation is an effective strategy for understanding the triboelectric properties of organic liquid dielectrics.



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#### 3.1.3. Gas Dielectric Materials

Triboelectric charge density on the solid dielectric surface is confined by the air breakdown effect.[127] Although gas dielectric materials have been widely used in industries such as power system, research on gas–solid CE is still in its infancy. Figure 7f compares the dielectric strength of various common gas materials.[136] Specifically, fluorine gases such as hexafluoroethane ($C_2F_6$), octafluoropropane ($C_3F_8$), trifluoroiodomethane ($CF_3I$), sulfur hexafluoride ($SF_6$) exhibit superior dielectric strength compared to that of $N_2$. The relationship between gas dielectric strength and its molecular structure has been investigated, and the specifics are available in references.[137,138] Besides, enhancing gas pressure or building a vacuum environment will also deliver better breakdown performance as the collision ionization process will be suppressed (Figure 7g).[139] The suppression of CE breakdown could be realized by employing the gas dielectrics as mentioned above or altering the environmental conditions, which will be discussed in Section 4.3.1.

### 3.2. CE Charge Transport for TENG

The TENG output depends significantly on the triboelectric charge dynamic equilibrium between generation and dissipation.[139–141] Attention should also be directed to the charge distribution of typical dielectric materials as it is a macroscopic representation of how triboelectric charges are generated and evolved and reflects the dynamic behavior of charges, including generation, migration, dissipation, and storage. In this section, we first introduced the charge distribution properties of dielectric materials after CE. Then, the specific charge dissipation mechanisms were outlined.

#### 3.2.1. Charge Distribution after CE

The distribution of surface charges on dielectric material after CE is essential for further understanding the macro properties and the TENG output performance, which could be obtained by dust diagram method,[142] Kelvin probe force microscope,[143] electrostatic probe method,[144] and Pockels effect method.[145] The dust diagram method utilizes the mutual attraction of opposite charges to sprinkle the charged powder on the dielectric, displaying the surface charge distribution with color differences. However, it cannot quantitatively measure the surface charges and belongs to the destructive measurement method. The Pockels effect method operates on the basis of the linear electro-optical effect, that is, the change of its refractive index is proportional to the magnitude of the applied electric field, to realize the measurement of surface charge distribution. This technique is limited to thin transparent dielectrics as it requires variations in light intensity to reflect electric field change. The Kelvin probe force microscope and electrostatic probe methods possess similar principles, which belong to the noncontact measurement of the surface potential of the dielectric surface. Notably, the measured surface potential is generated by the surface charges, and further calculation is required to obtain its distribution.

The surface charge distribution properties of solid–solid CE have been focused on and explored since 1940. Ideally, the two dielectrics will create spatially homogenous opposite-polarity charge distributions after CE, while the results that are actually obtained are spatially inhomogeneous. **Figure 8**a shows that each dielectric features a "mosaic" of positive and negative surface charges. This phenomenon was proven to be the inherent feature of CE and derives from the electrostatic discharges that occur between little separating gaps (Figure 8b).[146] It was concluded that the transferred charge upon solid–solid CE is not evenly distributed, and spontaneous charge polarity inversion exists. Meanwhile, CE could reflect the properties of dielectric materials as well as the gas medium surrounding them. The gas discharge phenomenon during contact separation must be considered for further investigation of the mechanism and high-performance TENGs.

Regarding liquid–solid CE, surface charges produced at the dielectric surface will generate an electrical field that further affects the subsequent droplet movement properties. As shown in Figure 8c, Sobolev et al. explored the spontaneous charging behavior of sliding droplets and found that the electrostatic force of the first droplet increases sharply and trends to saturation. As the accumulated surface charge density is relatively high, the electrostatic force increases with slide distance for the subsequent droplets.[146] Thus, the electrostatic force induced by the surface charges could significantly impact the droplet motion behavior, especially for the low-permittivity dielectrics. More recently, the charge density distribution of the dielectric surface was visualized. As illustrated in Figure 8d, a pixeled droplet TENG with a high-density electrode array was employed to in situ monitor the transferred charge.[147] It was found that the water droplet sliding on the dielectric film will generate nonuniform surface charge distribution along the paths, where the induced charges at the droplet center part are almost three times higher than that of the edge part. The transferred charge decreases and approaches saturation as the number of droplets increases, ascribed to the shielding effect of EDL formed during CE. Therefore, the formation of EDL or the adsorption of ions on the dielectric interface hinders the electron transfer process.

#### 3.2.2. Charge Dissipation Mechanism

In general, the dissipation and generation of dielectric surface charges occur simultaneously, and the dynamic equilibrium charge density determines the triboelectric output. Besides, the noncontact TENG relies only on relative motion to generate electricity.[148] Although this form of TENG has obtained more objective surface charge densities, the process of charge dissipation after separation significantly impacts its output characteristics.[90] Clarifying the charge dissipation mechanism on the dielectric surface is crucial to improving the stability and durability of TENG output. Generally, three mechanisms were considered responsible for surface charge decay: gas neutralization, bulk neutralization, and surface conduction, as shown in **Figure 9**a.[149-151]

Gas Neutralization—The contribution of "gas neutralization" describes the recombination and compensation of surface charges by charged carriers resulting from natural radiation. Being exposed to the electric field produced by the surface charges, some of the charged carriers can arrive at the


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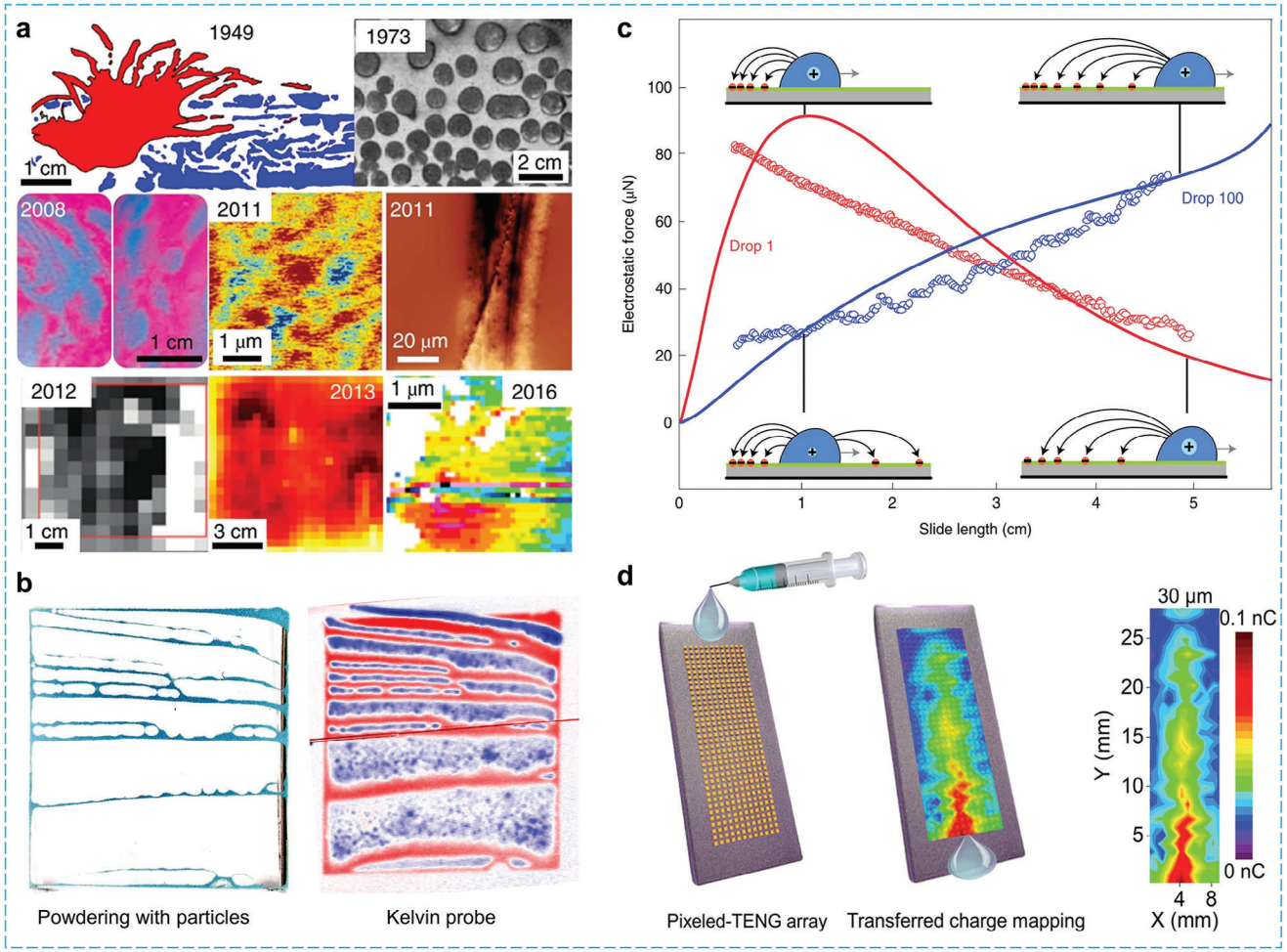


**Figure 8.** Surface charge distribution of CE. a) CE charge mosaics reported by previous literature. b) Surface charge distribution of a stamp visualized by particles and Kelvin probe. c). The electrostatic force of the first and 100th droplets upon liquid-solid CE. d) Surface charge distribution of the liquid-solid dielectric detected by TENG array. a) Reproduced with permission ref. [146]. Copyright 2022, the authors. b) Reproduced with permission ref. [146]. Copyright 2022, the authors. c) Reproduced with permission ref. [113]. Copyright 2022, the authors. d) Reproduced with permission ref. [147]. Copyright 2022, American Chemical Society.

dielectric surface and occupy the surface-free traps.[149,152,153] These charged particles interact with the deposited charges by screening them out or recombining with them so that the macroscopic effect is reflected as a decrease in the surface charge or surface potential. The gas neutralization process is affected by many factors, such as the concentration of charged carriers in the gas, the electric field strength in the vicinity of surface charges, and the energy level distribution of surface traps. Besides, air breakdown is an intense and short process that universally exists in CE or TENG operations. Once air breakdown occurs, the surface discharges will be neutralized immediately, substantially reducing both the charge density and output.

As the gas is easily affected by the instability factors, it is usually difficult to quantify the gas neutralization. Surprisingly, some ingenious methods describe the charge decay characteristics when the gas neutralizes the dominant charge dissipation, which guides the charge stability improvement of non-contact TENGs. As shown in Figure 9b, Zhang et al. investigated the effect of gas neutralization on surface charge dissipation by charging the surface through a needle electrode and then subjecting the sample to three environments. In the case of charge decay in the open air (case A1), the "bell-like" charge distribution is decaying by forming a "crater-like" shape. As to the samples stored in a grounded metallic cylinder (case A2), the rate of charge decay is much slower than that in case A1 and the "bell-like" distribution can be maintained for a long period. It is inferred that gas neutralization plays a dominant role in the dissipation of surface charges. In order to verify this inference, an ion blower was used to create a large number of positive and negative ions in the air (A3), and the "crater-like" distribution could be observed quickly. They also believed that the concentrated electric field in areas of high charge density would attract more free ions, leading to the occurrence of the crater structure.[154] Regarding air breakdown, the Townsend avalanche was used to describe the ionization process in gases, where the free electrons driven by an electric field will collide with gas molecules and release more electrons. Paschen's law could be employed as the critical


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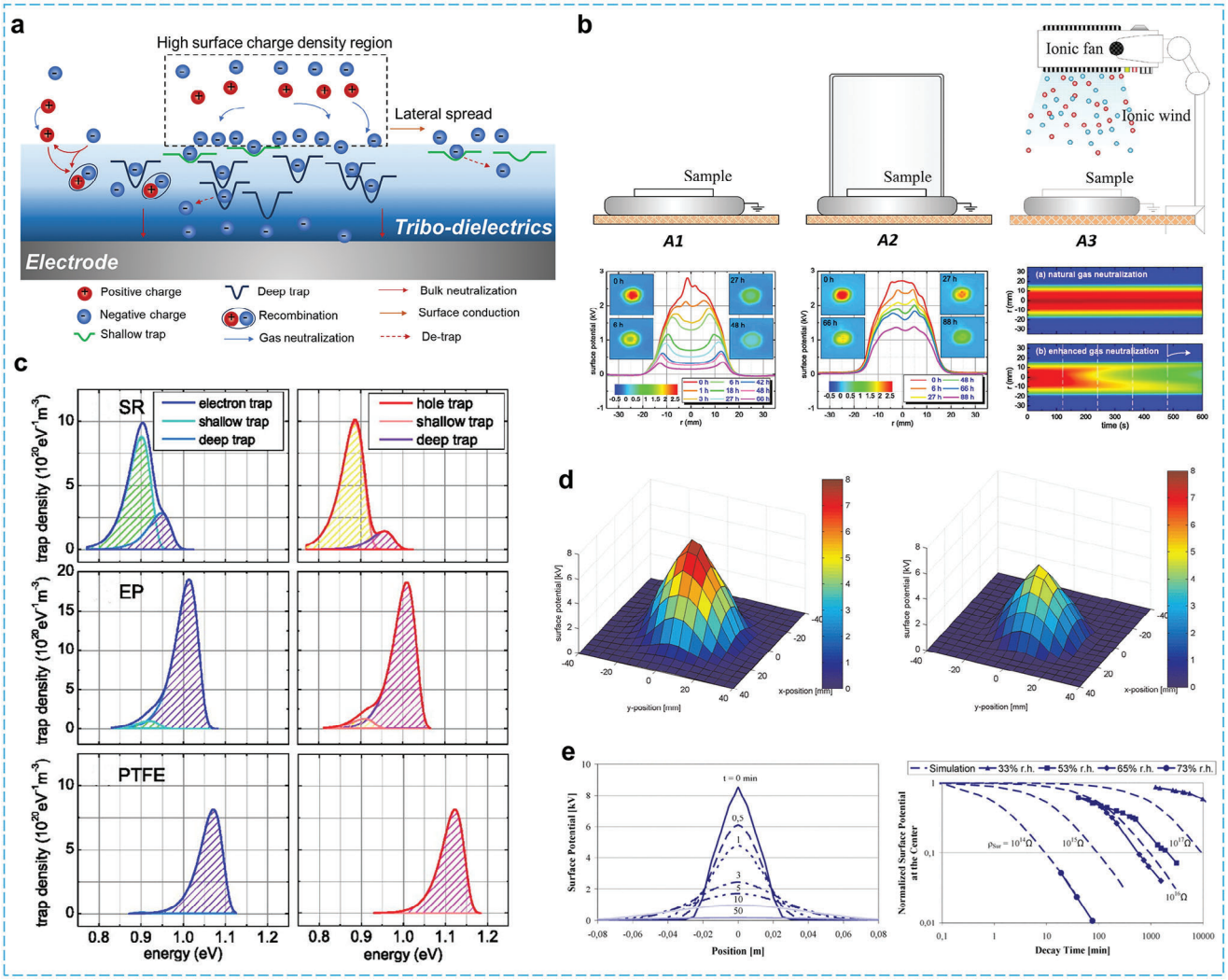


**Figure 9.** The surface charge decay mechanism of dielectric materials. a) The three charge decay mechanism on dielectric surface. b) The dynamic of surface charge decay in open air, in a grounded metallic cylinder and below the ionic fan. c) Surface trap distribution of different materials measured based on ISPD method. d) Surface charge dissipation characteristics when bulk neutralization dominates. e) The influence of surface conduction on surface charge dissipation and the dissipation characteristics under different humidity. a) Reproduced with permission ref. [154]. Copyright 2017, American Institute of Physics. b,c) Reproduced with permission ref. [154]. Copyright 2017, American Institute of Physics. d) Reproduced with permission ref. [149]. Copyright 2008, IEEE. e) Reproduced with permission ref. [165]. Copyright 2009, IEEE.

criterion for the surface charge density of dielectric materials, which will be discussed in detail in Section 4.3.1.

Bulk Neutralization—The bulk neutralization is determined by the intrinsic conduction of the dielectric material, which is closely related to the generation, recombination, drift, and diffusion.[155] Several theories have been developed for interpreting the surface charge decay kinetics due to intrinsic conduction., such as dipolar relaxation, induced conductivity, dispersive transport, trapping, and detrapping.[156] For the dielectric material, there are always more or less impurities (monomers, catalysts, additive residues, etc.) and defects, generating additional potential fields and changing the crystal lattice's original periodic potential field.[157] Moreover, it causes electrons or holes to be bound around impurities or defects, resulting in localized electronic states (trap), thus introducing corresponding impurity or defect energy levels in the forbidden band. If the charge carriers transit time is low compared to the mean characteristic detrapping time, the detrapping kinetics are the main factor determining the charge decay.[156]

Isothermal surface potential decay (ISPD) is currently the most widely used method for testing the energy level of surface traps on dielectric materials.[158] This method consistently maintains the Kelvin probe in close proximity to the surface center, and the probe has the same potential as the measured portion. Thus, it can be assumed that the electric field in the gap between the probe and sample surface is zero, making it impossible for free ions to be attracted to the surface under the probe. The electron and hole energy distributions of representative dielectric materials are shown in Figure 9c.[154] These trap parameters help us understand the materials' intrinsic charge dissipation properties, guiding us to propose effective material property tuning schemes for high-performance TENGs.


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Trap theory gives a material the ability to store charges from a microscopic point of view but cannot explain the dynamic behavior of charge decay when bulk neutralization is dominant. Kindersberger et al. developed a surface charge dissipation model that incorporates the three mentioned paths and pointed out that bulk dissipation results in consistent degradation at each location.[149] The same phenomenon can be observed experimentally on some materials with low bulk resistivity (e.g., silicone rubber), and the charges always present a bell-shaped distribution during decay even in an open environment (Figure 9d).[149] Thus, bulk neutralization has a consistent effect on charge dissipation, leading to charges being dissipated evenly along the path. More recently, the charge trapping failure obtained by regulating the carrier trap state of the of dielectric materials has been utilized to achieve the ultrahigh output for charge excitation TENGs.[159,160]

Surface Neutralization—Surface neutralization is the dissipation of charges along a sample's surface caused by the tangential electric field, which occurs unevenly when surface charges are spread apart. This process results in a lateral expansion of surface charges and can be quantified by means of surface conductivity, which strongly depends on external conditions (surface conditions, humidity, etc.)[161] Numerous studies have shown that surface conductivity of dielectric materials can be modified by plasma treatment,[162] surface coating,[163] and surface fluorination,[164] tunning the surface charge dissipation characteristics. Moreover, humidity has a substantial impact on the output of TENG by influencing the surface condition of triboelectric layers. Water molecules adsorbed on the dielectric surface at high humidity will create a water film, leading to an increase in surface conductivity.[161] Lutz et al. reported that surface conduction leads to lateral spreading of potential distribution in the case of an initial bell-shaped surface potential distribution. A higher surface conductivity causes a faster lateral spreading of the surface charges and a parallel shift of the surface potential decay curve towards shorter decay times (Figure 9e).[165] It is also worth noting that this process alone may only yield a local charge rearrangement while the total number of charges on the surface remains preserved. The surface neutralization mechanism provides a reference for CE charge transport and TENG performance modification in high-humidity environments. Strategies such as constructing hydrophobic coatings to reduce the surface conductivity of tribo-dielectrics to improve device moisture resistance have also been proposed.

## 4. Modification of Dielectric Materials for TENG

Modifying dielectric materials to achieve superior CE performance and diverse functionality effectively expands the triboelectric series and promising application scenarios. In this section, we highlighted the modification strategies of dielectric materials for high-performance TENGs, to clarify the gap between dielectric parameters, surface charge density, and triboelectric output performance. The principles, methods, and typical cases of various modification strategies, including surface, bulk modification, dielectric coordination, dielectric structure optimization were involved. Notably, we concentrate on solid dielectrics while also considering liquid and gas as dielectrics accommodating with the modification of solid materials as the dielectric properties of liquid and gas dielectrics are generally difficult to manage and the material selection for DC-TENGs have certain specificities (mentioned in Section 2.3.).

### 4.1. Surface Modification

#### *4.1.1. Surface Roughness Control*

Boosting the effective surface contact area or surface roughness of the solid dielectric is crucial for facilitating surface charge density. Highly roughened surfaces are mostly engineered by introducing micro-nano-scaled structures through surface patterning (structurization) techniques, such as bringing diverse geometric patterns (micro/nano pyramids, wrinkled structures, etc.).[166,167] The tribo-charges are more easily separated in the microstructured films, forming a larger dipole moment between the electrodes. Moreover, repeatable and cost-effective techniques for mass production were also widely studied, including mold/template method, laser surface patterning, lithography technique, electrospinning, plasma treatment, and electrochemical anodization.

Mold/template method is frequently utilized to manufacture materials with specific morphologies. It demonstrates the capacity to produce regular and uniform polymer patterned arrays, such as lines, cubes, and pyramids, etc., allowing for manufacturing replicas of patterned dielectric films from a single mold (**Figure 10**a(i)).[78] For example, Fan et al. prepared polymer patterned arrays with specific geometric structures and sharp tips. The triboelectric output of the pyramid-featured PDMS patterned device was much greater than that of the unstructured films and the devices with linear and cubic patterns.[78] Besides, the employment of metal molds is reliable for large-scale production and practical applications due to their superior mechanical properties and longevity, and the dielectric film modification procedure is also simple and low-cost.

Laser surface patterning involves the employment of precise laser beams with localized high-temperature processing to melt the dielectric surface, offering well-arranged micro-nano patterns with high spatial resolution and consistency (Figure 10a(ii)).[168] This noncontact method allows for more precise control over the surface structures, which preserves the integrity of changed surfaces and reduces the contamination and oxidation issues in TENG. It is also a flexible method for surface modification in energy harvesting applications since it may be used on various materials. However, the industrial applicability of laser surface patterning is occasionally limited because of the intricate relationship between the unique surface chemistry and nanoscale topography.

Lithography is one of the most flushing techniques for surface texturing in electronics, enabling the creation of micro-nano patterns on the photoresist-coated substrate. Diverse lithography techniques, such as photolithography, ion beam lithography, electron beam lithography, can be utilized for patterning both the triboelectric layer and electrodes. For example, Feng et al. employed the direct image lithography (DIL) technique to create surface microcones on PU with high precision and rapid prototyping (Figure 10a(iii)).[169] The microcones and the introduced surface fluorine from the fluorinated polyurethane (F-PU) layer,


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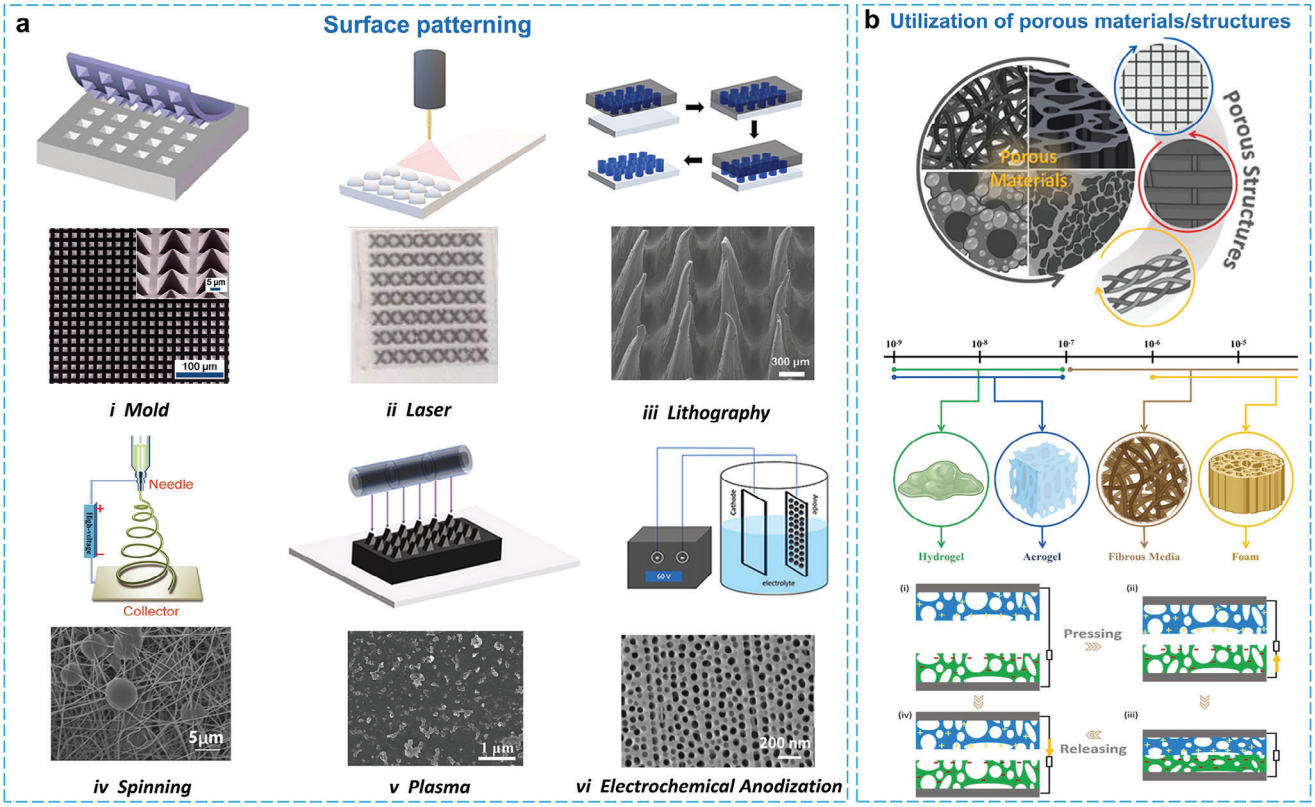


**Figure 10.** Surface roughness control methods for TENG dielectrics. a) Surface patterning techniques to create micro-nano structures on the dielectric surface: a(i) Mold/template method. a(ii) Laser surface patterning method. a(iii) Lithography technique. a(iv) Electrospinning. a(v) Plasma treatment. a(vi) Electrochemical anodization. b) Utilization of porous materials/structures to enhance the surface contact area (materials chosen and working mechanism). a(i) Reprinted with permission from ref. [78]. Copyright 2012, American Chemical Society. a(ii) Reprinted with permission from ref. [168]. Copyright 2020, Elsevier. a(iii) Reprinted with permission from ref. [169]. Copyright 2022, Elsevier. a(iv) Reprinted with permission from ref. [170]. Copyright 2019, Elsevier. a(v) Reprinted with permission from ref. [172]. Copyright 2022, Elsevier. a(vi) Reprinted with permission from ref. [173]. Copyright 2016, American Chemical Society. b) Reprinted with permission from ref. [174]. Copyright 2023, Elsevier.

contribute to the enhancement of triboelectric output by increasing the effective contact area and the electron affinity difference. However, there are still challenges in preventing UV radiation (particularly for photolithography) during the fabrication process, and the requirement for costly machinery as well as the time-consuming design of the prototype, may complicate the production.

Electrospinning is another efficient, cost-effective, and versatile method for creating rough surface structures, which provides nanofiber materials with random microarchitectures in large quantities (Figure 10a(iv)).[170] Introducing protrusions such as beads, micro-nano spheres from electrospinning will bring much opportunity to boost the tribo-layers' contact with the shallowly depressed region, leading to higher triboelectric outputs. Meanwhile, diverse structures such as beaded fibers, core-shell fibers, and hollow fibers can be fabricated by altering the polymer solution parameters (viscosity, volatility, conductivity, etc.), electrospinning parameters (applied high voltage, collection distance, spinning time, etc.), and environmental conditions (humidity, temperature, etc.) to achieve varying triboelectric performance. Additionally, engineering of the electrospinning nanofibers can further increase the surface roughness and the triboelectric output. For example, Sun et al. found that the surface roughness of ZnO/PAN nanofiber rose from 1.59 to 3.07 μm compared to the single PAN nanofiber membrane.[171]

Plasma treatment efficiently and environmentally alters the dielectric surface topography without impacting the interior structure, offering the dual advantage of functionalizing and etching the treated surface simultaneously based on the specific gas used. For instance, an in situ two-step Ar + $O_2$ reactive ion etching plasma treatment was employed to etch the PI film surface by Sun et al. (Figure 10a(v)), resulting in the generation of granular nanostructures and the enhancement of C—O and C═O content, upscaling the surface charge transport and the electron affinity of the dielectric.[172] The surface can be chemically altered using non-thermal plasma, such as dielectric barrier discharge (DBD), with slight physical alterations. However, long-term maintenance of the functionalization level attained on the dielectric surface is still tricky.

Electrochemical anodization is also suitable for dielectric surface patterning, which involves applying an appropriate voltage to a substrate in an electrolyte solution. The modified dielectric surfaces can be further grafted with additional molecular layers based on the intended application. For example, Lee et al. proposed a cylindrical water TENG with its inner surface patterned into superhydrophobic and hydrophilic parts using anodized


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aluminum oxide (AAO), as shown in (Figure 10a(vi)).[173] The diameter and thickness of aluminum oxide nanostructures can be optimized through the anodizing and widening times, thereby maximizing the triboelectric effect and electrostatic induction.

In addition to various processing methods applied to dense dielectric films for surface patterning, porous materials naturally possess high surface roughness, making them excellent dielectrics for increasing the charge density during CE.[174,175] Yoo et al. found that the porous structure may demonstrate a surface area fivefold higher than that of solid dielectric film, and the properties of large area-to-volume result in a larger contact area.[174] Porous materials for TENG mainly include hydrogels/aerogels, fibrous media, and foams, etc., which are typically fabricated through sol-gel process, electrospinning, and molding, respectively (Figure 10b).[174] Textiles and yarns can be assembled to expand TENG functions further. Surface modification of porous materials through physical, chemical, and biological strategies can further enhance the contact area. Generally, there are three categories for porous materials: macropore (pore size greater than 50 nm), mesopore (pore size between 2 and 50 nm), and micropore (pore size less than 2 nm). Diverse pore sizes can be achieved through different structural design and fabrication methods, achieving various desired application characteristics.

#### 4.1.2. Functional Group Modification

The chemical structure and electrical properties of the dielectric are closely related. Therefore, the surface functional groups that can donate or withdraw electrons in the polymer repeat unit can dictate the dielectric macroscopic electrification behavior (i.e., charge density and polarity). Several physical properties were employed to illuminate the correlation between the molecular structure and macroscopic CE behavior, including work function, interfacial barrier, electron affinities, energy band structure, crystallization, defects, etc.[126,176–178]

Regarding solid–solid CE, Li et al. elucidated the contributions of functional groups to CE by investigating polymer dielectrics with comparable leading chains but distinct functional groups on the side chain (**Figure 11a**).[179] It was found that the unsaturated groups (—CF═$CF_2$— and —CF═CF—) on the PTFE molecular chain groups exhibit powerful electron-withdrawing (EW) ability (the C═C bond can enhance the electronegativity of the whole functional group), which is much stronger than that of —$CF_2$—$CF_2$— group. As to the tribo-positive layer, polymer dielectrics containing NH and unsaturated N groups (such as PA) are commonly used (Figure 11b).[176] Furthermore, covalent organic frameworks (COFs), as a class of crystalline porous materials composed of organic building blocks connected through strong covalent bonds, exhibit superior properties such as high surface area and tunable porosity. COFs can serve as outstanding tribo-dielectric materials due to their advantageous periodic conjugate structure and adjustable functional group arrangement. For example, Shi et al. proposed a novel highly fluorinated COF Tp-TFAB as tribo-positive material for TENG.[180] Due to the greatly enhanced electron donating ability of fluorinated Tp-TFAB COF, the developed TENG with PVC and hybrid polyvinyl alcohol (PVA)/Tp-TFAB films illustrated the output of approximately sixfold higher than the conventional device.

As to liquid–solid CE, the role of functional group is more intricate. A fundamental issue is to identify the charge carriers involved (ion, electron transfer) in the CE process. Lin et al. investigated the effects of surface functional groups on the CE between $SiO_2$ surfaces and liquid.[181] Both electron and ion transfer occur during the CE process between modified $SiO_2$ and water, with electron transport being predominant in organic solutions. In addition, Li et al. functionalized PP obtained with amino (–$NH_2$), alkyl group (–$CH_2CH_3$), and fluorine group (–$CF_3$) and found that the ranking of corresponding liquid–solid CE output is as follows: APTES-PP < PP < OTS-PP < PFOTS-PP (Figure 11c).[182] Lin et al. found that the perfluorododecyl-, oxypropyl-, and dodecyl-modified $SiO_2$ samples received negative charges, while the aminopropyl-modified $SiO_2$ sample received positive charges after contact with DI water (Figure 11d).[181] It could be concluded that functional groups with a higher fluorine content impart superior EW ability to the dielectric surface.[179] Besides, the fluorine atoms on the surface can also lower the surface free energy to increase the hydrophobicity of the dielectric surface, while the –$NH_2$ groups impart more hydrophilicity and result in lower output performance.

The surface charge density could be modified by adjusting the quantity and density of functional groups with different electron-withdrawing/donating capabilities, namely chemically tailored molecular surface modification. For example, Shin et al. demonstrated an atomic-level chemical functionalization method, where the PET was functionalized with the molecules containing halogen (Br, F, and Cl) substituted phenyl moieties or the aminated molecules (**Figure 12a**).[79] In another work, Liu et al. chose silane coupling agents with identical main chains but different end functional groups to modify the surface of cellulose nanofibrils (CNFs). They observed that the triboelectric output of $CF_2CF_3$-CNFs and CN-CNFs decreases, while that of $NH_2$-CNFs and SH-CNFs increases (Figure 12b).[183] In addition, ion doping is also an efficient chemical surface modification strategy. Anions and cations can create additional electron-charged and unoccupied states in the solid polymer electrolytes, tuning the triboelectric polarity of dielectric material. For example, Ryu et al. found that the addition of phosphoric acid ($H_3PO_4$), calcium chloride ($CaCl_2$) to PVA brings tribo-negative and positive performance (Figure 3).[184]

Reactive ion etching plasma treatment is also performed by injecting reactive gases into the system for chemical modification.[173] For example, Zhang et al. employed an inductively coupled plasma etching method by adopting fluorocarbon (trifluoromethane ($CHF_3$) in the first step and octafluorocyclobutane ($C_4F_8$) in the second step) to modify PE (Figure 12d(i)).[185] Neutral beam (NB) technology is an advanced plasma-based etching and dielectric surface treatment method that utilizes an external radio frequency bias. During the neutralization process, the target dielectric surface experiences minimal formation of free radicals, charges, or electrical faults. Therefore, thin, dense oxides or nitrides can be formed through NB treatment, creating delicate surfaces with electrical stability. For example, Kim et al. chemically modified the surfaces of PDMS and thermoplastic polyurethane (TPU) by $N_2$ and $O_2$ gas-based NB process (Figure 12d(ii)).[186] It was proved to be an effective treatment for


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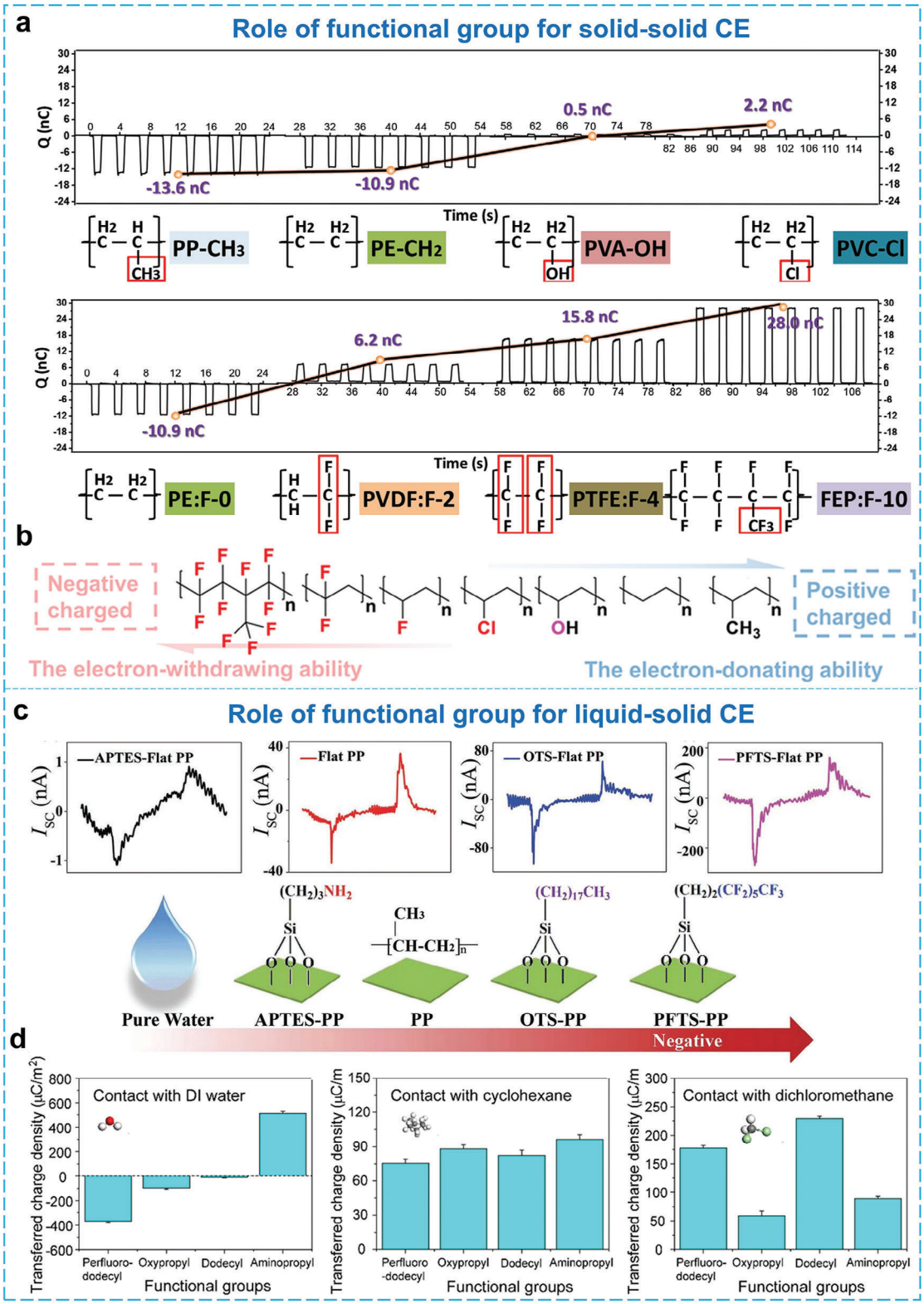


**Figure 11.** Role of dielectric surface functional group for CE. a,b) The solid–solid CE case. c,d) The liquid–solid case. a) Reprinted with permission from ref. [179]. Copyright 2020, WILEY-VCH. b) Reprinted with permission from ref. [176]. Copyright 2023, The Royal Society of Chemistry. c) Reprinted with permission from ref. [182]. Copyright 2019, WILEY-VCH. d) Reprinted with permission from ref. [181]. Copyright 2020, American Chemical Society.



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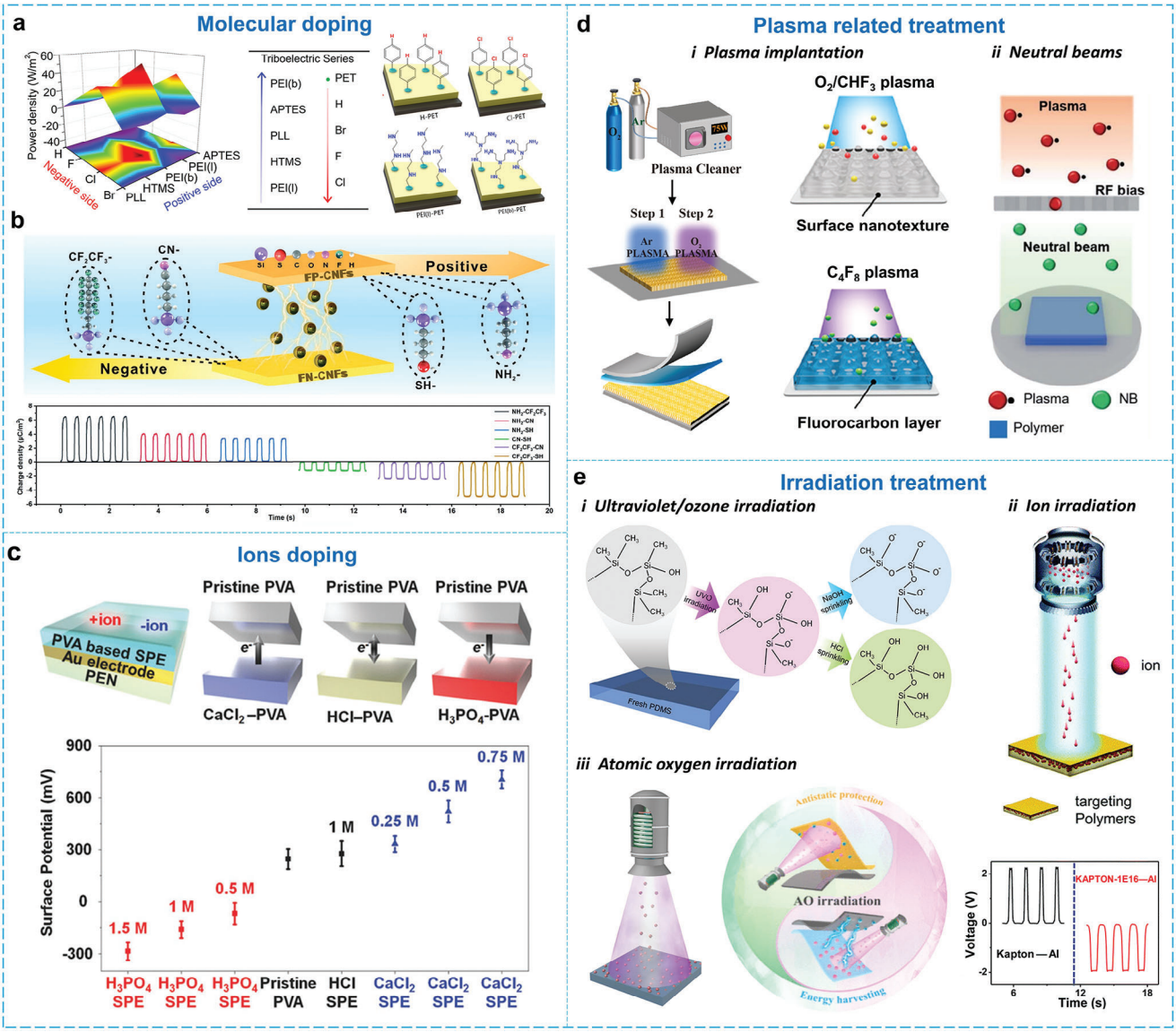


**Figure 12.** Surface functional group modification of dielectric materials for TENG. a,b) Molecular doping technique. c) Ions doping technique. Plasma-related treatment: d(i) Plasma implantation. d(ii) Neutral beams. Irradiation treatment: e(i) Ultraviolet/ozone irradiation. e(ii) Ion irradiation. e(iii) Atomic oxygen irradiation. a) Reprinted with permission from ref. [79]. Copyright 2017, American Chemical Society. b) Reprinted with permission from ref. [183]. Copyright 2021, Elsevier. c) Reprinted with permission from ref. [184]. Copyright 2017, WILEY-VCH. d(i) Reprinted with permission from ref. [172]. Copyright 2022, Elsevier. Reprinted with permission from ref. [185]. Copyright 2021, American Chemical Society. d(ii) Reprinted with permission from ref. [186]. Copyright 2019, The Royal Society of Chemistry. e(i) Reprinted with permission from ref. [187]. Copyright 2015, Elsevier. e(ii) Reprinted with permission from ref. [188]. Copyright 2020, The Royal Society of Chemistry. e(iii) Reprinted with permission from ref. [189]. Copyright 2021, American Chemical Society.

tribo-negative PDMS dielectric, while the $N_2$-NB treatment with high beam energy enhanced the properties of tribo-positive TPU dielectric.

Irradiation treatment is another surface modification with effectiveness and stability, such as ultraviolet/ozone irradiation, ion irradiation, and atomic oxygen irradiation (Figure 12e).[187–189] For example, Li et al. proposed a low-energy ion irradiation method to manipulate the chemical bonds and dielectric surface functional groups directly.[188] It primarily alters dielectric crystal structures through nuclear and electron energy loss, inducing defects while facilitating nanocrystal formation. Ion irradiation results in collisions that break chemical bonds, generating free radicals that react with ions to form new chemical bonds through structural rearrangement. The surface states for CE can be enhanced by these radiation-induced free radicals and chemical bonds, and the formation of functional groups can alter the electron binding energy of dielectric molecules. In one work, Luo et al. presented the atomic oxygen irradiation to modify carbon-based polymers, resulting in functional groups with C and H elements degrading into $H_2O$ and $CO_2$.[189] In contrast,


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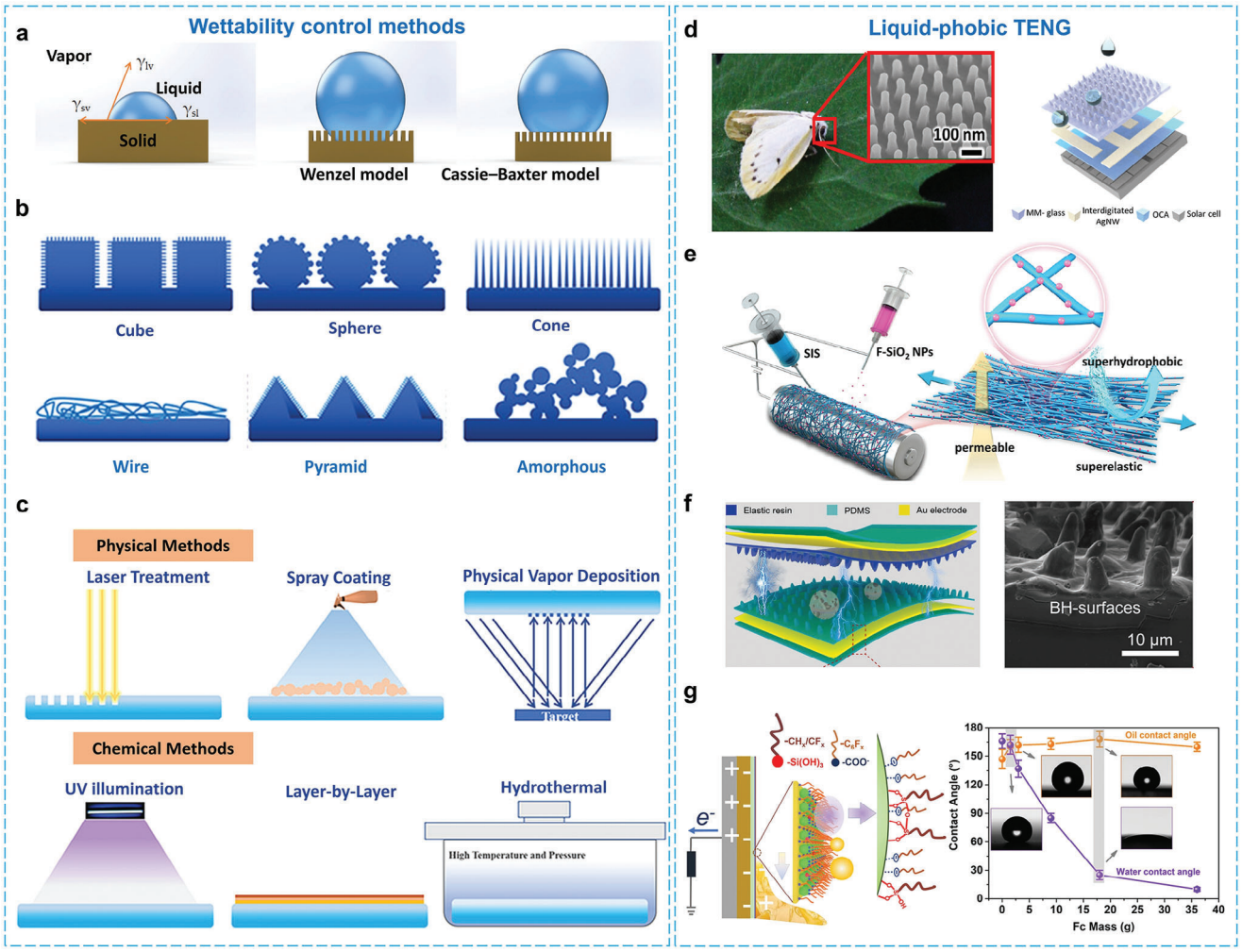


**Figure 13.** Wettability control of dielectrics. a) The Wenzel model and Cassie–Baxter model. b) Micro-/nanoscale structure towards the hydrophobic surface. c) Common super hydrophilic surface fabrication physical and chemical methods. d) Biomimetic antireflective TENG for water energy harvesting. e) Silk-inspired super hydrophobic nanofiber membrane for TENG. f) Inorganic coating-based hydrophobic interface for energy harvesting. g) Oleophobic interface for energy harvesting and intelligent monitoring. a) Reprinted with permission from ref. [190]. Copyright 2023, the authors. b,c) Reprinted with permission from ref. [191]. Copyright 2020, the authors. d) Reprinted with permission from ref. [80]. Copyright 2019, Elsevier. e) Reprinted with permission from ref. [192]. Copyright 2022, Elsevier. f) Reprinted with permission from ref. [193]. Copyright 2022, the authors. g) Reprinted with permission from ref. [194]. Copyright 2022, the authors.

silicon-based polymers can withstand atomic oxygen irradiation because the oxide layer forms on the dielectric surface. Consequently, it can effectively boost electron-donating functional groups to improve the CE performance of PDMS by transitioning from Si–C bonds to Si–O bonds.

### *4.1.3. Surface Wettability Control*

A suitable surface wettability or hydrophobicity (neither too high nor too low) is crucial for liquid-based TENG as it affects the charge transfer properties. Specifically, it has been observed that ion transfer is the predominant mechanism at the hydrophilic dielectric interface, whereas electron transfer is primary for the hydrophobic interface.[117] Meanwhile, the hydrophilic interface reduces the charge transfer and CE response due to the shielding effect of ions in residual water stains. On the other hand, dielectric surface wettability control is also necessary for TENG with self-cleaning capabilities.

Generally, the wettability is determined by surface roughness and surface free energy. The dielectric interface with low surface free energy tends to be hydrophobic, and the standard optimization strategy is employing fluoro-substances via chemical reactions, self-assembly, or coating. The relationship between surface roughness and contact angle (CA) could be explained by the Wenzel model and Cassie—Baxter model, as illustrated in **Figure 13a**.[190] The construction of various micro-nanostructures is a practical approach to enhance the surface roughness (Figure 13b).[191] Specifically, physical techniques like lithography and laser treatment can create regular structures (cubes, spheres, pyramids) (Figure 13c). Various regular micro-nano dual-scale surface structures can also be obtained via


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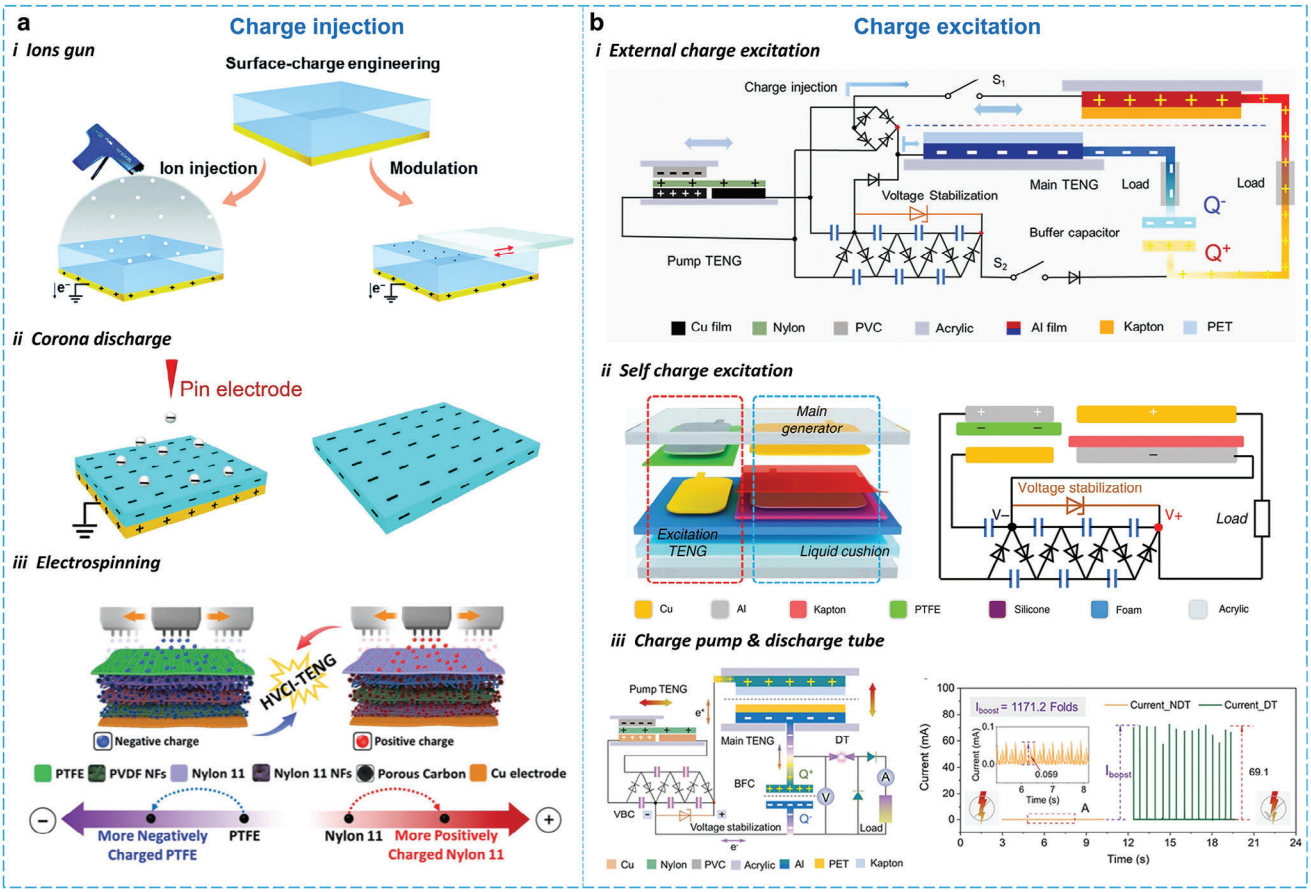


**Figure 14.** Dielectric surface charge density control. Charge injection: a(i) Ions gun. a(ii) Corona discharge. a(iii) Electrospinning. Charge excitation: b(i) External charge excitation. b(ii) self-charge excitation. b(iii) Charge pump and discharge tube. a(i) Reprinted with permission from ref. [195]. Copyright 2021, The Royal Society of Chemistry. a(ii) Reprinted with permission from ref. [196]. Copyright 2016, Copyright 2016, Tsinghua University Press and Springer-Verlag Berlin Heidelberg. a(iii) Reprinted with permission from ref. [199]. Copyright 2022, Elsevier. b(i) Reprinted with permission from ref. [200]. Copyright 2021, Wiley-VCH. b(ii) Reprinted with permission from ref. [201]. Copyright 2019, the authors. b(iii) Reprinted with permission from ref. [202]. Copyright 2022, Wiley-VCH.

sol–gel, hydrothermal, layer-by-layer, and electrodeposition methods. As to micro-/nanowires or fibers, electrospinning or electrospraying is commonly used.

Bioinspired design of superhydrophobic surfaces has also been proposed. For example, Yoo et al. fabricated a biomimetic TENG with outstanding light transmittance and hydrophobicity properties, as shown in Figure 13d.[80] Inspired by silk, Li et al. reported an in situ interlocked superelastic microfibers with superhydrophobicity by simultaneous electrospinning of styrene-isoprene (SIS) and electrospraying of fluorinated silica nanoparticles.[192] As shown in Figure 13e, the low surface energy nanoparticles were in situ embedded into high roughness microfibers through the solvation welding effect, realizing the superior comprehensive performance. In another work, Li et al. designed a hierarchical micro/nanostructure replicated from a lotus leaf with self-cleaning and superhydrophobility.[193] TENG demonstrates superior humidity-resistance performance with ≈11% output loss at 80% R.H. (Figure 13f).

Additionally, the wettability control for other liquid dielectrics, such as oil is also documented for high-performance TENGs. For example, Wang et al. reported a self-powered, highly selective, and long-durable oil-solid TENG to detect impurities in lubricating oil.[194] The 1*H*, 1*H*, 2*H*, 2*H*-Perfluorodecyltriethoxysilane and hydrophilic nano-$SiO_2$ coating were employed to prepare the oleophobic and hydrophobic dielectric interface (Figure 13g). The controlled surface-wetting properties endow the device with a superior ability to monitor contaminants in oil. Thus, by controlling surface wettability, it is possible to achieve triboelectric characteristics that match those of liquid and solid dielectric materials, which in turn aids in constructing high-performance TENGs.

### 4.1.4. . Surface Charge Density Control

The triboelectric output was strongly correlated with the surface charge density on the dielectrics. Charge injection during fabrication or charge excitation can be used to alter the surface charge density of dielectric materials to a certain extent. Specifically, charge injection could be realized by ionization gun, corona discharge, and electrospinning. As shown in **Figure 14**a(i), Wang et al. achieved the maximum surface charge density for FEP (630 μC $m^{-2}$) through air-ionization gun.[81] It was demonstrated that


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ionic charges are more stable than electronic charges, even under extreme humidity conditions.[195]

Another helpful strategy is corona discharge, which uses the high voltage to cause collision ionization of air to produce polar ions. Generally, the needle-plate electrode was employed to provide a highly non-uniform electric field for local gas ionization under DC circumstances. Then, the positive or negative ions produced were accelerated and injected into the dielectric surface, as shown in Figure 14a(ii). Zhou et al. presented the corona discharge injected PTFE-based TENG, exhibiting an output voltage of 2.45 kV, transferred charge density of 240 μC $m^{-2}$ (seven times higher than the untreated group).[196] In another work, Sun et al. designed an ionic/molecular and pH regulated chitosan film with corona charge injection, delivering an output voltage 25 times higher than the initial film.[197] The stability of corona charge injected ions was also verified, sustaining ≈92.6% of its initial performance after 200 000 operational cycles.[198]

Electrospinning has been widely adopted for preparing high-performance micro/nanofiber membranes. In addition to allowing in situ polarization, the high DC voltage used during electrospinning also introduces charges into the membrane. Generally, the surface potential of the electrospun nanofiber membrane is higher than that of the ordinary film, presenting superior output performance. Besides, Cha et al. achieved the triboelectric series control based on the electrospun gradient charge-confinement layer.[199] As shown in Figure 14a(iii), the dispersion of mesoporous carbon spheres into the PVDF or Nylon enhances high-voltage charge injection efficiency during electrospinning, where mesoporous carbon spheres with good charge-transport and charge-confinement capabilities are positioned deep and on the surface of the membrane. As a result, the surface charge densities of PTFE and Nylon 11 were enhanced by 1.5 and 1.8 times, respectively.

Charge excitation is another effective strategy to inject surface charges into dielectric materials, mainly conducted at the device level. For example, Xu et al. reported the first external charge-pumping TENG, where a charge pump TENG was designed to pump charges into the floating layer for storing charges in electrostatic induction.[106] The device delivered an ultrahigh surface charge density of 1020 μC $m^{-2}$. Further, the charge pumping structure for sliding-mode TENG was also proposed. The voltage booster and switch circuit incorporated were utilized for charge density elevation and high output voltage reduction(Figure 14b(i)). As a result, the electricity output was doubled, with a 43.4% enhancement in current and a 63.9% reduction in voltage fluctuation.[200] Moreover, the self-charge excitation strategy was also proposed by Liu et al.[201] The surface charges were directly excited on the electrode, and charge transfer between the primary TENG and ceramic capacitor produced the AC output (Figure 14b(ii)). A Zener diode was adopted to stabilize the output voltage and prevent breakdown and the surface charges were up to 1.25 mC $m^{-2}$.[201] Recently, the decoupled charge pump and discharge tube were proposed to realize a thousandfold boosting instantaneous current of TENG.[202] Specifically, the pump and main TENG operate in the freestanding and CS mode, as shown in Figure 14b(iii). The maximum charges obtained from pump TENG were injected into the main TENG, bringing a relatively high power density of 21.25 W $m^{-2}$.[202] Thus, the maximum release of the dielectric materials was achieved by adopting Zener diode, discharge tube, and external capacitor, further pushing the charge excitation efficacy.

### 4.2. Bulk Modification

#### *4.2.1. Polarization Modification*

According to the TENG parallel-plate capacitor model described in Section 2.1., the charge density $\sigma$ is connected to the capacitance $C_d$ of the dielectric layer in the following manner

$$\sigma = \frac{C_d V}{S} \tag{21}$$

$$C_d = \frac{S\varepsilon_r\varepsilon_0}{d} \tag{22}$$

Obviously, the capacitance $C_d$ is a crucial component that can raise the surface charge density $\sigma$ in TENG, and the primary way of raising capacitance in TENG dielectrics as well as the triboelectric output is to modulate the relative permittivity $\varepsilon_r$ or the thickness $d$ of the dielectric layer.[46] Increasing the relative permittivity constant is a more reliable method because thin dielectric materials are prone to breakdown. On the one hand, the relative permittivity of dielectric materials can be enhanced by introducing dielectric additives (such as high-$k$ nanoparticles) and diverse structural designs. On the other hand, polarization can be induced in the dielectric through external means, including corona polarization, quenching polarization, self-polarization, etc.

From a microscopic perspective, the polarization of ideal dielectric materials mainly includes electronic displacement polarization, ion displacement polarization, and dipole orientation polarization. Various forms of polarization, including interfacial, space-charge, and thermionic polarization, are typically present in dielectric materials. The polarization of dielectric materials requires a particular amount of time to establish under electric fields, and the duration of this period varies depending on the type of polarization. For example, electronic displacement polarization usually takes around $10^{-15}$ s (fast polarization, producing resonance response), whereas interfacial polarization requires ≈$10^{-10}$ to $10^{-2}$ s (slow polarization, producing relaxation response), as shown in **Figure** 15a.[203] Generally, dielectrics used in TENGs are typically subject to different polarizations because of the impact of the electric field produced by tribo-charges.

Surface charge and polarization charge are the primary charges found in dielectric materials. High-voltage corona polarization could introduce charges into polymer dielectrics, commonly used for TENG.[204] As shown in Figure 15b, the DC high voltage between the needle electrode and the grounded plate electrode deflects the dipole moment of the polymer dielectric molecules during the corona polarization process.[205] Subsequently, the high-voltage electrode ionizes the air above the polymer dielectric, introducing negative charges into the dielectric surface. For example, Liu et al. proposed a repeated rheological forging process to enhance the charge storage capacity of FEP by applying corona polarization.[205] It should be noted that corona polarization is limited to nonpolar polymer


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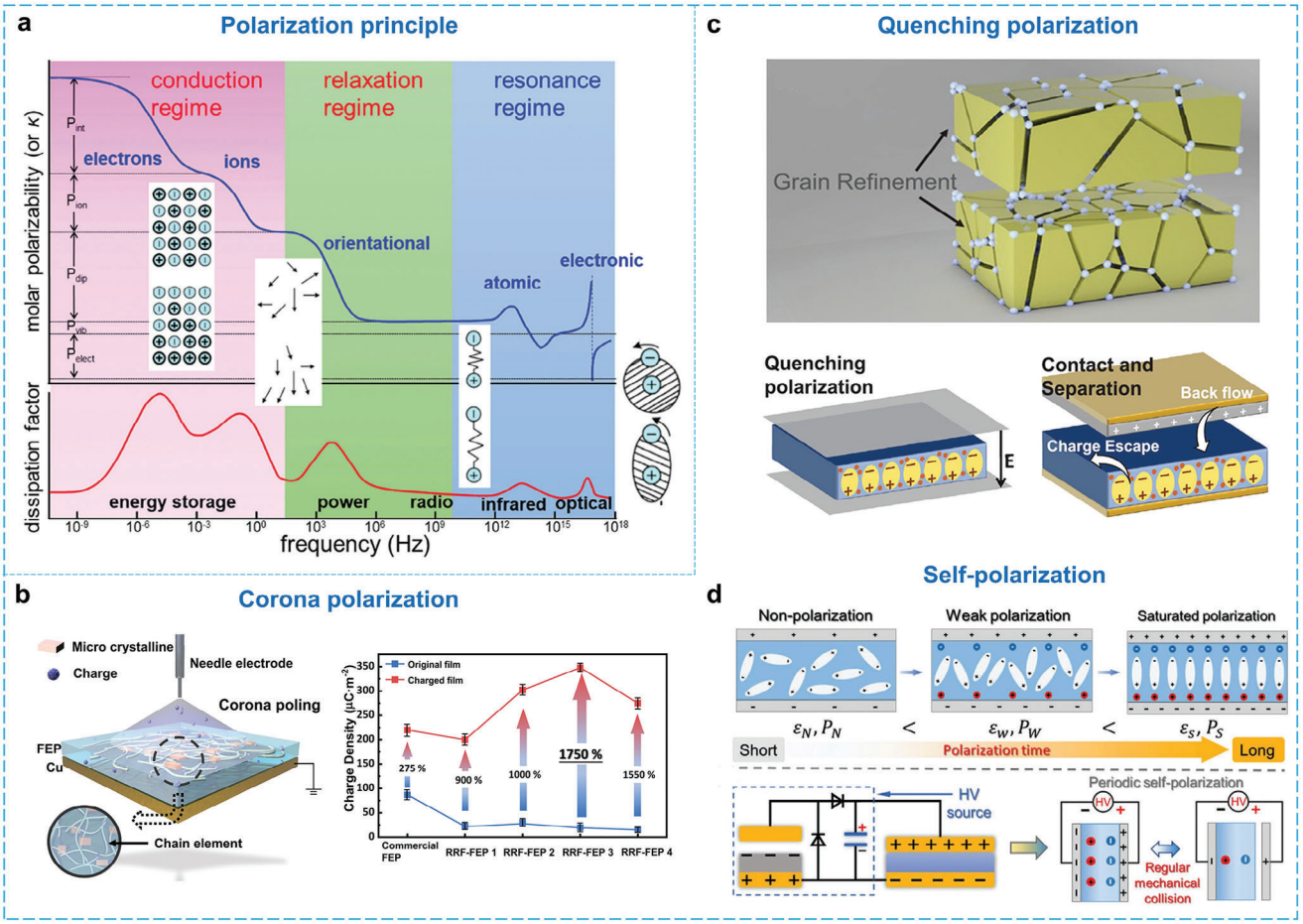


**Figure 15.** Polarization modification of solid dielectric material for TENG. a) Polarizationbility and dissipation as the function of frequency in a dielectric with diverse polarization mechanisms. b) Corona polarization process and the performance enhancement with repeated rheological forging treated FEP. c) The mechanism of quenching polarization for performance enhancement. d) The mechanism of self-polarization for performance enhancement. a) Reprinted with permission from ref. [203]. Copyright 2023, Wiley-VCH. b) Reprinted with permission from ref. [205]. Copyright 2021, the authors. c) Reprinted with permission from ref. [82]. Copyright 2023, Wiley-VCH. (d) Reprinted with permission from ref. [207]. Copyright 2022, Wiley-VCH.

dielectrics with strong electrical insulating properties, as the injected space charges demonstrate poor persistence. Furthermore, the dielectric surface potential distribution is highly irregular, typically exhibiting a bell-shaped pattern along the dielectric radial direction.[206]

As to weak-polar polymer dielectrics, the quenching polarization method is efficient in injecting charges with more stability. Specifically, high temperature and high DC electrical field are required to polarize the polymer dielectric, and liquid nitrogen is employed to quickly cool down the polarization system to freeze the charges inside the dielectrics (Figure 15c).[82] Accordingly, the grains of weak-polar polymer dielectrics can be refined through the treatment process, considerably boosting the interfaces between the crystalline and amorphous regions of dielectric (deep traps), which facilitates the charge injection into the dielectric bulk area and results in high charge density with extended persistence. For example, Liu et al. treated the ethylene chlorotrifluoroethylene copolymer by quenching polarization, offering an ultrahigh surface charge density of 391 μC $m^{-2}$. However, this technique requires more intricate procedures and equipment and cannot be applied to nonpolar dielectrics.

The self-polarization technique could also achieve high charge density and triboelectric output. This strategy involves using the high electric field produced by the TENG to align the dipole moment in the polar dielectric material with the electric field direction, as illustrated in Figure 15d.[90] It efficiently enables the dielectric to reach the saturated polarization state, enhancing relative permittivity. For example, Wu et al. introduced the self-polarization method to the high-relative-permittivity lead zirconate titanate (PZT) powder-filled PVDF dielectric film, achieving an ultrahigh output of 3.53 mC $m^{-2}$.[207] Wang et al. proposed a double-layer structured TENG that shares a single floating metal electrode. The TENG contains two PVDF composite films filled with $BaTiO_3$, demonstrating an extremely rapid self-polarization effect due to the strong electrical field produced by the floating electrode. The polarization speed reaches saturation approximately three times faster than the external-charge excitation method mentioned earlier. Therefore, self-polarization exhibits tremendous potential in future applications because


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of its high output performance and straightforward dielectric treatment.[83]

#### 4.2.2. Dielectric Parameter Modification

Modifications of dielectric parameters, such as the relative permittivity, trap energy level, dielectric microstructure, dielectric loss, and dielectric strength, could effectively regulate the charge transport behavior of dielectric materials, thereby enhancing the triboelectric performance. The basic principles are summarized as follows.

Relative Permittivity Enhancement—Relative permittivity quantifies a dielectric's ability to store capacitive energy in an electric field at a microscopic level, which can be regulated through polar phase induction, high-*k* micro-nanofiller doping. The polar phase in dielectric material exhibits higher relative permittivity due to its stronger electrostatic potential, more delocalized LUMO, and stronger dipole–dipole interaction.[208] Ferroelectric polymers, such as PVDF, PVDF-HFP, and PVDF-TrFE, etc. are commonly utilized as tribo-negative dielectric materials, which possess five distinct phases ($\alpha$, $\beta$, $\gamma$, $\delta$, and $\epsilon$). Among them, $\beta$-phase is the electroactive polar phase, where dipoles are oriented in the all-trans (TTTT) planar zig-zag conformation. Meanwhile, thermoplastic polymers such as PA6, PA66, and PA11, etc. are commonly employed as tribo-positive dielectric materials with four phases ($\alpha/\alpha'$, $\beta$, $\gamma$, and $\delta''$), where the $\delta''$-phase enables dipole alignment (**Figure 16**a(i)).[209]

Incorporation of piezoelectric fillers (ZnO,[211] $BaTiO_3$,[210] and PZT,[207] etc.), conductive fillers such as (graphene,[212] CNT,[214] MXene,[76] etc.), and semi-conductive fillers ($MoS_2$,[211] $ZnSnO_3$,[213] etc.) has been proven to be an effective strategy to enhance the polar phase content. The electroactive polar phase-dominated crystallization process is driven by interfacial interactions, specifically hydrogen bonding/electrostatic interactions (van der Waals attractive force) between polymer molecular chains and nanofillers, resulting in secondary polarization. During such procession, nanofillers act as nucleating agents for the perfect alignment of crystalline polymer. Accordingly, the higher the content of surface functional groups in nanofillers, the stronger the interfacial interaction forces, leading to a more pronounced dipole orientation in the dielectric polymer.[215] For example, Bhatta et al. developed a 2D siloxene-PVDF composite nanofibrous membrane as the tribo-dielectric for TENG, paired with Nylon 6/6 membrane, as shown in Figure 16a(ii).[216] The siloxene functional groups (–O and –OH) can interact with the PVDF polymer chain (–$CF_2$–$CH_2$–), creating a stable hydrogen bond and facilitating the in situ formation of the electroactive $\beta$-phase with an organized —$CF_2$—$CH_2$— chain. However, a high filler concentration can lead to agglomeration, which hampers the dipole moments of different crystalline phases.

Metal–organic frameworks (MOFs) with nanopores of different sizes is capable of acting as nanofillers to induce polar phases and exhibit a distinct self-poling effect (SPE), as shown in Figure 16a(iii).[217] As a proof of concept, Sohn et al. proposed MIL-101(Cr)-structured MOFs embedded in PVDF nanofibers to elucidate the SPE mechanism. It was found that MOFs adsorb and trap a portion of *N*, *N*-dimethylformamide (DMF) solvent during electrospinning. When the DMF gradually evaporates and escapes at ambient temperature, PVDF nanofibers are in contact with the solvent to trigger the creation of the $\beta$-phase by aprotic hydrogen bonding, similar to solvent annealing. The SPE has significant benefits as it can facilitate the electroactive polar phase content and corresponding TENG output performance without changing the morphology of nanofibers.

Polymer dielectric materials demonstrate merits such as flexibility, wear resistance, ease of processing, etc., while they exhibit low dielectric constant (typically $\epsilon_r < 10$). In contrast, inorganic ceramic dielectric materials (such as barium titanate, barium strontium titanate, lead zirconate titanate, etc.) exhibit relatively high dielectric constants but low impact-resistance and dielectric strength, rendering them unsuitable for use in triboelectric materials subjected to extended mechanical forces.[83] Incorporating high-*k* nanoparticles into the polymer matrix is the most straightforward method for increasing the relative permittivity, achieving complementary advantages of different types of dielectrics, as shown in Figure 16b.[218,219]

In high-*k* or conductive nanofillers doped dielectric materials, carriers within the nanofillers redistribute at the interface due to the electric field from tribo-charges. This redistribution causes surface charge accumulation and interfacial polarization, as demonstrated in the parallel-plate capacitor structure in Figure 16c(i).[220] Furthermore, when nanofillers are dispersed uniformly within the dielectric film appropriately, the configuration can be reckoned as a series of microcapacitors embedded within a large capacitor (Figure 16c(ii)).[46] The expressions for the equivalent capacitance of doped dielectric can be described as follows

$$C_1 = \frac{\varepsilon_0 \varepsilon_r S}{d} \tag{23}$$

$$C_2 = C_1 + \varepsilon_0 \varepsilon_r mn\pi \left(\frac{l}{2}\right)^2 \left(\frac{1}{d-kl} - \frac{1}{d}\right) \tag{24}$$

where the capacitances of the undoped and doped dielectric are denoted by $C_1$ and $C_2$, respectively. Thus, the dielectric equivalent capacitance increases due to the presence of microcapacitors that act as multiple charge trapping sites.[222] The doping type, shape (pore size), quantity, and uniformity can be inferred to impact dielectric performance significantly, which offers a theoretical basis for doping modification.[221]

For example, Ghosh et al. proposed a sponge-like hierarchical TPU/BTO/MXene composite film as the tribo-positive dielectric.[223] The significant boost in the relative permittivity of the TPU/MXene nanocomposites ($\epsilon_r \approx 30$ at 1 kHz) can be ascribed to the formed microcapacitor by the neighboring MXene conducting sheets within the insulating TPU dielectric (Figure 16c(iii)). In addition, multiple components doping is also considered an effective strategy to modulate nanofiller dispersion, constructing a more uniform microcapacitors network.[224,225] Hyun et al. reported that by aligning and encircling the 2D h-BN, MWCNT could create a denser microcapacitor network in the TPU matrix. It suggested that the composite film with 6 vol% h-BN and 1.05 vol% MWCNT demonstrates the permittivity of 232 (Figure 16c(iv)).[225]

However, the direct inorganic doping modification method encounters difficulties due to inadequate filler dispersion, causing


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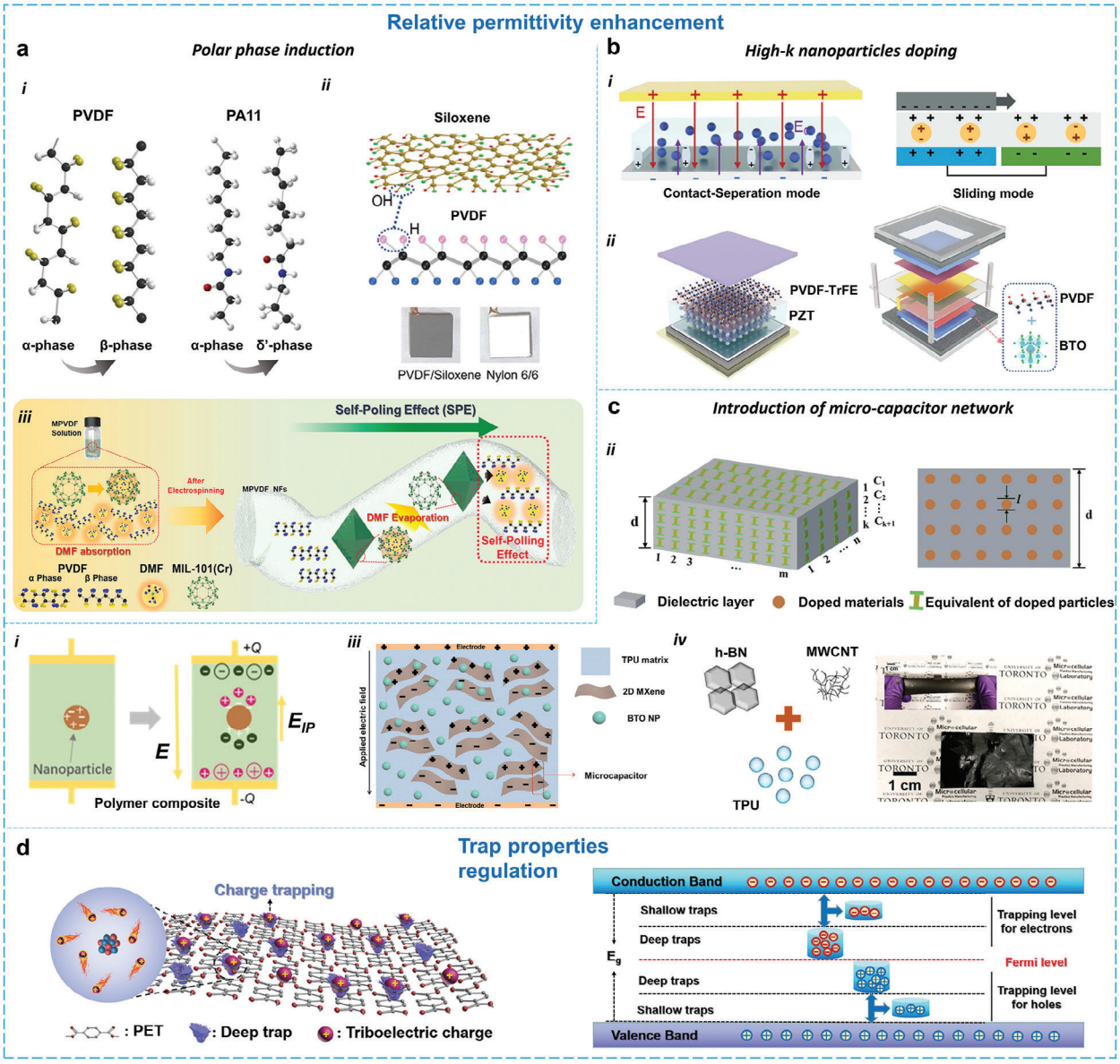


**Figure 16.** Relative permittivity and trap energy level enhancement. Polar phase induction: a(i) Schematic of the transition from nonpolar to polar phase. a(ii) 2D siloxene-PVDF composite nanofibrous membrane paired with Nylon 6/6 membrane. a(iii) Self-poling effect of MIL-101(Cr)-structured-MOFs embedded PVDF nanofibers. High-*k* nanoparticles doping: b(i) Schematic of the high-*k* nanoparticles inducing bound charges. b(ii) PZT/BTO doped dielectric. Introduction of microcapacitor network: c(i) Schematic of the nanofillers increasing the charge storage capability. c(ii) Microcapacitor model. c(iii) TPU/BTO/MXene nanocomposites forming microcapacitors. c(iv) TPU/h-BN/MWCNT nanocomposites forming microcapacitors. Trap energy level enhancement: d) KH556@PET dielectric with deep traps. a(i) Reprinted with permission from ref. [209]. Copyright 2022, Elsevier. a(ii) Reprinted with permission from ref. [216]. Copyright 2022, Wiley-VCH. a(iii) Reprinted with permission from ref. [217]. Copyright 2023, Elsevier. b(i) Reprinted with permission from ref. [83]. Copyright 2022, Wiley-VCH. Reprinted with permission from ref. [219]. Copyright 2020, Wiley-VCH. b(ii) Reprinted with permission from ref. [218]. Copyright 2023, Wiley-VCH. Reprinted with permission from ref. [83]. Copyright 2022, Wiley-VCH. c(i) Reprinted with permission from ref. [220]. Copyright 2022, Elsevier. c(ii) Reprinted with permission from ref. [46]. Copyright 2023, Elsevier. c(iii) Reprinted with permission from ref. [223]. Copyright 2022, American Chemical Society. c(iv) Reprinted with permission from ref. [224]. Copyright 2024, Elsevier. d) Reprinted with permission from ref. [226]. Copyright 2023, Wiley-VCH.

aggregation because of substantial disparities in physical and chemical properties, such as surface energy and chemical bonds, between the filler and the organic polymer matrix. This results in inadequate compatibility and adhesion at the nanofiller/polymer interface, a prevalent issue in many previous studies. The effective medium theory states that a substantial amount of inorganic particles (>40 vol% for spherical particles with percolation) is required to produce a significant change in relative permittivity. Nanoparticles are challenging to scatter evenly and tend to aggregate together, leading to increased current leakage density and a significant rise in dielectric losses. Recently, the conjugated conductive domains, such as oligoaniline (OANI), oligothiophene


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(OTH), polyacetylene (PA), have been considered as additives for high-$k$ polymer dielectrics.[203] These oligomers exhibit controllable structure, superior compatibility, large interfacial area, and adjustable conductance, which have been widely used in engineering dielectrics with noteworthy $k$ and low dielectric loss. Constructing high-performance tribo-dielectrics through oligomer modification might be an effective strategy.

Trap Properties Regulation—Local states, known as traps, are unavoidably present in the energy band structure of dielectrics. For example, the interface between the polymer crystalline region and the amorphous region, the composite filler and the polymer molecular chain, the folding (entanglement), fracture, and impurities of the chain can all form traps. Physical defects usually result in shallow traps, while chemical defects lead to deep traps. Carriers confined in deep traps have reduced migration propensity compared to shallow traps, resulting in enhanced stability. Accordingly, dielectric trap regulation is another crucial route to modulate triboelectric performance.

Charge storage capabilities of dielectrics can be customized by regulating trap energy levels and densities. For example, doping nanoparticles with high electrostatic potential or electron affinity will create numerous interface regions between nanofillers and polymer matrix, providing additional locations for carriers to charge storage. Recently, Wang et al. employed the (*N*, *N*-dimethyl-3-aminopropyl) trimethoxysilane (KH556) to introduce the deep traps in PET fabric.[226] As illustrated in Figure 16d, KH556 creates structural defects that function as deep carrier traps and bring an enhancement of 1300% in triboelectric output. Besides, dielectrics with invalid charge trapping ability is also reported to prevent self-charge excitation failure.[159,160] Hu et al. utilized the carbon powder to reduce the trap level of PVDF film and the shielding effect of surface charge was hindered, reaching a high charge density of 4.13 mC $m^{-2}$.[159]

*Dielectric Loss Suppression*: Dielectric loss is energy dissipation from relaxation polarization and internal leakage current in dielectric materials when subjected to an alternating electric field, categorized into relaxation and conduction losses. The former primarily refers to the energy consumed in reorienting dipole moments, whereas the latter refers to losses from charge carriers moving within the dielectric along the electric field, which is more critical in TENG. High dielectric losses will cause surface tribo-charges to flow into the dielectric material instead of the output circuit, reducing the output performance and dielectric strength. Typical dielectric conduction mechanisms, including Poole–Frenkel emission, hopping conduction, and Schottky emission, have been proposed.[227-229]

Generally, percolation is a typical physical phenomenon in nano-composite dielectrics when the physical properties of the dielectric material experience a significant change due to the formation of a percolative network by secondary components like nanofillers. Percolation significantly affects the electrical conductivity of dielectrics. As the volume percent of the filler component nears the percolation threshold ($f_c$), the electrical conductivity of the composite material experiences a significant rise. Thus, the trade-off between high-$k$ and dielectric loss should be considered. Besides, enhancing the interfacial interaction between nanofillers and polymers through interface modification could balance dielectric losses and dielectric constants. For instance, Peng et al. introduced a third phase (a layer of amorphous carbon shell) at the particle (silver nanoparticles, AgNPs)-polymer (PDMS) interface, as shown in **Figure 17a(i)**.[84] The fabricated Ag@C@PS nanocomposite system achieved 50% decrease in dielectric loss while maintaining high relative permittivity (2.6-fold increment), which can be attributed to carbon shell with low surface free energy, moderate permittivity, and low conductivity, resulting in a homogeneous space electric field as well as the attenuated electron path between nanofillers. In another work, Yang et al. proposed core-shell structured $BaTiO_3$ nanoparticles by grafting two types of fluoroalkyl acrylate monomers via surface-initiated reversible addition-fragmentation chain transfer (RAFT) polymerization.[230] It was observed that the organic surface modification of fluoro-polymers effectively improves the dispersion of BT nanoparticles and enhances the interfacial adhesion of nanocomposite dielectric.

Applying insulating layer(s) on the outer surface of conductive nanofillers has been shown to limit long-distance charge transfer, effectively tackling the issue of dielectric loss without decreasing relative permittivity.[231] For example, Zhou et al. clarified the polarization mechanism of Zn@ZnO/PVDF composite with high secondary component ratio, as shown in Figure 17a(ii).[232] It was determined that electron transit through filler clusters, which involves domain polarization rather than behavior at individual filler or filler/polymer interfaces, is the crucial factor influencing the dielectric characteristics of this percolative composite material. Therefore, optimized shell electrical characteristics can be used to build a percolative composite dielectric with a larger cluster size and reduced intercluster electron transport.

Dielectric Strength Enhancement—The maximum breakdown charge density ($\sigma_{DB}$) of dielectric materials significantly impacts the surface charge density yet is frequently overlooked.[233,234] Although the surface charge density can be increased by reducing the thickness, the composite dielectrics will inevitably introduce defects, especially when the amount of doped nanofillers approaches the "percolation threshold". The dielectric physical properties will suddenly change and encounter breakdown owing to the weaker interface interactions and substantial local electric field distortions at the interfaces (Figure 17b(i)).[235] Thus, dielectric strength maintenance or improvement should be considered during modification. For example, nanofillers with excellent insulating and voltage-withstanding properties, such as boron Nitride Nanosheets (BNNS) with high electrical breakdown strength (up to 800 kV $mm^{-1}$), can be incorporated.[236] By modifying the distribution, orientation, and form of nanofillers, it is feasible to change the carrier migration paths and enhance the critical breakdown electrical field. For instance, Luo et al. proposed a hybrid dielectric material with $BaTiO_3$ nanoparticles embedded in boron nitride (BN) nanosheet that filled in PVDF dielectric film (Figure 17b(ii)).[237] An electric breakdown strength of ≈580 kV $mm^{-1}$ (1.76 times of PVDF film), was achieved by adding 5 wt% filler. In this case, BN serves as a host for incorporating ferroelectric nanoparticles to enhance insulation and polarization when exposed to a high electric field. In another work, Zhang et al. proposed large-aspect-ratio $BaTiO_3$@$TiO_2$ composite nanofibers with interior hierarchical interfaces, where the orientation of nanofibers perpendicular to the external electric field can significantly enhance the dielectric strength (Figure 17b(iii)).[238]


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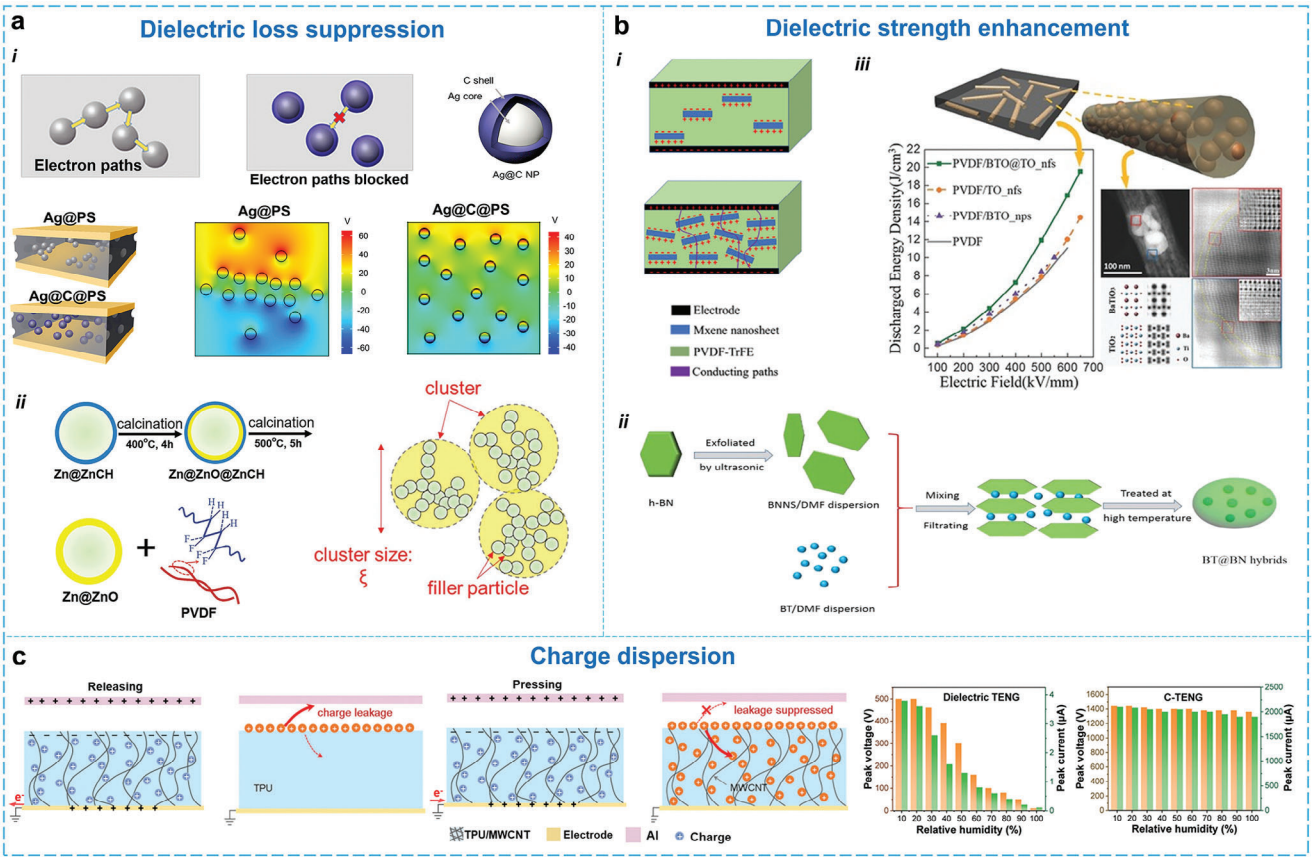


**Figure 17.** Dielectric loss suppression and dielectric strength enhancement. Dielectric loss suppression: a(i) Ag@C@PS nanocomposite with attenuated electron path between nanofillers. a(ii) Zn@ZnO core–shell nanoparticles embedded PVDF matrix. Electrical strength enhancement: b(i) Schematic of the "percolation threshold". b(ii) Fabrication process of $BaTiO_3$ nanoparticles embedded in BN nanosheet. b(iii) $TiO_2$ nanofibers embedded with $BaTiO_3$ nanoparticles in PVDF-HFP matrix. c) Conductive network of MWCNTs embedded in the dielectric layer of TPU. a(i) Reprinted with permission from ref. [84]. Copyright 2022, American Chemical Society. a(ii) Reprinted with permission from ref. [232]. Copyright 2022, WILEY-VCH. b(i) Reprinted with permission from ref. [235]. Copyright 2021, American Chemical Society. b(ii) Reprinted with permission from ref. [237]. Copyright 2018, WILEY-VCH. b(iii) Reprinted with permission from ref. [238]. Copyright 2014, WILEY-VCH. c) Reprinted with permission from ref. [85]. Copyright 2023, WILEY-VCH.

Charge Dispersion—Recently, a unique charge dispersion strategy was reported, where the generated tribo-charges are dispersed into the interior of triboelectric layers. Specifically, a conductive network of multiwall carbon nanotubes (MWCNTs) was introduced into the TPU (Figure 17c).[85] The CE surface charges were effectively imported to the interior dielectric TPU layer, boosting the output current several orders of magnitude. Furthermore, the imported charges are less likely to migrate to the air in high-humidity environments, endowing the device with excellent resistance. This work demonstrates that simultaneously using both leakage current and electrostatic induction charge is a viable method to enhance triboelectric output.

### 4.3. Dielectric Coordination

#### *4.3.1. Gas Dielectric Coordination*

As mentioned in Section 3.1.3., the surface charge density of tribo-dielectric materials is limited by the air breakdown. Generally, the triboelectric charges are composed of the residual surface charges and the breakdown charges, where the former determines the induced transferred charge amount (as illustrated in **Figure 18a**).[127] It is apparent that the induced charges are lower than the triboelectric charges. Figure 18b demonstrates the output energy density of various types of TENG considering the air breakdown effect. The sliding freestanding TENG possesses the highest energy density with increased charge density, while the saturation trend exists for the CS-TENG owing to the breakdown effect.[239] To this end, the control of gas composition/pressure, or gas dielectric coordination strategy, was proposed.

According to Figure 18c, the dielectric strength of gas dielectric materials follows Paschen's law. Specifically, decreasing or increasing the air pressure could enhance the air breakdown voltage. The breakdown voltage rises when the air pressure drops from atmospheric pressure to vacuum, as both the gas density and collision ionization probability decrease. On the contrary, the increase in gas pressure brings higher gas density. A stronger electric field is needed to make the kinetic energy gained by each molecule for the occurrence of self-sustaining discharge. Thus, the air breakdown effect could be successfully prevented by varying the air pressure. For example, Liu et al. found that


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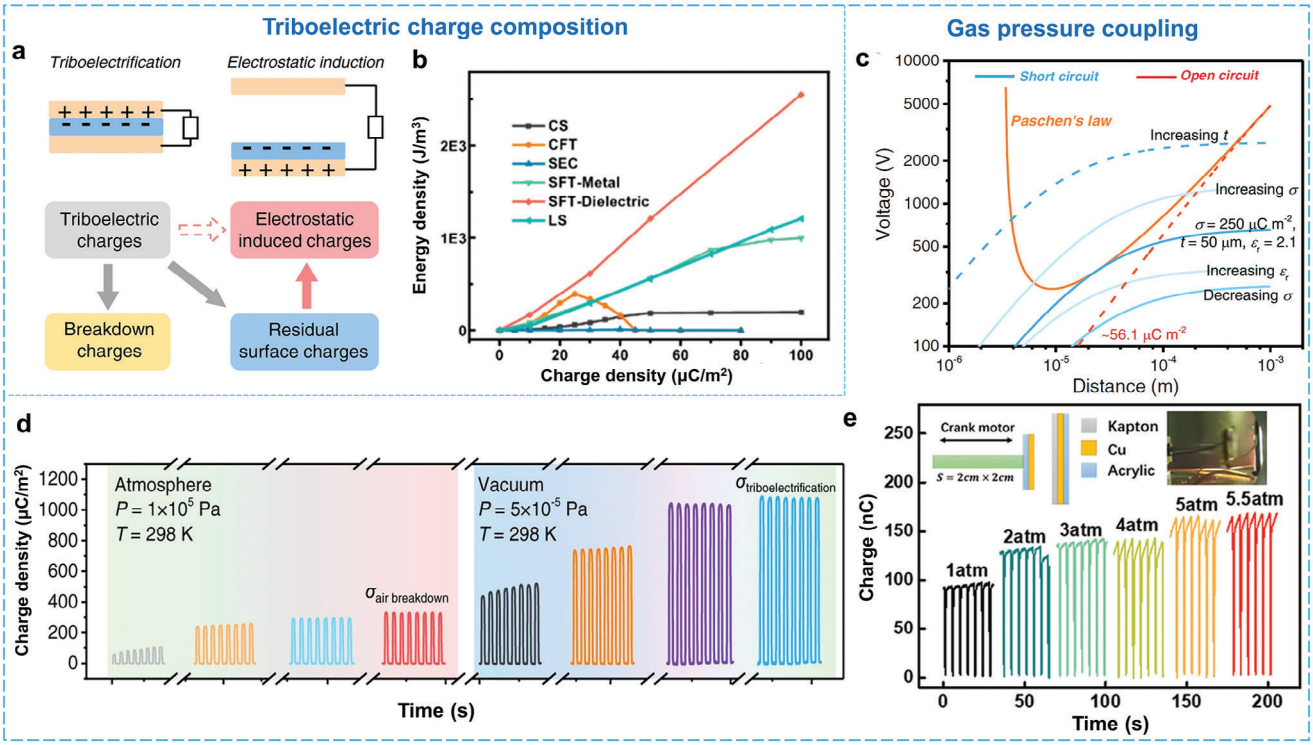


**Figure 18.** Gas dielectric coordination for high-performance TENGs. a) The composition of triboelectric charge. b) Relationship between the energy density and charge density of various types of TENGs. c) Relationship of breakdown voltage and gap distance. d) Surface charge density of PVC at atmosphere and vacuum conditions. e) The influence of gas pressure on triboelectric output performance. a,c,d) Reprinted with permission from ref. [127]. Copyright 2022, the authors. b) Reprinted with permission from ref. [239]. Copyright 2019, American Chemical Society. e) Reprinted with permission from ref. [240]. Copyright 2020, the authors.

the surface charge density of PVC gradually stabilized in the atmosphere while keeps increasing in a vacuum condition, and the highest surface charge density of 1250 μC $m^{-2}$ was achieved (Figure 18d).[127] In another work, Fu et al. found that the charge density of TENG at 10 atm was increased by 1.6 times compared to the atmosphere condition (Figure 18e).[240] Although applying vacuum or high pressure can enhance the TENG charge density and prevent interface breakdown, the practical usage of these methods is restricted due to the hostile gas environment.

Accordingly, gas dielectric materials with superior insulating performance were employed. In one work, Han et al. investigated the influence of atmosphere ($N_2$, $O_2$, $CO_2$, Ar, He) on the triboelectric performance of PTFE, Kapton, and PET by a piston-structured TENG.[89] As shown in **Figure 19**a(i), it was found that gas dielectric with a smaller electron yield per ion ($\gamma$) delivers a higher transferred charge. The triboelectric performance in the $CO_2$ atmosphere reaches the highest and lowest in He condition. Further, Lv et al. designed an enclosed cubic TENG and investigated the role of gas dielectrics (He, Ar, Air, $N_2$, $CO_2$, $CHF_2Cl$, $SF_6$) in the triboelectric performance.[241] According to Figure 19a(ii), the device filled with $SF_6$ demonstrated the highest output, and the voltage, current were increased by 67%, 120% compared to the air. The gas-enhanced mechanism was ascribed to the superior dielectric strength (voltage endurance capability) of $SF_6$ that inhibits the breakdown in the gap. The utilization of fluorine gases ($SF_6$) could enhance the output performance effectively.

#### 4.3.2. Liquid Dielectric Coordination

Liquid dielectric coordination can also enhance the TENG output. On the one hand, interface lubrication based on liquid dielectric can improve a device's wear resistance and service life by reducing the wear of solid dielectric during operation. On the other hand, liquids perform better dielectric strength than gas, which can suppress the air breakdown to a higher surface charge density. As shown in Figure 19b(i), air breakdown usually happens in the microgaps between the triboelectric layers.[242] Using liquid lubrication with superior dielectric strength could enhance the output performance and durability of the sliding mode AC-TENG and DC-TENG. In another work, the role of different liquid lubricant dielectrics on the triboelectric performance was further clarified. As shown in Figure 19b(ii), the permittivity and dynamic viscosity of liquid dielectrics mainly determine the coupling effect.[243] Liquids with larger permittivity and higher viscosity deliver inferior triboelectric output performance, and heptane with low permittivity, viscosity demonstrated the highest outputs. Meanwhile, liquid lubricants could prevent transfer film formation and ensure the solid–solid effective contact area.

More recently, the utilization of dielectric oil to enhance the charge storage capacity of the floating layer was also reported. As shown in Figure 19b(iii), Lei et al. developed the external charge-excitation or self-excitation based high-performance noncontact TENG.[90] The dielectric oil was filled between the solid dielectric films, delivering the maximum charge density of 260.15 μC $m^{-2}$


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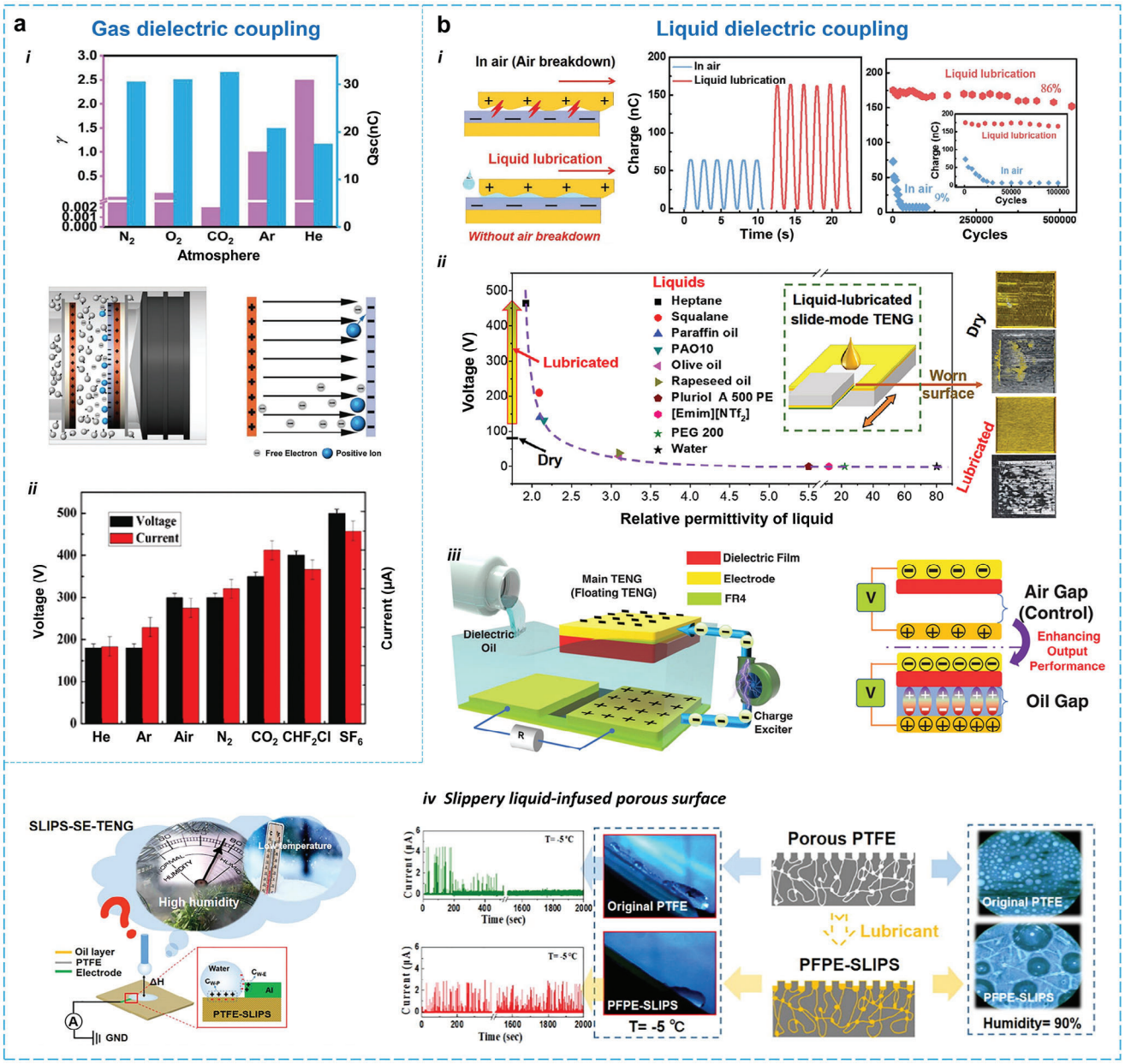


**Figure 19.** Gas and liquid dielectric coordination for high-performance TENGs. Gas dielectric coupling: a(i) The effect of gas dielectrics on the output performance. a(ii) The gas enhanced effect on triboelectric output. Liquid dielectric coupling: b(i) Power density and durability enhancing effect of interface liquid lubrication. b(ii) The influence of liquid relative permittivity on triboelectric performance. b(iii) Liquid lubrication enhanced noncontact TENG. b(iv) Slippery liquid-infused porous surface based TENG. a(i) Reprinted with permission from ref. [89]. Copyright 2019, Wiley-VCH. a(ii) Reprinted with permission from ref. [241]. Copyright 2019, Elsevier. b(i) Reprinted with permission from ref. [242]. Copyright 2020, Wiley-VCH. b(ii) Reprinted with permission from ref. [243]. Copyright 2019, Elsevier. b(iii) Reprinted with permission from ref. [90]. Copyright 2022, Elsevier. b(iv) Reprinted with permission from ref. [246]. Copyright 2022, Elsevier.

(4.41 times higher than the air gap). Besides, the dielectric liquid-based self-operating switch TENG was also reported by regulating the triboelectric layer's field emission through the movement of dielectric liquid that acts as a self-operating switch.[244] In another work, a nonpolar liquid lubricant submerged TENG was presented, where the rolling electrode effectively improved the operation durability.[245] It is generally possible to efficiently suppress interface discharge, lower friction and wear, and enhance output performance by utilizing low dielectric constant and viscosity liquid dielectrics. By correctly implanting liquid dielectric microspheres in solid dielectrics, it will be possible to release lubricating fluid as necessary to clear interface wear debris and preserve high-performance output.

Additionally, liquid lubrication can create a superhydrophobic dielectric surface and reduce water droplet friction to improve the performance of liquid–solid TENGs. The hydrophobic surface is


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known as a slippery liquid-infused porous surface (SLIPS), inspired by Nepenthes Pitcher plants. SLIPS can be created by capturing a layer of lubricating oil on a porous medium surface, like a porous PTFE/PDMS membrane, via capillary wicking. This results in superhydrophobicity, which is more durable than typical hydrophobic surfaces that trap air bubbles. On the SLIPS, droplet can slide smoothly and continuously without pinning, bouncing, or separating, leading to a consistent CE process. For example, Chen et al. employed perfluoropolyether (4 μL $cm^{-2}$) as a lubricant for porous PTFE to create high-performance SLIPS-TENG (Figure 19b(iv)).[292] The device achieved an instantaneous output short-circuit current of 3 μA and charge density of 8.8 nC $cm^{-2}$, demonstrating stable output under freezing temperatures or high humidity conditions.

However, SLIPS exhibits an inherent weakness in dust resistance. Interfaces such as slippery covalently attached liquid (SCAL) have been developed to address the issue of hydrophilic dust adhesion. The type and volume of the lubricating oil infused are crucial factors for optimizing SLIPS-TENG performance. Excessive lubricating oil will lead to a higher proportion of liquid–liquid CE and the exposure of solid surfaces, reducing the output performance due to lubrication failure. Furthermore, because of its transparent properties and capacity to withstand severe weather conditions, SLIPS-TENG can also be integrated with solar panels to develop high-performance energy harvesting devices for outdoor environments.

### 4.4. Dielectric Structure Optimization

Single modification of a dielectric surface or bulk material typically has a maximum charge capacity. In order to adapt to the dynamic transport characteristics of triboelectric charges, that is, efficient generation, stable storage, or ordered migration, dielectric structure optimization strategies have been proposed. In this section, we summarized the principles and typical cases of dielectric structure modification currently.

#### *4.4.1. Charge Transport, Trapping, and Blocking Layer*

Generally, the generated tribo-charges will undergo migration, dissipation, and storage processes, eventually achieving a dynamic equilibrium.[247] Charges on the dielectric surface will drift towards the electrode under the driving force of the built-in electric field. Some will disperse due to the charge concentration gradient, ultimately neutralizing with opposite induced charges on the electrode. Introducing deep traps in tribo-dielectrics via techniques such as nanofiller doping is a frequent method to slow down charge decay, as discussed in Section 4.2.2. However, the density of deep traps is constrained by the percolation threshold, potentially resulting in mechanical fragility and loss of charge. To this end, dielectric structural design, including incorporating the dielectric charge trapping layer, charge blocking layer, has been proposed to maximize the utilization of tribo-charges.[248]

The dielectrics for charge trapping layer typically possess more defects (physical and chemical), including crosslinking points, residual polar groups, crystal grid imperfections, etc., that enhance charge trapping capability (**Figure 20**a(i)).[88] For example, PS and PI offer abundant charge trapping sites thanks to the aromatic rings present on the molecular chains. 2D materials exhibit a controllable bandgap and distinctive electrical properties, making them appropriate for charge trapping layers.[248] For example, a 50-nm-thick $MoS_2$ layer as the charge trapping layer could achieve an improved output performance (1000 μC $m^{-2}$), as shown in (Figure 20a(ii)).[249] In another work, Shrestha et al. proposed $MoS_2$-incorporated laser-induced graphene as the charge trapping layer, and the synergistic effect between them helps to boost the dielectric property and polarization (Figure 20a(iii)).[250]

Moreover, Lv et al. found that the charge trapping depth is significantly influenced by the volume conductivity of the tribo-dielectric (Figure 20b).[251] In the absence of a PI layer and MWCNTs, tribo-charges are trapped in shallow traps. Comparatively, the PI/TPU double layer allows for trapping tribo-charges in deep traps with a sluggish escape rate. Introducing MWCNTs into TPU without the charge trapping layer leads to straightforward charge detrapping and significant charge loss. By using MWCNTs/TPU as the tribo-dielectric and PI as the charge storage layer, the increased volume conductivity facilitates the movement of additional tribo-charges, enabling tribo-charges in shallow traps to reach deep traps more effectively. Thus, optimizing the volume conductivity of the tribo-dielectric can boost the intermediate layer's charge trapping ability synergistically.

Besides, an additional charge-transport layer can be inserted as an intermediary layer between the tribo-dielectric and charge trapping layer.[248] The three-layer dielectric boosts electron transfer efficiency and increases the induced charges on the electrode. The electrostatic induction, electron drift velocity, and trap site volume should all be considered during the design process. For example, Li et al. proposed a three-layer tribo-dielectric system consisting of a negative triboelectric layer of polyethersulfone (PES), a charge transport layer of PS @ carbon black, and a charge trapping layer of PS, as shown in Figure 20c.[252] The output charge density was increased seven times after inserting the PS layer and further rose to 11.2 times when an additional charge transport layer was included.

Using high-$k$ dielectric as the charge blocking layer has attracted significant interest, which could effectively block the migration or diffusion of tribo-charges from the tribo-dielectric layer to the electrode. Effective high-$k$ electron-blocking layers should possess characteristics such as high relative permittivity, low defect density, large conduction band offset, large bandgap width, and strong moisture resistance. For example, Xie et al. embed a $La_{0.1}Zr_{0.9}O_x$ based high-$k$ charge blocking layer to establish an energy barrier ($\Delta E > 1.3$ eV) between ITO (electrode) and PDMS (triboelectric layer), as shown in Figure 20d.[253] The low level of leakage oxygen vacancies of $La_{0.1}Zr_{0.9}O_x$ successfully decreases the leakage path formation, thereby ensuring the high output performance of 215 V, 96.3 mA $m^{-2}$, and 243.3 μC $m^{-2}$. Recently, Yoo et al. reported a Janus membrane as the bifunctional charge trapping-blocking dielectric layer, through inducing gravitational sedimentation of cobalt MOFs (CoMOFs) in the styrene-ethylene-butylene-styrene (SEBS)-toluene solution (Figure 20e).[254] The proposed CoMOFs dense layer functions as the charge trapping layer, while the CoMOFs free layer acts as the blocking layer, delivering a 327% increase in output voltage. This approach prevents delamination or cracking caused by multilayer structures


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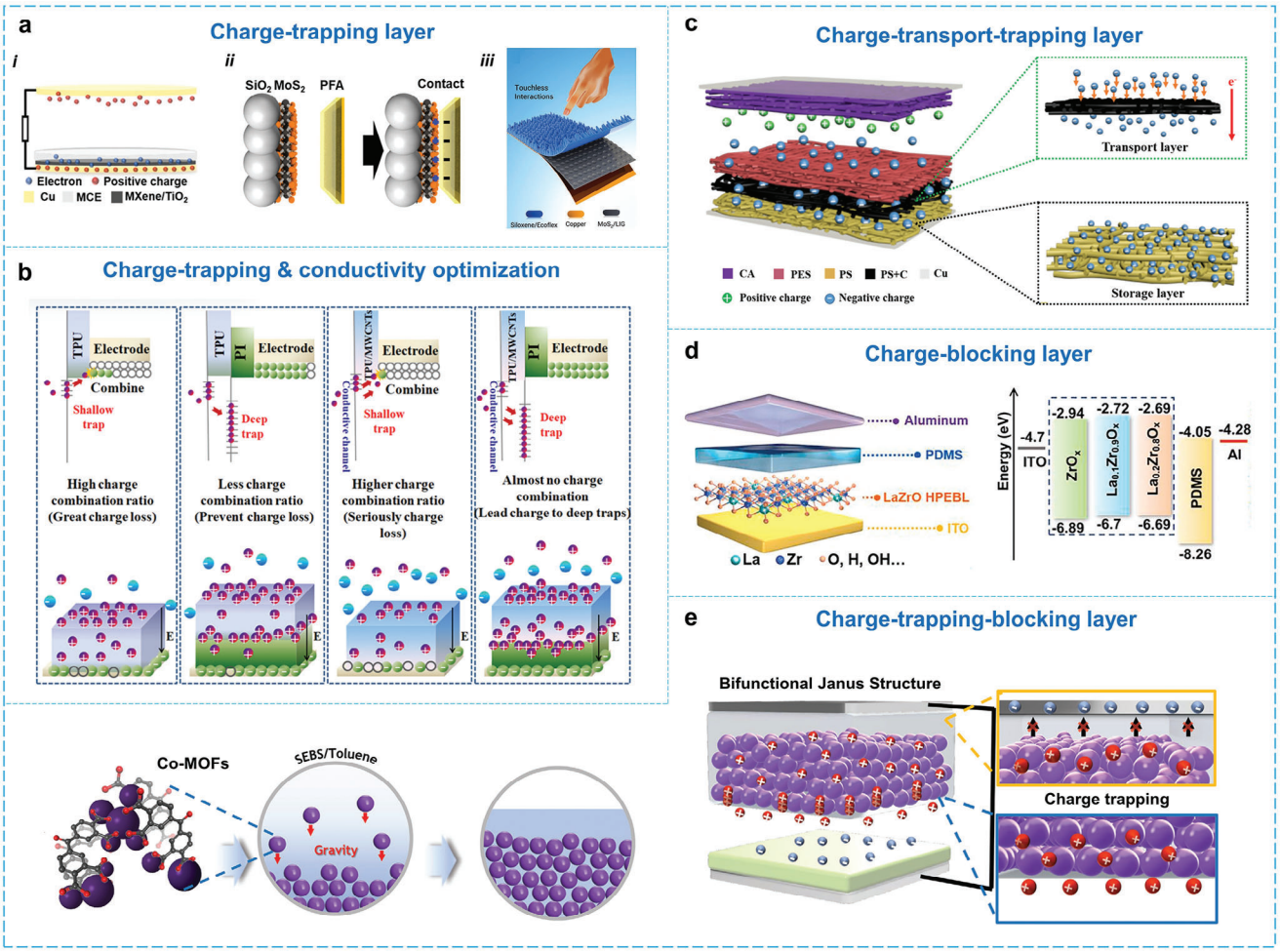


**Figure 20.** Charge transport, trapping, and blocking layer of TENG. Charge trapping layer: a(i) Schematic of the charge trapping layer. a(ii) 50-nm-thick $MoS_2$ layer as the charge trapping layer. a(iii) $MoS_2$ incorporated LIG as the charge trapping layer. b) Mechanism of the volume conductivity influencing the depth of charge trapping and the degree. c) Schematic of the charge-transport-trapping layer. d) Schematic of the charge blocking layer. e) Schematic of the charge trapping-blocking layer. a(i) Reprinted with permission from ref. [88]. Copyright 2022, Elsevier. a(ii) Reprinted with permission from ref. [249]. Copyright 2023, The Royal Society of Chemistry. a(iii) Reprinted with permission from ref. [250]. Copyright 2022, Wiley-VCH. b) Reprinted with permission from ref. [251]. Copyright 2021, American Chemical Society. c) Reprinted with permission from ref. [252]. Copyright 2018, Elsevier. d) Reprinted with permission from ref. [253]. Copyright 2022, Elsevier. e) Reprinted with permission from ref. [254]. Copyright 2024, the authors.

during stretching operations, exhibiting flexibility, and long-term stability.

#### 4.4.2. Multilayered Dielectric Structure

In addition to the microscopic interfaces between nanofillers and the polymer matrix in dielectrics, there is another form of contact seen in multilayer-structured dielectric films: the dielectric/dielectric interfaces.[203] Similar to the interface polarization between nanofillers/polymer matrix, charge accumulation also occurs at the interface with large differences in conductivity or relative permittivity (dielectric heterogeneous interfaces). This leads to alterations in the local electric field and the creation of induced dipole moments. The interface polarization at the interlayer interfaces has a significant impact on the overall dielectric performance of the composite film by providing more sites for charge trapping. The Maxwell-Sillar-Wagner (MWS) model is typically employed to illustrate the mechanism (**Figure 21**a(i)).[220] Assuming $\sigma_f$ and $P_f$ represent the charge density and polarization strength at the interface, their expressions are as follows

$$\sigma_f = \frac{(\sigma_1\varepsilon_2 - \sigma_2\varepsilon_1)U}{(\sigma_1 + \sigma_2)d} \tag{25}$$

$$P_f = \frac{(\sigma_1\varepsilon_2 - \sigma_2\varepsilon_1)U}{2(\sigma_1 + \sigma_2)d} \tag{26}$$

where $U$ denotes the voltage generated by tribo-charges; $\sigma_1$, $\sigma_2$ denote the conductivities of the upper and lower layers; $\epsilon_1$, $\epsilon_2$ denote the relative permittivity of the upper and lower layers. When $\sigma_1\epsilon_2 = \sigma_2\epsilon_1$, there is no charge accumulation at the interface, and the interface polarization strength is 0. When $\sigma_1\epsilon_2 \neq \sigma_2\epsilon_1$, the interface charge density and polarization strength increase with the


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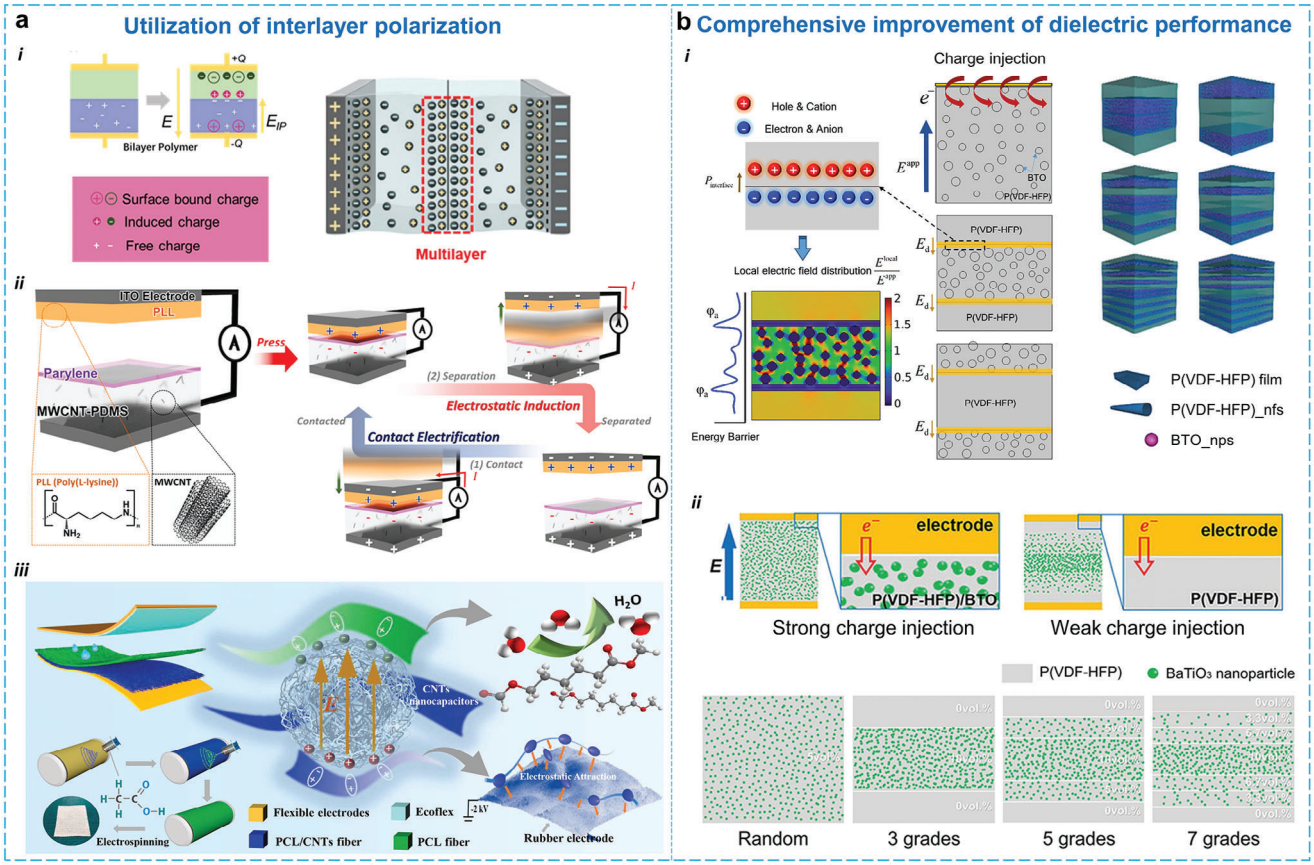


**Figure 21.** Multilayered dielectric structure for TENG. Utilization of interlayer polarization: a(i) Schematic of the Maxwell–Sillar–Wagner model. a(ii) Soft bilayer film of parylene derivatives and PDMS embedded with MWCNTs. a(iii) Bilayer PCL nanofiber comprising outer PCL and inner PCL/CNTs nanofibers. Comprehensive improvement of dielectric performance: b(i) Stacking layers of P(VDF-HFP) and P(VDF-HFP)/$BaTiO_3$. b(ii) P(VDF-HFP)/$BaTiO_3$ dielectric films arranged with different doping levels. a(i) Reprinted with permission from ref. [220]. Copyright 2022, Elsevier. Reprinted with permission from refs. [203]. Copyright 2023, Wiley-VCH. a(ii) Reprinted with permission from ref. [255]. Copyright 2023, Elsevier. a(iii) Reprinted with permission from ref. [220]. Copyright 2022, Elsevier. b(i) Reprinted with permission from ref. [258]. Copyright 2019, Elsevier. b(ii) Reprinted with permission from ref. [87]. Copyright 2019, WILEY-VCH.

increasing magnitude of $|\sigma_1\epsilon_2 - \sigma_2\epsilon_1|$. Accordingly, asymmetric permittivity manipulation with a multilayered dielectric structure is also an efficient strategy to improve the whole relative permittivity of the dielectric film and the surface charge density. Noteworthy, the dielectric losses will rise simultaneously with the accumulated charges at the interface, which may seriously degrade the TENG performance.

Kim et al. proposed a soft bilayer film composed of two layers: the parylene derivatives with different functional groups and the composite layer of PDMS embedded with MWCNTs, as shown in Figure 21a(ii).[255] The bilayer dielectric film can efficiently generate interfacial polarization due to the difference in permittivity, which greatly enhances the output of the TENG. In another work, a bilayer PCL nanofiber membrane comprising outer PCL and inner PCL/CNTs nanofibers was developed as the tribo-positive dielectric, as shown in Figure 21a(iii).[220] The TENG achieved a transmitted charge of 84 μC $m^{-2}$ with a doping level of 3 wt% CNT and a bilayer thickness ratio of 1:1, which was 8.4 times higher than that of the PCL-based TENG.

Generally, dielectric heterogeneous interfaces demonstrate a blocking effect on the growth of electric trees during dielectric failure (breakdown). The rationally designed multilayered dielectric can integrate the suppressed effects of dielectric/electrode and dielectric/dielectric interfaces on charge injection/migration, creating interface barrier effects that effectively block breakdown paths, thereby improving the breakdown strength of multilayered dielectric films.[256] Moreover, the high controllability of multilayered dielectric film is manifested in the ability to manipulate the electric field distribution by adjusting the structure and properties between different dielectric layers, allowing for flexible design for the desired performance. Detailly, each layer can serve as either a high $E_b$ layer ("hard" layer) or a high $\epsilon_r$ layer ("soft" layer). The former helps to suppress the formation of conductive channels by incorporating high-insulating fillers, while the latter provides polarization strength under high electric field by integrating high-$k$ fillers. By utilizing various stacking configurations of "hard" and "soft" layers, it is feasible to boost a single property or enhance dielectric performance comprehensively. For example, a three-layer structured dielectric with "hard"–"soft"–"hard" layers can decrease leakage current density effectively. The interfaces between the layers facilitate the spread of electric trees in the in-plane direction, enhancing breakdown


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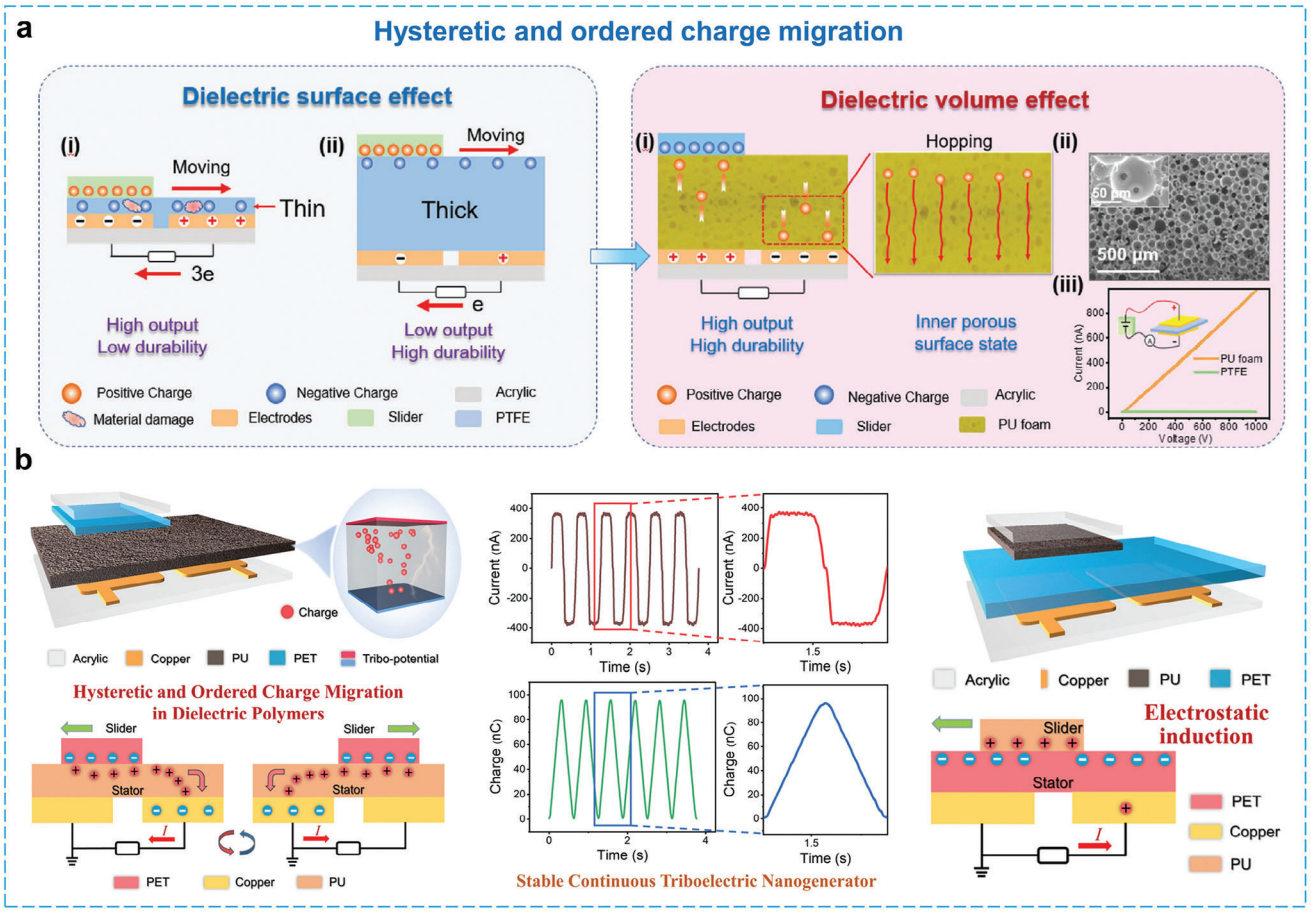


**Figure 22.** Hysteretic and ordered charge migration strategy. a) Schematic of transforming the dielectric surface effect into the 3D volume effect. b) Acquiring stable and continuous current. a) Reprinted with permission from ref. [86]. Copyright 2023, Wiley-VCH. b) Reprinted with permission from ref. [260]. Copyright 2023, The Royal Society of Chemistry.

strength.[257] In one work, Jiang et al. optimized the number (up to 16 layers) and order of the stacking layers (PVDF-HFP) and PVDF-HFP/$BaTiO_3$) in the multilayered dielectric, delivering improved breakdown strength up to 862 MV $m^{-1}$, as shown in Figure 21b(i).[258] The polarized ions at the dielectric/dielectric interfaces will induce a strong depolarized field $E_d$ opposite to the applied electric, reducing the local electric field $E_{local}$ and increasing the potential barrier height for the charge migration.

Multilayered dielectric structures based on gradient-doping strategies have also been proposed. In a vertical 3-layered gradient dielectric structure, as the relative permittivity of each layer increases from top to bottom, the electric field in each layer decreases due to the electric field produced by tribo-charges. If the electrical breakdown is caused by charge carriers migrating from bottom to top, the electric field strength distributed in the bottom layer is the lowest, thereby facilitating the overall breakdown strength of the multilayered dielectric.[259] For example, Jiang et al. created gradient multilayered dielectric films by arranging PVDF-HFP/BaTiO3 dielectric films with varying doping levels in sequence, as shown in Figure 21b(ii).[87] The construction of multilayered dielectrics based on gradient-doping strategies has a relatively small impact on the whole relative permittivity. However, the increase in Young's modulus and improved interfacial effects have greatly enhanced the breakdown strength and leakage current density, successfully regulating the electric field gradient distribution by continuously varying electrical parameters. All these strategies provide a reference for building high-performance tribo-dielectric materials.

### 4.4.3. Charge Migration Control

Generally, TENGs tend to choose thinner dielectric material for a high surface charge density, while surface charges can migrate easily in specific dielectric films such as PU, PA, Nitrile, etc. One effective structural design method involves transforming the dielectric surface effect into a 3D volume effect by integrating dielectrics with charge migration properties, thereby enhancing the output performance. To this end, the millimeter-thick dielectric film can be employed for practical application without output degradation, as shown in **Figure** 22a.[86] For instance, Fu et al. utilized the 1 mm PU foam dielectric film with an inner porous network structure that possesses higher conduction than that of same-thick PTFE (by two orders of magnitude), and tribo-charges can easily move throughout the surface state of the network. The dielectric volume effect TENG works on the combination of


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triboelectrification, charge migration, and electrostatic induction, exhibiting a high performance of 20.7 W $m^{-2}$ $Hz^{-1}$ [86].

Additionally, the dielectric films with weak charge constraint ability and high surface tribo-potential will demonstrate the hysteretic and ordered charge migration behavior under the traction of the triboelectric electric field. Therefore, they can be further employed in freestanding TENG to obtain sustained output by changing the short pulse electrical signals into stable and continuous output.[260] As shown in Figure 22b, a strong triboelectric electric field is generated between part of the slider area and the bottom electrode as the slider moves back and forth. This enables the tribo-charges to migrate spontaneously, producing a stable and continuous AC electrical signal in the external circuit with a high power density of 9.4 W $m^{-2}$ $Hz^{-1}$. In summary, utilizing the ordered migration of surface charges within dielectric materials is an emerging approach to facilitating triboelectric output stability.

## 5. Dielectric Materials for TENG Application

Dielectric materials based TENGs have demonstrated excellent application value in micro power sources, high power sources, self-powered healthcare, self-powered sensing, and blue energy. In this section, we highlight TENGs with advanced dielectric materials that perform better overall for particular application scenarios, aiming to provide new inspiration for future materials, devices, applications, and industrialization in the field.

### 5.1. Micro Power Sources

TENGs that serve as microdistributed power sources through multiple microenergy harvesting have made significant progress in power density, durability, and stability, where the dielectric materials play a crucial role in ensuring high output performance. TENGs as micro power sources are restricted by the high output impedance, low current, and high voltage intrinsic characteristics, which cannot match most load impedance characteristics of low voltage and large current. These deficiencies can be overcome by implementing power management circuits or modules, and inherent characteristics control is also desirable. Recently, Ren et al. proposed a strategy for optimizing the inherent features of free-standing TENG by coupling the device to a capacitor auto-conversion circuit.[261] As shown in **Figure 23a(i)**, the TENG is combined with synchronous periodical mechanical switches, which could reduce the output impedance and deliver the power density of 124 W $m^{-2}$ and energy density of 252 mJ $m^{-2}$ under 470 and 150 Ω. The high DC voltage and low impedance characteristics allow for powering 5000 LEDs and two lamps (220 V, 50 W), offering a new method to reduce external load impedance and improve output. In another work, the ultrahigh instantaneous power density of 10 MW $m^{-2}$ was achieved by opposite-charge-enhancement effect and transistor-like device design.[92] The device is similar to a complementary transistor pair, consisting of a stator substrate and slider. The stator comprises four electrodes and coplanar tribo-surfaces with opposite-charge polarization (Figure 23a(ii)). The charges can flow between the "source" and "drain" during operation, bringing an instantaneous power density of 10 MW $m^{-2}$ at 1 Hz that could light a 180 W commercial lamp or 30 W LEDs wirelessly.

Regarding high-performance DC-TENG for mechanical energy harvesting, challenges are concentrated in enhancing the power density. Traditional metal-dielectric based devices demonstrate poor CE performance, and integrating multiple dielectric–dielectric CE remains challenging as the discharge paths from neighboring tribo-layers on one electrode are opposite. Recently, Shan et al. proposed a ternary dielectric electrification TENG utilizing Kapton/electrode/PTFE as the slider and PTFE powder-modified PU as the stator (Figure 23a(iii)).[262] The optimal ternary CE material delivers a codirectional charge collection on a single electrode. The enhanced charge collecting design efficiently gathers the produced charges, leading to a charge density of 5.5 mC $m^{-2}$ and an average power density of 12.4 W $m^{-2}$. This work highlights the significance of dielectric material optimization in DC-TENG. Furthermore, the proposal for cooperative use of CE, electrostatic discharge, and electrostatic induction in a single device was realized to create a dual-functional TENG.[263] The square-wavelike AC output was generated under back-and-forth reciprocal motion, and the constant DC output can be obtained in the case of unidirectional motion. The device delivers a constant current density and power density of 1.51 mA $m^{-2}$ and 398 mW $m^{-2}$, realizing the convertible shift of AC/DC output with little limitation on CE material selection.

Wind energy harvesting belongs to the further expansion of solid–solid TENG. Reducing the cut-in wind speed and increasing the power density, durability, and environmental adaptation have been the main topics of recent research. Specifically, effective strategies such as the utilization of soft contact separation,[264] free-standing operation mode,[265] structural wind speed adaptation,[266,267] and EMG coupling have been proposed.[268–270] For example, Yong et al. designed a dual-rotation shaft TENG for broadband wind speed adaptation (2.2–16 m $s^{-1}$) and harvesting (Figure 23b).[271] Li et al. proposed a leaf-like TENG to realize damped forced vibration and prevent the electrostatic adsorption between CE layers, delivering the low cut-in speed of 0.2 m $s^{-1}$ and peak power density of 3.98 mW.[272] Further optimization of the self-cleaning, hydrophobic, moisture-proof, and corrosion-resistant qualities of dielectric materials is highly anticipated to significantly advance wind energy harvesting in harsh outdoor situations for industrial applications.

Another significant subset of micro power sources is the water droplet energy harvesting, and the output performance can be improved by optimizing the interface wettability (hydrophobicity), surface charge density of the dielectric layer. In 2020, Xu et al. reported a pioneering breakthrough device structure for high instantaneous droplet energy harvesting, as illustrated in Figure 23c(i).[273] The TENG consisted of a glass substrate, ITO electrode, PTFE triboelectric layer and Al electrode. The incorporation of Al electrode transforms the interfacial effect into a bulk effect by building a closed-loop electrical system, delivering an open-circuit output voltage of 143.5 V, a short-circuit current of 270.0 μA, and an instantaneous peak power density of 50.1 W $m^{-2}$. From the circuit model, the spreading droplet and PTFE can be treated as a resistor and capacitor ($C_p$), respectively. The capacitor at the interfaces between water/PTFE ($C_1$) and water/aluminum interface ($C_2$) exists when the aluminum electrode is employed, turning the initial open circuit


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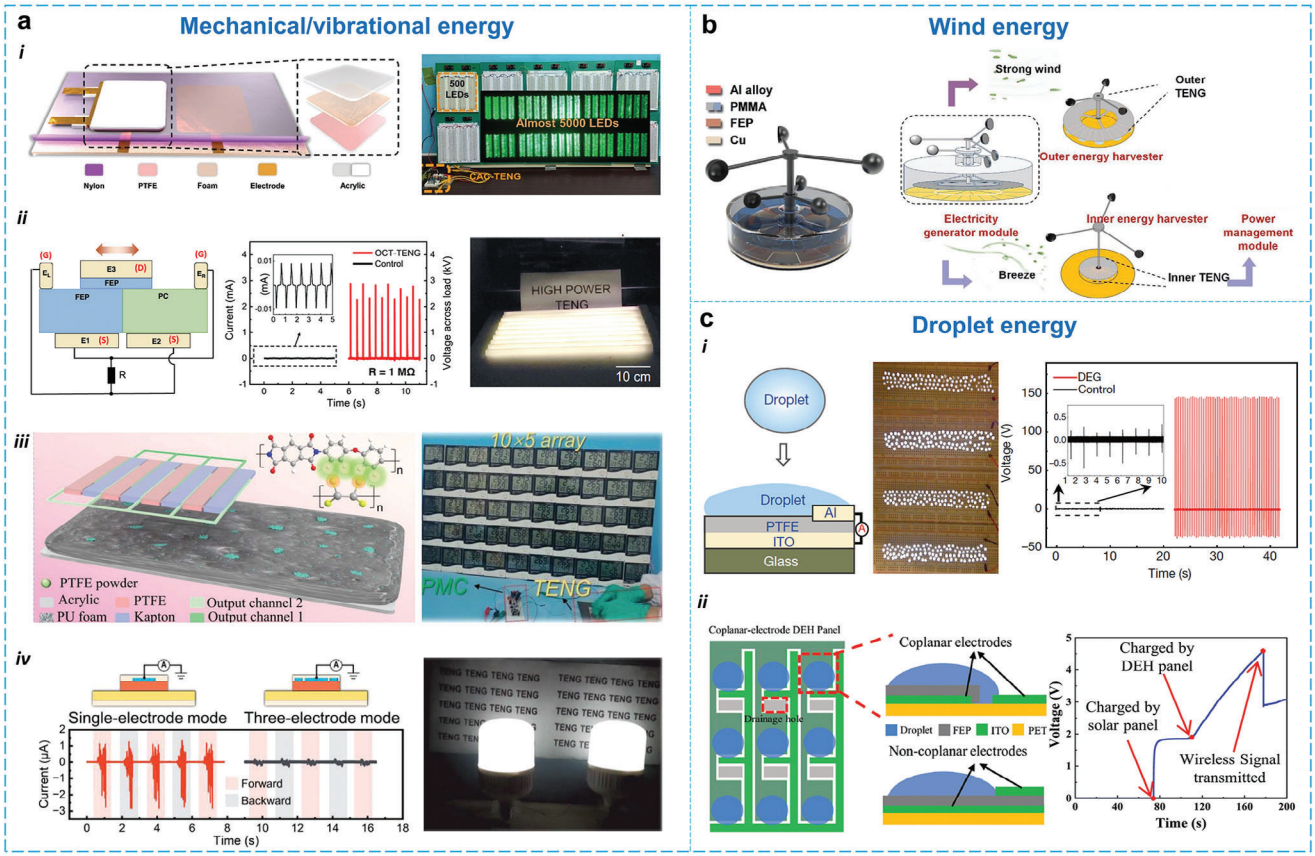


**Figure 23.** TENGs as micro power sources. Mechanical/vibrational energy. a(i) Schematic and the output of capacitor auto-conversion circuit-TENG. a(ii) Structure and the output performance of the opposite charge enhanced transistor like TENG. a(iii) Structure diagram and the output of the ternary dielectric electrification-TENG. a(iv) The output of AC/DC dual-functional TENG. b) Structure and working mechanism of the dual-rotation shaft TENG for wind energy harvesting. c(i) Droplet based high instantaneous power TENG. c(ii) Droplet energy harvesting panel. a(i) Reprinted with permission from ref. [261]. Copyright 2023, Wiley-VCH. a(ii) Reprinted with permission from ref. [92]. Copyright 2021, the authors. a(iii) Reprinted with permission from ref. [262]. Copyright 2023, Wiley-VCH. a(iv) Reprinted with permission from ref. [263]. Copyright 2023, Wiley-VCH. b) Reprinted with permission from ref. [271]. Copyright 2021, Wiley-VCH. c(i) Reprinted with permission from ref. [273]. Copyright 2020, Springer Nature. c(ii) Reprinted with permission from ref. [274]. Copyright 2022, The Royal Society of Chemistry.

into a closed circuit upon operation. Considering the capacitance $C_P$ is negligible compared to the $C_1$, $C_2$ owing to the PTFE thickness being several orders of magnitude higher than the DEL, the voltage across $C_P$ is several orders higher when the droplet contacts the Al electrode. Thus, the ultrahigh output can be obtained. Based on the transistor-inspired bulk effect, scholars have designed droplet energy-harvesting panels for large-scale applications. As shown in Figure 23c(ii), a transparent, high-performance droplet harvesting panel was designed to deliver superior collaborative integration with the solar pane for hybrid rain-solar energy harvesting.[274]

As to micro power sources, traditional high-performance dielectric materials based triboelectric pairs demonstrate flexible compatibility and various options with optimized unique device structures. Dielectric materials can be effectively tailored to enhance energy harvesting efficiency and device longevity in various application settings. For instance, the corrosion resistance, oxidation resistance, and self-cleaning properties of dielectric materials are beneficial for outdoor energy harvesting. Interface wettability can aid in specific situations such as droplet energy harvesting, whereas the transparency of certain dielectric materials enhances the operation of droplet-TENGs in conjunction with solar cells. Regarding DC TENGs, the significance lies in the dielectric materials' coupling and wear resistance. The development of micro and nano power sources is expected to be enhanced by overcoming barriers to using high-performance dielectric materials and advanced topologies.

### 5.2. High Voltage Power Sources

Dielectric material based TENGs can serve as a high-voltage power source in the industrial sector because of its inherent high voltage, low current, and high internal resistance characteristics. The benefits include cost-effectiveness, superior safety, and reduced energy consumption compared to commercial high-voltage power supplies. It can also be utilized for plasma modification or characterizing high-performance dielectrics. In 2019, Cheng et al. introduced the first triboelectric microplasma enabled by TENG.[91] As shown in


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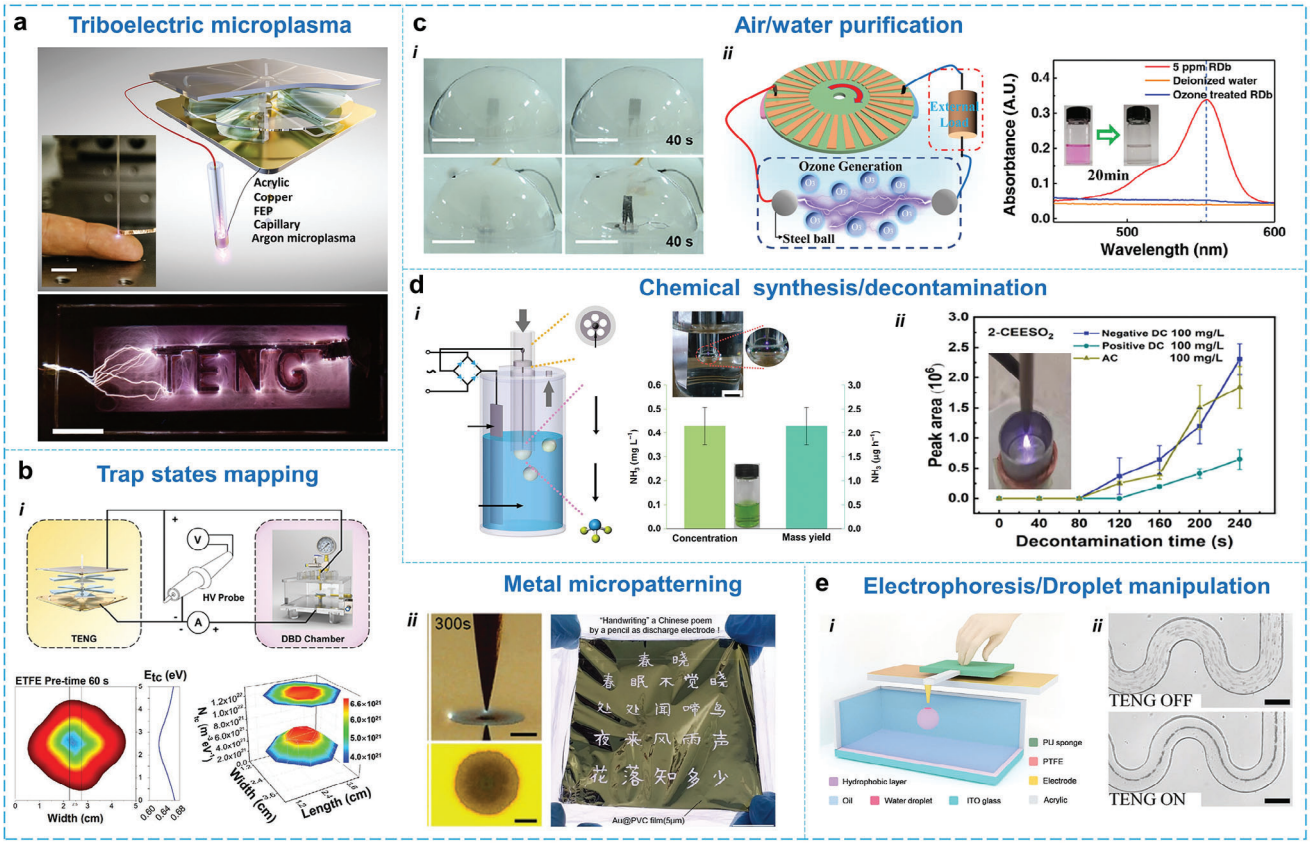


**Figure 24.** TENGs as High voltage power sources. a) Triboelectric microplasma generated by TENG. b(i) The electron trap states measurement system driven by TENG. b(ii) Metal micropatterning driven by TENG. c(i) Negative air ions generated by TENG for air purification. c(ii) Water purification system driven by TENG. d(i) TENG based self-powered ammonia synthesis. d(ii) Chemical warfare agents decontamination through tribo-plasma. e(i) 3D human-droplet interaction driven by TENG. e(ii) Dielectrophoretic microparticle manipulation driven by TENG. a) Reprinted with permission from ref. [91]. Copyright 2019, the authors. b(i) Reprinted with permission from ref. [275]. Copyright 2021, Wiley-VCH. b(ii) Reprinted with permission from ref. [91]. Copyright 2021, Wiley-VCH. c(i) Reprinted with permission from ref. [276]. Copyright 2023, Wiley-VCH. c(ii) Reprinted with permission from ref. [277]. Copyright 2021, Elsevier. d(i) Reprinted with permission from ref. [279]. Copyright 2021, Springer Nature. d(ii) Reprinted with permission from ref. [281]. Copyright 2022, Elsevier. e(i) Reprinted with permission from ref. [282]. Copyright 2023, Wiley-VCH. e(ii) Reprinted with permission from ref. [283]. Copyright 2022, Wiley-VCH.

**Figure** 24a, the FEP-based triboelectric layer successfully generated four classical atmospheric-pressure microplasma sources, including DBD, corona discharge, microspark discharge, and atmospheric-pressure nonequilibrium plasma jets (APNP-J) were successfully triggered by mechanical stimuli. In another work, Wang et al. achieved the electron trap states mapping of dielectric materials by the high-voltage dielectric based TENG.[275] The detrapping currents were extracted from the DBD parameter based on the proposed discharge model with dynamic carriers' evolution, and then the quantitative calculation of energy levels and densities of trap states was realized (Figure 24b(i)). Recently, Wang et al. developed a triboelectric spark discharge-induced metal micropatterning strategy (Figure 24b(ii)).[91] The high voltage TENG generates the controllable spark discharge, and the electrical discharge energy is transformed into thermal energy to realize high-temperature melting and evaporation for metal etching. The spark discharge parameters such as voltage, current, and frequency could be flexibly controlled to adapt to different pattern materials, overcoming the difficulties of metal micropatterning on thin, flexible, or elastic substrates.

The high voltage produced by dielectric based TENG may also be utilized for air and sewage purification.[276–278] This approach offers the advantages of low cost, low carbon, and simplicity in structure, demonstrating vast potential for industrial pollutants removal, automobile exhaust treatment, etc. For example, Liu et al. proposed a single electrode TENG with ultra-high voltage output (12.7 kV) by optimizing the surface charge density and packing layer thickness.[276] The device can generate negative air ions by corona discharge for dust removal, as shown in Figure 24c(i). Based on the electrostatic adsorption effect, the active particles quickly dilute air pollutants. In another work, Lei et al. proposed a "like charges repel" TENG to generate ozone for water purification.[277] As shown in Figure 24c(ii), the maximum output voltage of 30.7 kV endows the 30 mg $m^{-3}$ ozone generation, which could degrade Rhodamine B by 97.6% in 20 min.

Besides, chemical synthesis or decontamination based on high voltage TENG is also reported.[279–281] In one work, self-powered ammonia synthesis was achieved via $N_2$ discharge driven by high-voltage TENG.[279] $N_2$ molecules were excited and ionized under a high electrical field to form $N^+$ and $N^{2+}$, which further react with


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$H_2O$ to generate $NH_3$ (Figure 24d(i)). Wong et al. designed a high-voltage TENG (output of 1300 V) for self-powered nitrogen fixation. The generated microplasma drives the conversion of $N_2$ into nitrogen dioxide, nitric acid, and the $NaNO_3$ fertilizer produced by human walking stimuli is confirmed and demonstrated for crop cultivation.[280] In addition, the self-powered chemical warfare agent decontamination was also reported by Bai et al.[281] As shown in Figure 24d(ii), a double-layer paper-strip rotary TENG was employed for excited air microplasma generation, resulting in more than 99% degradation efficiency of 2-chloroethyl ethyl sulfide (2-CEES) within 2 min.

The coulomb force generated by high voltage TENG demonstrates promising application in high-precision control of small objects/fluids, electroosmosis, electrostatic spraying, and inkjet printing. Recently, Sun et al. achieved a pipette in an oil medium via tribo-electrophoresis by rational design of TENG- electrostatic tweezer.[282] The generated electric field charges the droplet and enhances its maneuverability, delivering a new strategy for human–droplet interactions (Figure 24e(i)). Moreover, the dielectrophoretic microparticle manipulation through a rotary free-standing TENG was reported by Zhou et al., where a nonuniform electric field was built within the microchannel by coupling the rectifier/filter circuit.[283] As shown in Figure 24e(ii), the randomly distributed particles converge in a row, and different particles are separated by the distinct DEP forces when the TENG is on in the serpentine channel. Other typical applications, such as self-powered electrospinning system, electrokinetic microfluidic, micro-flow pumps have also been realized.[284-286]

The diverse application potential of HV-TENG has been shown using several triboelectric couples such as FEP/copper, PTFE/PMMA, or Nylon, Kapton/Nylon, highlighting the wide range of options and cost-effectiveness. Humidity has a notable effect on discharge, and humidity-resistant dielectric materials can assist high-voltage sources in operating consistently in humid conditions. Creating a high-voltage power source using TENG requires designing and combining high-performance dielectrics and metal electrodes, combined with external mechanical force, usually without additional sophisticated components. As a result, HV-TENG stands out from other high-voltage power supplies due to its outstanding safety features and energy efficiency. Even at thousands of volts, its low current brings little hurt to living things. Further enhancement of the output performance could be achieved by utilizing high-performance dielectric materials and creating innovative architectures, which is advantageous for reducing device size and enabling portable applications.

### 5.3. Self-Powered Healthcare

The versatile adjustability of dielectric materials allows TENG to be utilized as a wearable device for self-powered healthcare, offering benefits such as affordability, uncomplicated design, independence from power sources, excellent biocompatibility, and suitability for implantation.[287] In addition to possessing outstanding triboelectric characteristics, the dielectric material should fulfill worn or implantable requirements, including flexibility, breathability, stretchability, nontoxicity, biodegradability, biocompatibility, etc. Advances have been achieved in the following scenes.

The high voltage and low current pulse output of TENG make it appropriate for antimicrobial treatments, disinfection, and bioelectrical stimulation therapy.[288] For example, Zhang et al. developed a "genetically engineered" biofunctional TENG utilizing recombinant spider silk proteins (RSSP) for in vivo therapeutic operation.[177] The RSSP patch creates an electric potential difference between bacteria and a charged surface, leading to bacterial morphological destruction and triggering a burst of reactive oxygen species, resulting in a high antibacterial rate of 67.4% (**Figure 25a(i)**). In another work, Huo et al. presented the self-powered microbial disinfection by the triboelectric electric field, comprising a vibration-driven TENG, a three-electrode disinfection filter and a power management system.[289] The positive electrode modified by $Cu_3PNW$ could create an ultrahigh electric field ($>10^7$ V m$^{-1}$) to realize electroporation inactivation, as illustrated in Figure 25a(ii).

Meanwhile, bioelectrical stimulation therapy such as infected wound healing, improved epidermal growth, drug loading, and cancer catalytic were also reported.[290–296] For example, Zhang et al. developed a TENG based self-powered microneedle system for infected wound treatment and conductive drug delivery with daily cloth and Mxene/alginate/glycerol as tribo-pair (Figure 25b(i)).[291] In another work, Wang et al. proposed a knee joint movements-driven hybrid tribo/piezoelectric nanogenerator for bone repair by increasing intracellular $Ca^{2+}$ levels. The device was inserted into the subcutaneous tissue of the thigh to produce two-phase electric pulses, creating a bone-forming environment in the damaged area using a conductive injectable hydrogel (Figure 25b(ii)).[292] Regarding drug loading, Du et al. introduced a TENG patch featuring surface-engineered electrodes for precise drug loading and release control based on the localized electrical stimulation (Figure 25c(i)), where the Mg–Al layered double hydroxide was used as a smart drug container and friction layer.[293] Yang et al. improved pharmacodynamics of epidermal growth via microneedle-based self-powered transcutaneous electrical stimulation system, where a sliding free-standing TENG and microneedle patch was employed (Figure 25c(ii)).[296] Besides, a self-powered reactive oxygen species synthesis system for cancer catalytic therapy was also reported.[297] The peroxidase-like activity of 1D ferriporphyrin covalent organic framework (COF)-CNT treated by triboelectric field brings a superior catalytic effect for malignant 4T1 breast carcinoma.

TENG biomechanical sensors or electronic skin can convert human movements into analog signals to monitor physiological indications continuously, such as respiratory activity, cardiovascular status, etc.[298] Regarding respiratory monitoring, Fang et al. proposed a permeable and moisture-proof TENG textile sensor with two twisted triboelectric yarns of PVDF and epoxy, as shown in Figure 25d(i).[93] Then, a computational fluid dynamics-assisted mask sensor network was constructed to achieve high-precision respiratory signal recognition. In another work, the electrospun PVDF/Nylon triboelectric pair was utilized for respiratory healthcare, as shown in Figure 25d(ii). An extended service lifespan of 60 hours and filtration efficiency of 95.8% (0.3 μm particles) was achieved.[299] As to cardiovascular healthcare, TENG enables self-powered monitoring of pulse-wave signals without bulky monitors or external wires. Progress has been made in


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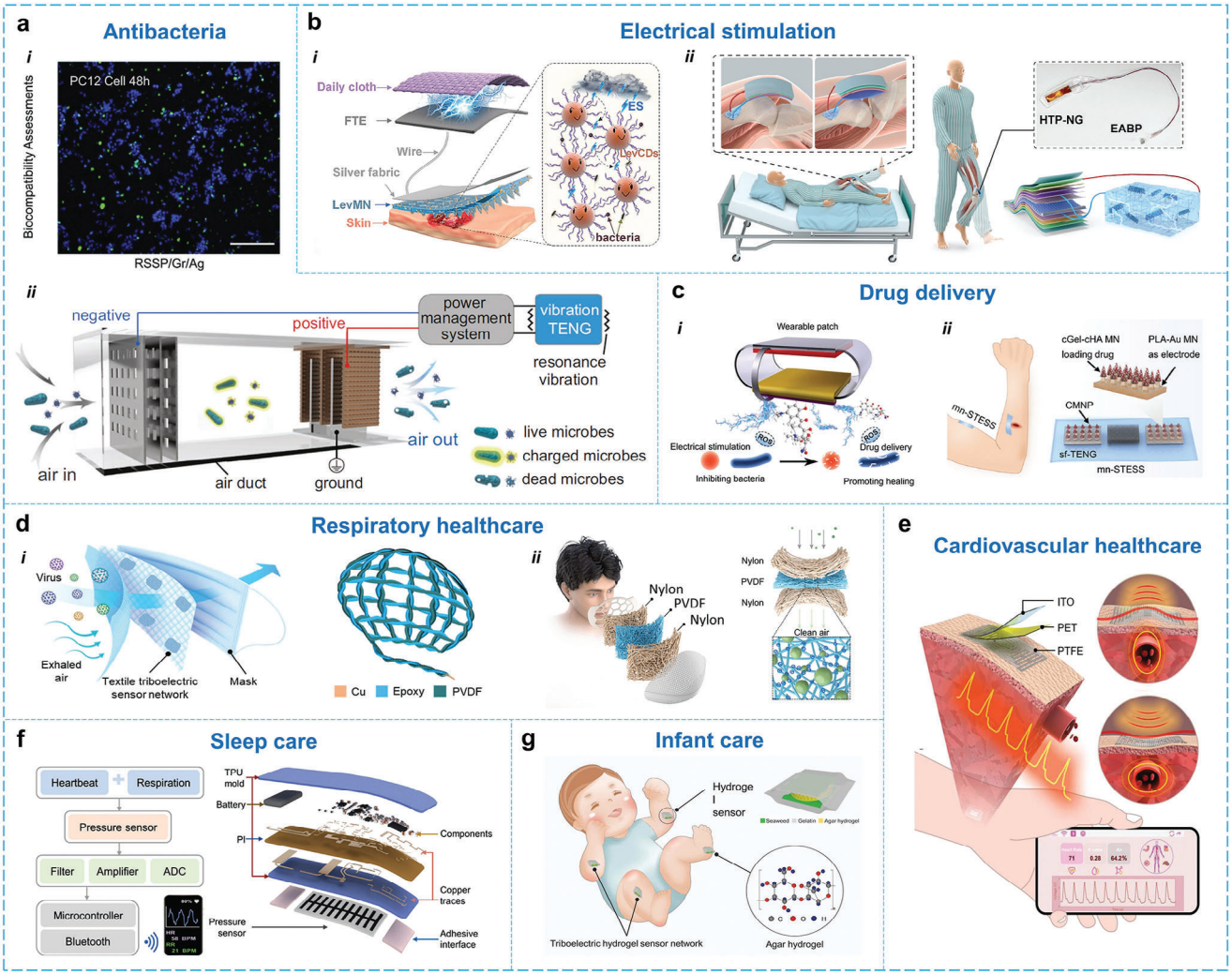


**Figure 25.** TENGs for self-powered healthcare. Antibacterial activity: a(i) Demonstration of antibacterial rate. a(ii) Self-powered microbial disinfection enabled by TENG. Electrical stimulation: b(i) Self-powered microneedle system for treating infected wounds enabled by TENG. b(ii) Pulsed electrical stimulation to speed up bone healing enabled by TENG. Drug delivery: c(i) Self-powered drug delivery system enabled by TENG. c(ii) Improved pharmacodynamics of epidermal growth enabled by TENG. Respiratory healthcare: d(i) TENG textile sensor with two twisted triboelectric yarns of PVDF and epoxy. d(ii) Self-charging electrostatic face mask enabled by TENG. e) Self-powered cardiovascular healthcare system enabled by TENG. f) Self-powered sleep care system enabled by TENG. g) Self-powered infant care system enabled by TENG. a(i) Reprinted with permission from ref. [177]. Copyright 2018, Wiley-VCH. a(ii) Reprinted with permission from ref. [289]. Copyright 2021, the authors. b(i) Reprinted with permission from ref. [291]. Copyright 2023, Elsevier. b(ii) Reprinted with permission from ref. [292]. Copyright 2024, the authors. c(i) Reprinted with permission from ref. [293]. Copyright 2021, Elsevier. c(ii) Reprinted with permission from ref. [296]. Copyright 2022, the authors. d(i) Reprinted with permission from ref. [93]. Copyright 2022, Wiley-VCH. d(ii) Reprinted with permission from ref. [299]. Copyright 2022, the authors. e) Reprinted with permission from ref. [94]. Copyright 2022, Wiley-VCH. f) Reprinted with permission from ref. [301]. Copyright 2023, Wiley-VCH. g) Reprinted with permission from ref. [302]. Copyright 2022, Wiley-VCH.

self-powered cardiac monitoring, therapeutic devices, and power sources for cardiac pacemakers, etc.[300] For example, Meng et al. developed a kirigami-inspired TENG for wireless cardiovascular monitoring, as shown in Figure 25e.[94] The 3D vertical kirigami-inspired notches were designed to enhance the device's adaptability to vertical pulse waves, flexibility, and breathability. Wang et al. proposed a soft bioelectronic patch for sleep care, and continuous breathing and heartbeat monitoring throughout the night can be realized for obstructive sleep apnea-hypopnea syndrome diagnosis, as shown in Figure 25f.[301] Additionally, soft and edible dielectrics can also be applied for TENG sensors in infant care applications. Guo et al. proposed an edible and flexible gelatin hydrogel as the triboelectric layer to resemble a sandwiched structure TENG (Figure 25g).[302] Since every component of the TENG sensor is edible, the possibility of a baby accidentally ingesting it and the associated risks may be eliminated. Then, a body area network was established to detect abnormal movement patterns in infants with deep learning algorithms, enabling comprehensive monitoring of infant mobility.

Briefly, dielectric materials like nanofiber, and textiles possess elasticity, stretchability, breathability, washability, which offer efficient energy harvesting and comfort for wearable electronics


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adapted to skin, organs. It allows for self-powered monitoring of health factors like pressure, temperature, auditory signals, walking, and respiration. Simultaneously, high-performance dielectrics can generate electrical stimulation for medical purposes such as wound healing, and self-powered drug delivery. The biocompatibility, degradability, and antibacterial properties of dielectric materials enable the potential for in-body implantation and associated applications. Additionally, specific dielectric materials, known for being edible and biologically safe, can be used to expand self-powered healthcare for newborns and young children. Overall, the distinctive characteristics of these dielectric materials bestow a broad spectrum of applications in self-powered healthcare.

### 5.4. Self-Powered Sensors

TENG-based self-powered sensors in detecting physical, chemical parameter with outstanding performance have received widespread achievements in recent years. On the one hand, TENG can address the distributed sensing power supply issue by acting as a micro/nano power source for conventional low-power sensors, as outlined in Section 5.1. On the other hand, feature information sensing without an external power supply can be accomplished by utilizing the correlation between the output signal of TENG (voltage, current, and charge) and external stimulus inputs. Regarding chemical parameters, solution detection, gas composition, etc. have been reported. For example, Luo et al. designed a variable frequency liquid–solid TENG for nanofluid dispersion measurement.[303] As shown in **Figure 26**a(i), the FEP tube is employed as the structural framework and triboelectric sensing layer, with a copper film serving as the electrode. Combined with the quantum genetic and levenberg–marquardt hybrid algorithm optimization method, the particle size and spacing of nanoparticles in the solution were obtained. Recently, Liu et al. reported a triboelectric probe based on FEP for sucrose solution concentration sensing.[96] The detection mechanism is ascribed to the sucrose that destroys the hydrogen bond network and changes the interface EDL properties, resulting in alterations in the liquid–solid CE performance and delivering a sensitivity of $-0.0038\%^{-1}$ (Figure 26a(ii)).

In addition, high-precision monitoring of micro-droplet parameters through liquid–solid TENG was achieved by Liu et al.[304] The microfluidic structure consists of a glass slide and PDMS sheet for fluid constrain, and the TENG based on a PDMS triboelectric layer and copper electrode (Figure 26a(iii)). The frequency, length, and velocity of the liquid can be determined using the TENG output pulse frequency and duration. Advancements were also made in the self-powered liquid–solid TENG for industrial settings. For example, Zhu et al. presented a shaftless turbine intake flow sensor composed of ball-bearing TENG and magnetic gear EMG.[305] As illustrated in Figure 26a(iv), the device delivers a sensing sensitivity of 0.71 Hz $L^{-1}$ $min^{-1}$ in the 300–900 L $min^{-1}$ range with superior linear relationship. In another work, Zhao et al. realized the worn debris detection in lubricating oil by employing 1*H*, 1*H*, 2*H*, 2*H*-perfluorodecyltriethoxysilane and hydrophilic nano-$SiO_2$ functional layer to enhance the triboelectric output and sensitivity, resulting in the detection limit of 0.01 wt% of worn debris and 100 µL $L^{-1}$ of water in oils.[194] It motivates us that sensing layer tuning is necessary for liquid component monitoring to get better performance.

As to self-powered gas sensing, TENG also demonstrates exciting advantages and features. Generally, the impedance or resistance of the resistive gas sensor will vary depending on the gas composition. When employing TENG as a micro power source to enable self-powered gas sensing, the voltage response to various impedances will alter noticeably. Consequently, it is possible to further enhance the gas sensor's response sensitivity by leveraging the "impedance matching effect" between TENG and the gas sensor. For example, Wang et al. designed a latex/PTFE based TENG combined with $Ti_3C_2T_x$/metal–organic framework-derived copper oxide gas sensor for self-powered room temperature $NH_3$ detection.[306] An excellent response of $V_g/V_a$ = 24.8 @ 100 ppm was achieved compared to the resistive response ($R_g/R_a$) of 1.8 (Figure 26b(i)). Besides, TENG can also directly utilize gas-sensitive materials as triboelectric layers. The basic principle of this strategy is that the gas composition will affect the surface charge density of the triboelectric active sensing layer, resulting in the output signal change (voltage or current). That is, the triboelectric-sensitive layer functions as a probe to detect the chemical composition. For example, Su et al. presented an alveolus-inspired membrane sensor (AIMS) for room temperature $NO_2$ detection.[307] The tungsten trioxide ($WO_3$) composite film was employed as tribo-active layer with latex as the other triboelectric film, delivering high sensitivity (452.44%), outstanding linearity (0.976), and superior selectivity of $NO_2$. The detection mechanism is attributed to the adsorbed $NO_2$ molecules that withdraw the electrons in the $WO_3$ layer and cause depolarization field reduction to a higher triboelectric output (Figure 26b(ii)). Thus, the selection and optimization of the tribo-dielectric active layer are crucial for this type of self-powered sensor (Figure 26b(iii)), and more detailed information can be obtained from the previous reviews.[308,309]

Besides, there have been plenty of application scenarios for TENG based self-powered sensing in detecting physical parameters, including industrial equipment status monitoring, human–machine interaction, soft-robotics, etc. For example, the all-in-one self-powered sensing system was proposed for vibration monitoring of power transmission and distribution equipment such as generators, transformers, gas-insulated switchgear (**Figure 27**a(i)).[310] Some other progress was also reported for vibration, swing monitoring of transmission lines, electric motor, while traditional high-performance dielectric materials are mainly used for adaptive equipment structure.[311–314] TENG also helps build smart transportation systems such as intelligent vehicle perception, self-powered road signs. Recently, He et al. reported a self-excited liquid suspension TENG with dual dielectric tribo-layers (Nylon, oil) for security identifier, as shown in Figure 27a(ii).[315] The creative dielectric coupling offers advantages such as a small driving force, long service life, high output, and robust performance.

As to human–machine interaction and soft-robotics, TENG could be utilized as the pressure sensor,[216] touch panel,[316] tactile perception,[95] electronic skin,[317] active eye tracking,[318] acoustic sensing,[222] etc., as illustrated in Figure 27b(i). Changes have been made to the tribo-dielectric sensing layer to better meet the requirements of application scenarios. For


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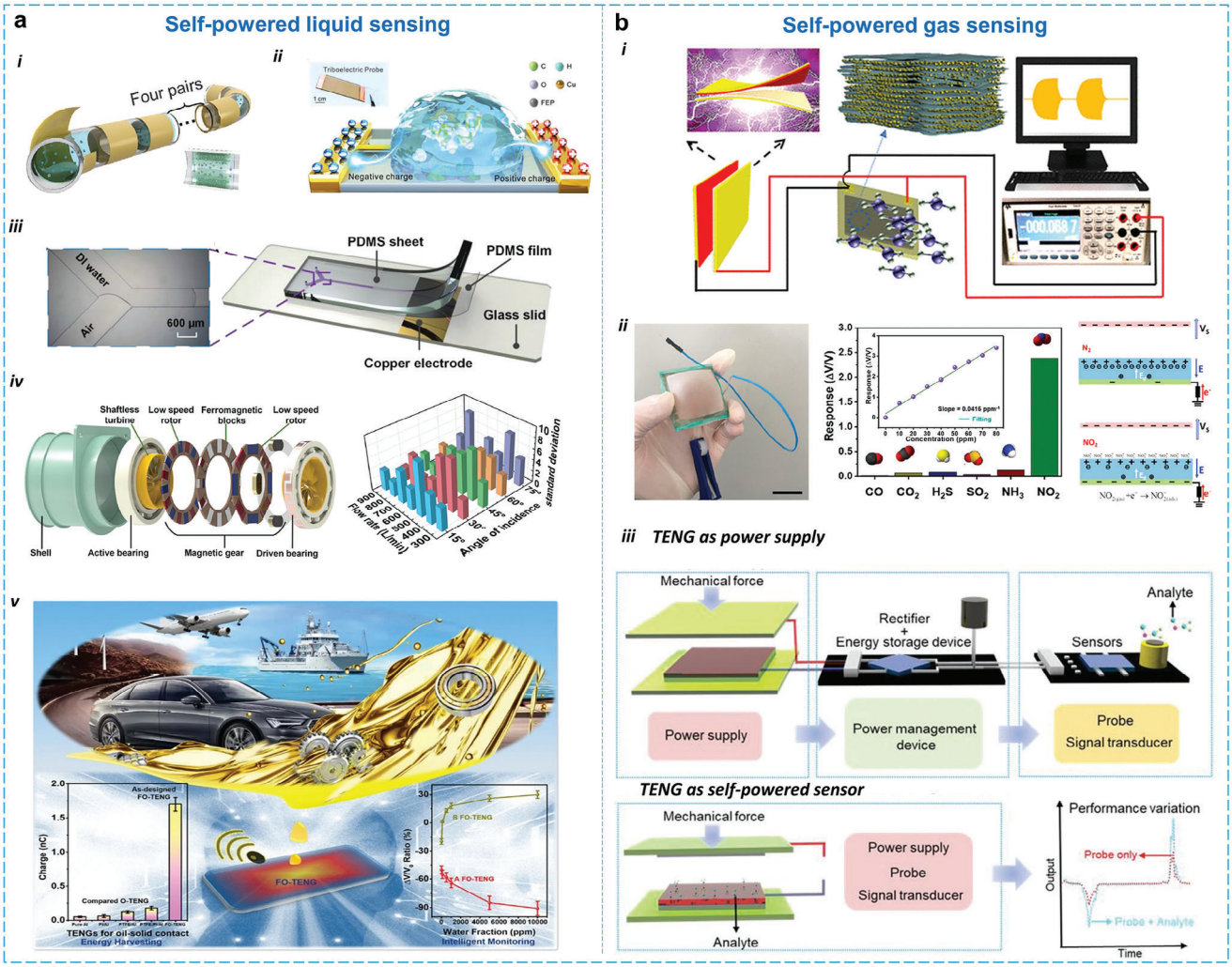


**Figure 26.** TENGs as self-powered sensors for chemical parameter detection. Self-powered liquid sensing: a(i) In situ nanofluid dispersion monitoring enabled by TENG. a(ii) Real-time monitoring of sucrose fluid status through TENG. a(iii) Micro-droplet parameters monitoring enabled by TENG. a(iv) Self-powered flow sensing enabled by TENG. a(v) Lubricating oil condition monitoring enabled by TENG. b(i) TENG driven self-powered ammonia sensor. b(ii) Alveolus-inspired membrane based self-powered sensor for $NO_2$ detection. b(iii) Schematic diagram of self-powered sensing system. a(i) Reprinted with permission from ref. [303]. Copyright 2022, Wiley-VCH. a(ii) Reprinted with permission from ref. [96]. Copyright 2023, Wiley-VCH. a(iii) Reprinted with permission from ref. [304]. Copyright 2023, Wiley-VCH. a(iv) Reprinted with permission from ref. [305]. Copyright 2023, Elsevier. a(v) Reprinted with permission from ref. [194]. Copyright 2022, the authors. b(i) Reprinted with permission from ref. [306]. Copyright 2021, American Chemical Society. b(ii) Reprinted with permission from ref. [307]. Copyright 2020, American Chemical Society. b(iii) Reprinted with permission from ref. [309]. Copyright 2021, Wiley-VCH.

example, 2D siloxene-PVDF/PA66 electrospun dielectric membrane was designed for hybrid pressure sensing owing to membrane compressibility, bringing excellent dynamic and static pressure sensitivities.[222] In another work, the conductive hydrogel was exploited to construct the antifreezing, self-adhesive, self-healing, transparent, stretchable, and degradable self-powered touch panel.[316] Moreover, the hierarchically engineered spongy thermoplastic PU composite coupling with ferroelectric BTO/$Ti_3C_2T_x$ nanosheets was fabricated for high-performance energy conversion, pressure sensitivity, and acoustic sensing.[222] For electronic skin, the breathability, stretchability and biocompatibility of functional materials and devices are of utmost necessity to facilitate heat dissipation and sweat evaporation. Shrestha et al. designed a siloxene/SEBS based breathable TENG as electronic skin for pressure sensing and gesture recognition.[317]

Besides, the ultrahigh surface charge density on dielectric material also endows the noncontact perception. For example, Shi et al. proposed a transparent, flexible, ultrapersistent TENG for active eye tracking.[318] The unprecedented charge storage capability was realized thanks to the triple-layer structure of polychlorotrifluoroethylene (PCTFE)/PDMS, Ag nanowires and PDMS. The composite dielectric bilayer was precharged and delivered an electrostatic charge density of 1671.10 μC $m^{-2}$ after 1000 noncontact operations. An in-depth understanding of the functional properties of dielectrics and microstructural engineering has effectively promoted self-powered HMI and soft robotics. Further, the creative intelligent HMI could be built with


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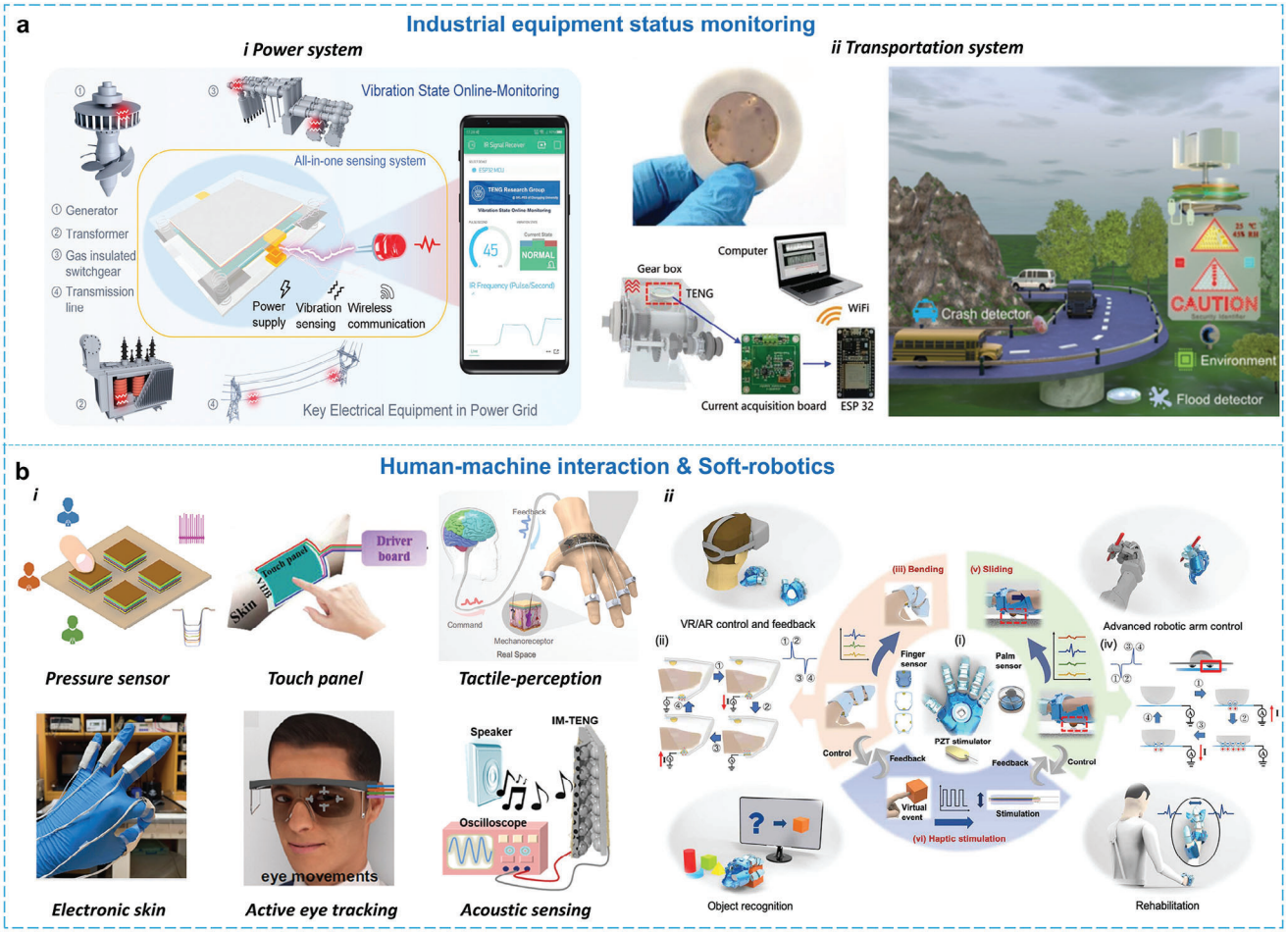


**Figure 27.** TENGs as self-powered sensors for physical parameter detection. Industrial equipment status monitoring. a(i) All-in-one sensing system for electrical equipment. a(ii) TENG based smart transportation system. Human-machine interaction and Soft-robotics: b(i) TENG employed as pressure, touch panel, tactile-perception sensor, electronic skin, active eye tracking, and acoustic sensing. b(ii) TENG based smart HMI system. a(i) Reprinted with permission from ref. [310]. Copyright 2023, Wiley-VCH. a(ii) Reprinted with permission from ref. [315]. Copyright 2023, Wiley-VCH. b(i) Reprinted with permission from ref. [95]. Copyright 2022, the authors. Reprinted with permission from ref. [216]. Copyright 2022, Wiley-VCH. Reprinted with permission from ref. [222]. Copyright 2022, American Chemical Society. Reprinted with permission from ref. [316]. Copyright 2022, Wiley-VCH. Reprinted with permission from ref. [317]. Copyright 2023, Wiley-VCH. Reprinted with permission from ref. [318]. Copyright 2023, the authors. b(ii) Reprinted with permission from ref. [319]. Copyright 2020, the authors.

the help of AI algorithms and machine learning.[319–322] For instance, Zhu et al. proposed a haptic-feedback smart glove for virtual/augmented reality applications.[319] Figure 27b(ii) shows that the smart glove based on a triboelectric signal is competent for multiple application scenarios, including motion detection, VR/AR control/feedback, robotic arm control, object recognition, and rehabilitation.

Overall, the sensitivity to physical or chemical parameters has been effectively improved by constructing various high-performance dielectric-sensitive materials, although stability and cross-selectivity advancements are still possible. Self-powered sensing relies on interface charges' exceptional reactivity to detect unknown chemical components, particularly for chemical parameters. When gas, liquid, and solid coexist, the distribution of tribo-charges or electric double layer (EDL) varies compared to when only one component is present. Moreover, certain substances can undergo chemical reactions with tribo-dielectric (such as chemical adsorption), affecting the dielectric parameters (such as relative permittivity and carrier concentration). Consequently, the high-performance tribo-sensitive layer (coupling certain sensitive materials with tribo-dielectric) bestows TENG with highly flexible self-powered sensing capabilities. As a result, in accordance with methods suggested earlier, TENG-based self-powered sensing technology is expected to make substantial advancements.

### 5.5. Blue Energy

TENG is capable of converting disordered, low-frequency wave energy into electricity, providing a new route to blue energy vision. Researchers have explored the potential of harnessing kinetic energy from ocean currents, waves, updrafts, and coastal impacts using various TENG structures, which can be broadly


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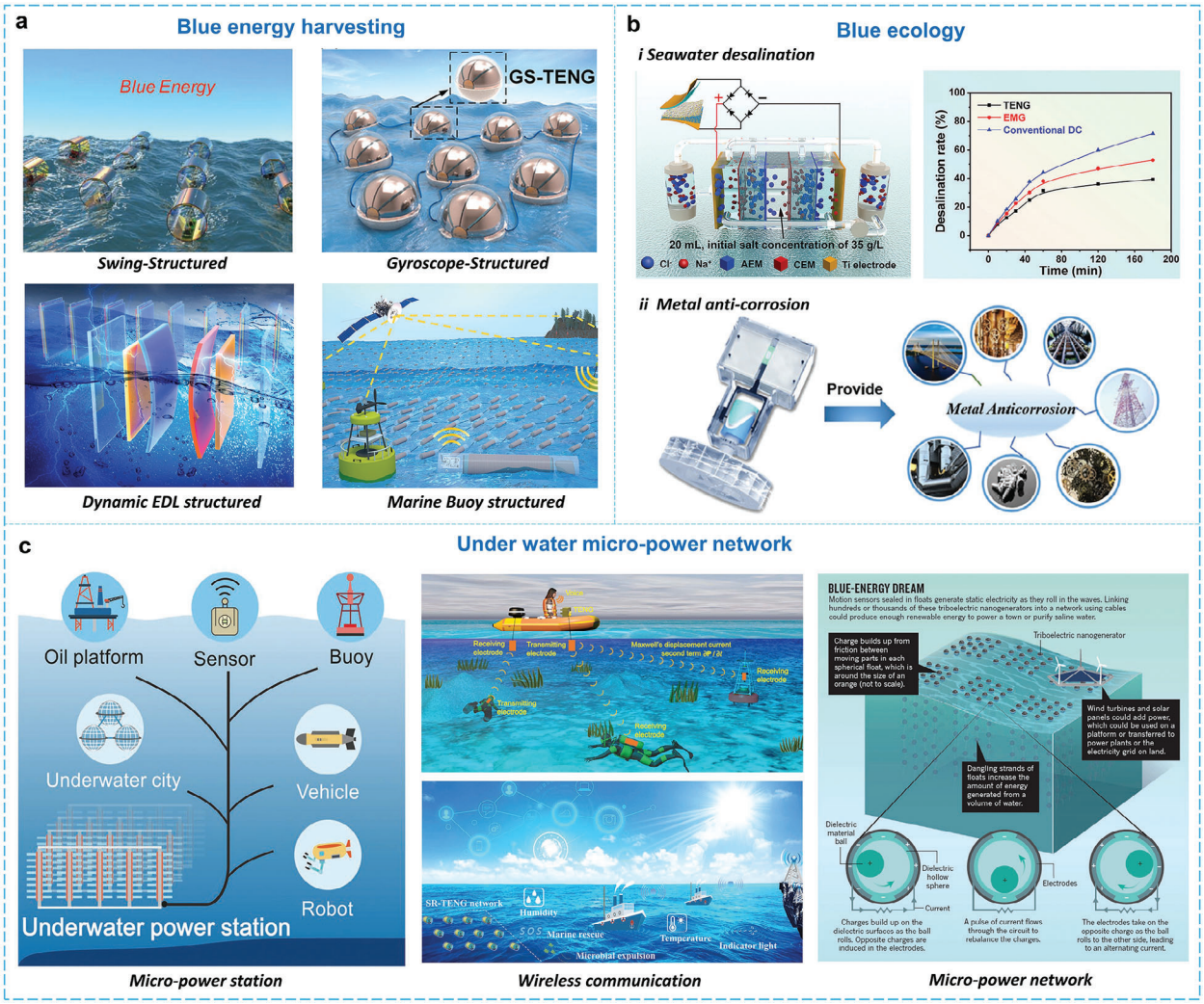


**Figure 28.** TENG enabled blue energy. Blue energy harvesting: a) Swing-structured, gyroscope-structured, dynamic edl structured and marine buoy structured TENG for blue energy. Blue ecology: b(i) TENG enabled seawater desalination, b(ii) TENG enabled metal anti-corrosion. Underwater micro-power network: c) TENG based micro-power station, wireless communication and micro-power network. a) Reprinted with permission from ref. [97]. Copyright 2020, Wiley-VCH. Reprinted with permission from ref. [323]. Copyright 2022, American Chemical Society. Reprinted with permission from ref. [324]. Copyright 2023, Wiley-VCH. Reprinted with permission from ref. [325]. Copyright 2023, Wiley-VCH. b(i) Reprinted with permission from ref. [328]. Copyright 2023, Wiley-VCH. b(ii) Reprinted with permission from ref. [329]. Copyright 2022, Wiley-VCH. c) Reprinted with permission from ref. [326]. Copyright 2022, Wiley-VCH. Reprinted with permission from ref. [330]. Copyright 2022, Wiley-VCH. Reprinted with permission from ref. [331]. Copyright 2022, the authors. Reprinted with permission from ref. [98]. Copyright 2017, Springer Nature.

classified into two types: sealed waterproof structure and liquid–solid interface contact. The basic principle of sealed waterproof structure is similar to that of solid–solid TENG, which employs wave energy as a mechanical drive, and the dielectric material selection of triboelectric layer is flexible and diverse. In terms of the liquid–solid interface contact structure, higher demands are placed on tribo-dielectric materials and electrodes as the device output relies on the seawater-solid CE. Hydrophobic or superhydrophobic dielectric layers are typically exploited to increase the output. Simultaneously, the issue of corrosion and durability must be taken into account. Some representative TENGs with swing, gyroscope, dynamic EDL and marine buoy structures have been proposed, as shown in **Figure 28a**.[97,323–326] Furthermore, combining TENG with other energy-capture technologies like solar, EMG could further enhance the harvesting efficiency. For example, Zhang et al. proposed a shadow-enhanced TENG for wave and solar energy harvesting, where an AC CE and DC shadow-effect (SE) output was generated under dark and illumination conditions. As a result, the peak power density of 718 μW cm$^{-2}$ was realized by wave and light stimulation.[327]

Besides, TENG also demonstrates promising application potential in self-powered seawater desalination and metal corrosion protection for the blue ecology. For example, Yang et al. designed a PAN/PVA-$CaCl_2$ composite nanofiber membrane as the tribo-positive layer with superior water uptake speed, swelling resistance, and triboelectric performance. Then, a high-performance PAN/PVA-$CaCl_2$/PTFE TENG was fabricated for seawater desalination, as shown in Figure 28b(i). The self-powered system


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demonstrated an energy consumption of 0.19 kWh $m^{-3}$, which is relatively lower than the EMG and DC strategies.[328] In another work, a multilayer stacked TENG was presented for metal anticorrosion (Figure 28b(ii)).[181] The open-circuit and corrosion potentials were decreased by 477.3 mV, −1.218 V when the steel was connected to the TENG, delivering the superior anticorrosion effect.[329]

To achieve high-power energy harvesting in a wide range, TENG devices must be connected to the grid in order to realize the creation of a blue energy system. Emerging visions, such as underwater micro power stations, underwater wireless communications, underwater power grids, etc., have been proposed in recent years.[329–331] For instance, Zhao et al. achieved underwater wireless communication via Maxwell's displacement current.[331] The acoustic waves were converted into electrical signals through an FEP-based single electrode TENG, and an electric field $E_0$ was induced in the water. Then, a polarization electric field $P_0$ was generated by the move reciprocal motion of ions, which will induce a current in the receiving electrode. The proposed method demonstrates superior anti-interference performance, delivering superior transmission performance of texts and images at 16 bits $s^{-1}$. In another work, a macroscopic self-assembly network of encapsulated TENG was developed for large-scale wave energy harvesting.[332] The self-assembly, self-healing and facile reconfiguration performance was achieved with the help of self-adaptive magnetic joints, and the system could generate an average power density of 8.69 W $m^{-3}$.

Overall, blue energy relies heavily on high-performance dielectrics that have hydrophobic surfaces and are resistant to seawater corrosion. Gas dielectrics with high pressure/vacuum or great insulation strength can enhance the output performance of sealed waterproof structural TENGs. However, the TENG microgrid connection and ocean energy harvesting efficiency remain challenging at present. Advancements in power generation and communication methods, device structures, and high-performance dielectric materials are expected to promote the expansion of this application area.

## 6. Conclusion and Perspectives

### 6.1. Challenges

#### *6.1.1. High-Performance Tribo-Dielectric Design Theory*

High-performance tribo-dielectric materials are the basis and central component of TENG. First principles calculations, molecular dynamics, and numerical simulations combined with advanced microscopic characterization methods (AFM, etc.) have helped reveal the physical causes of CE in different dielectrics, particularly the role of electron transfer. Additionally, it is found that a larger HOMO–LUMO gap energy level can provide more local states for electron transfer. Halogen-rich groups with strong EW or amino groups with ED ability can be used in high-performance tribo-dielectrics, relying on their strong polar effects. However, there is still a lack of a unified and comprehensive understanding of the relationship between polymer chemical structure and triboelectric properties. The tools for dielectric design are quite restricted, which complicates the correlation between atomic-level microscopic parameters and macro performance outcomes. Furthermore, experiment-based modification efforts lack exploration into the mechanisms underlying their performance enhancements. Further exploration is needed to understand the structure–activity relationship between dielectric characteristics, trap parameters, and triboelectric properties, as well as the trade-offs involved in modifying these factors during the high-performance tribo-dielectric materials design.

#### *6.1.2. Modification and Characterization of Tribo-Dielectrics*

Various physical and chemical modification methods have been proposed for advanced tribo-dielectrics. The nondoped intrinsic or organic doping modification of polymer material with superior CE performance is desired since it does not pose the risk of dopant leakage during wear. It has been reported that the relative permittivity of polymer dielectrics can be increased by incorporating transition metal atoms (such as Sn, Zn, and Cd, etc.) into polymer chains through covalent bonds or adjusting the dipole position/polymer chain conformation. The employment of conjugated conductive domains by oligomers such as oligoaniline (OANi), polyacetylene (PA) with superior compatibility, customizable chemical, and electrical properties have been reported to prepare high-*k* dielectrics, demonstrating application potential in TENGs. Furthermore, there are still shortcomings in the characterization of tribo-dielectrics. For instance, regarding the interface charge traps in polymer nanocomposite dielectrics, characterization is primarily conducted through indirect means (such as depolarization currents in TSDC), with limited involvement of direct measurement methods. The guidance method for designing and modifying high-performance polymer tribo-dielectrics still needs further exploration. The coupling contribution and mechanism of different modification methods to the triboelectric properties of polymer tribo-dielectrics are still unclear, and the joint efforts of physics, chemistry, and materials scholars are still needed.

#### *6.1.3. Multidimensional Fusion Control of CE Performance for Tribo-Dielectrics*

The surface charge density and dynamic behavior are directly linked to the tribo-dielectrics as well as high-performance TENGs. In addition to considering the dynamic equilibrium between the production and dissipation of surface charge, the air breakdown effect must also be inhibited. In the case of CS-TENG, a combination of surface modification, bulk modification, functional layer introduction, device structure adaptability optimization based on the regulation principles of polarization and dielectric parameters can realize the synergistic enhancement of the charge transfer, capture, and storage capabilities, and subsequently ensure the high surface charge density and dynamic balance of the device.

Regarding non-CS TENG or self-powered sensors, attention should be paid to improving surface charge injection, storage capacity, and response stability to ensure the long-term stability and longevity of the device output. Furthermore, there is still a need for high-performance tribo-positive


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dielectric materials because the majority of current studies concentrate on tribo-negative ones. Achieving a closed-loop R&D and integration matching between advanced tribo-dielectric materials and high-performance devices is generally desired.

#### 6.1.4. Application Scenario Adaptability of Tribo-Dielectric Materials

TENG as a power source or self-powered sensor has grown significantly for various application scenarios in recent years. The unique requirements of specific situations must be considered in addition to guaranteeing optimal triboelectric performance. For instance, wearable or implantable TENGs must have biocompatibility, good safety, breathability, and stretchability for the tribo-dielectric material. Additionally, improving the hydrophobicity and corrosion resistance of the dielectric interface is essential for liquid energy harvesting, such as ocean, raindrop, water flow, and wave energy harvesting. For emerging insulating, lubricating oils and other fluids, exploring the construction of oleophobic interfaces and strategies to improve triboelectric properties is necessary. Giving the tribo-dielectric interface wind-sand resistance, low-temperature resistance, and self-cleaning capabilities is essential for wind energy harvesting in harsh environments to ensure service dependability and save maintenance costs.

Regarding the high-reliability of self-powered sensors, it is especially crucial to maintain the tribo-dielectric surface charge density and enhance the sensitivity and lifespan of the tribo-sensing layer. In order to improve durability, reliable packaging can be used to avoid the effects of environmental changes such as dust and humidity. Besides, the intermittent working method could minimize interface wear by utilizing the residual charge of tribo-dielectric surface to provide long-term stability and, in certain cases, enable non-contact operation. Employing suitable liquid dielectric to accomplish solid interface contact wear and gap discharge suppression has been advocated recently, realizing simultaneous enhancement of durability and output performance.

#### 6.1.5. Large-Scale Production and Commercial Application of Tribo-Dielectric Materials

Significant strides in high-performance tribo-dielectric materials have been made at the laboratory level. In the future, efforts should be made to develop durable, efficient, and low-cost large-scale material preparation techniques to support the commercialization of TENG. One possible solution is 3D printing. The layer-by-layer printing is capable of creating a variety of macro/microscopic structures. Scaling up the processing of tribo-dielectric materials is also anticipated to be possible through electrospinning, roll-to-roll, techniques. Regarding the networking application requirement of TENG, further critical challenges for facilitating network reliability and energy collection efficiency, including the interunit redundancy design, high-performance matching of tribo-dielectric materials, still need to be addressed.

### 6.2. Perspectives

#### 6.2.1. R&D of Tribo-Dielectric Materials Assisted by Big Data and AI

The design, development, and practical preparation of advanced tribo-dielectric materials are anticipated to enter the intelligence age along with the growth of big data and AI technology. Integrating multiscale calculation (atomic, molecular, mesoscopic, macro scale) and AI algorithm endow the design of high-performance tribo-dielectric materials. Machine learning databases can be built with the fundamental support provided by massive tribo-dielectric material testing data and the standardization of tribo-electric series. First principle calculations and molecular dynamics can be performed on basic materials with functional group substitution, element replacement, and doping modification to clarify the role of molecular structure and electronic orbital parameters on the CE mechanism. Further, the prior design and selection of novel dielectric materials can be realized by regulating the parameters, such as affinity energy, dipole moment, interfacial binding force, chemical bond parameters, etc. In the case of multiple dielectric coupling, the charge transfer and migration paths at the atomic level could be explored by first principle calculations, while the interface behavior and the overall dielectric behavior of multi-component dielectrics need to be considered through multiphysics analysis to ensure the accuracy.

Then, AI technology can be used to investigate the high-dimensional correlation characteristics between multiple variables from existing experimental and theoretical data. Consequently, it will be possible to explore the profound correlation between the structure, properties, and contact electrification characteristics of dielectric materials and the interface charge transport behavior in the future with the help of big data and AI (machine learning, deep learning, neural networks, etc.), directing the design and development of advanced tribo-dielectrics at high throughput, saving costs, and time.

#### 6.2.2. Utilization of Dielectric and Interfacial Behavior

The matrix/particle interface in composite polymer dielectrics is suggested to have a significant impact on bulk performance. On one hand, the dielectric and trap properties of polymers are closely related to triboelectric properties. The strong electric field induced by CE surface charge will promote the dipole, interface polarization process that affects the electrostatic induction and output performance. The trap properties determine the charge transport behaviors, including charge migration, storage, and dissipation. The stable storage of triboelectric charges contributes to more efficient electrostatic induction, and the utilization of the charge migration, bulk effect could also enhance the output performance. The utilization of dielectric and interfacial behavior should be focused. On the other hand, suitable filler concentrations are required to minimize particle aggregation and promote large-scale production. Further exploration should be focused on nanofillers, polymer matrices, and efficient processing methods that can significantly improve dielectric performance at low filler concentrations. Besides, it is essential to investigate dielectric responses using finite element simulations and phase field models, considering the dynamic behavior of tribo-charges


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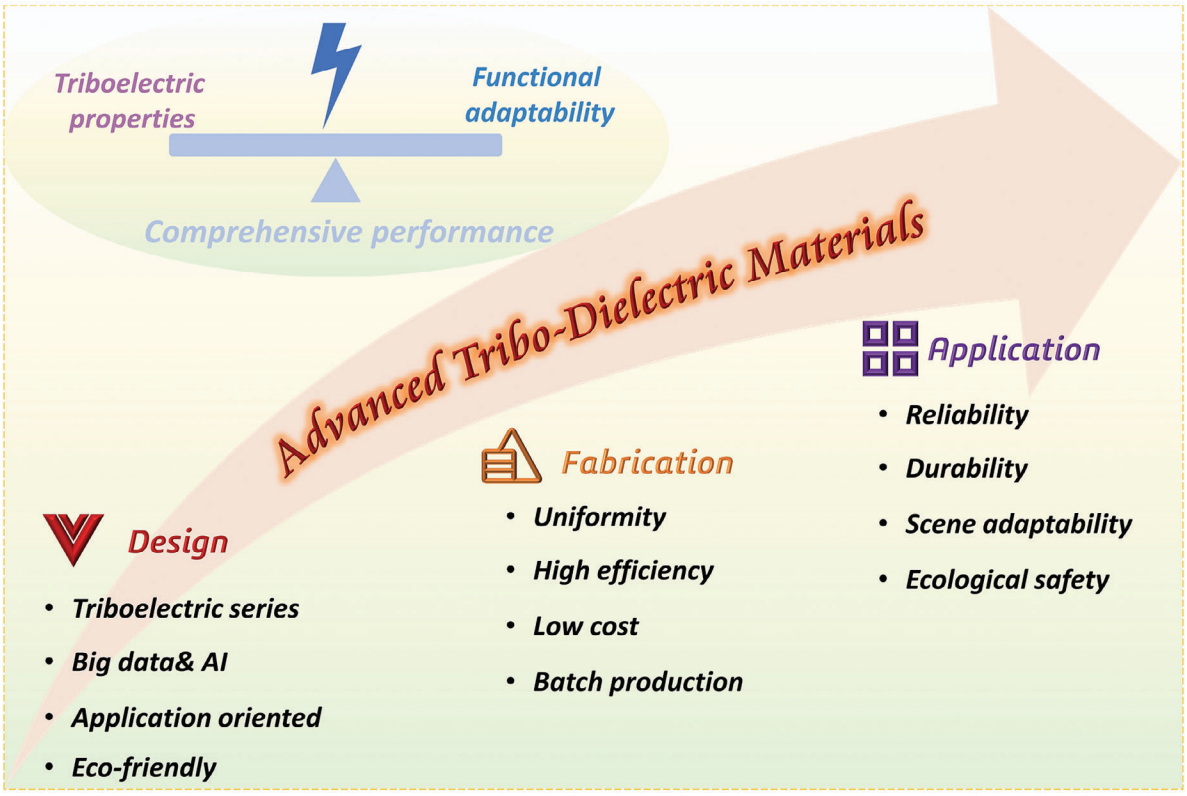


**Figure 29.** The roadmap of advanced tribo-dielectric materials for TENGs.

in TENG. This will help elucidate the interaction mechanisms between various nanofillers, polymers as well as the role of local electric field.

#### *6.2.3. Multidisciplinary Integration and Cross-Promotion of Original Applications*

By virtue of its strong structural adaptability, multiple working modes, and multiple material selection, TENG has garnered attention from multidisciplinary researchers in materials science, physics, mechanics, electrical engineering, biology, and chemistry, exhibiting wide application potential in various types of energy harvesting and self-powered sensing. However, some innovative devices employ traditional tribo-dielectric as the functional layer, while the application potential of some modified high-performance tribo-delectric materials has not yet been fully explored. Thus, it is anticipated that the integration of various disciplines will result in the "chain-like" integrated development of CE mechanisms, novel TENG structures, high-performance tribo-dielectric materials, as well as numerous application scenarios, achieving vertical R&D and development from mechanism to materials and devices.

#### *6.2.4. Safety and Eco-Friendly Development of Tribo-Dielectric Materials*

Currently, high-performance tribo-dielectric materials are mainly prepared or modified using strongly polar fluorine-rich (fluorinated) polymers. However, fluorinated materials are permanent compounds that fall within the category of PFAS (per- and polyfluoroalkyl substances). Certain PFAS have been shown to accumulate in the environment and human tissue due to their strong stability and difficulty in decomposition, which raises concerns for both biological health and the ecological environment. The amount of PFAS compounds under worldwide supervision has increased recently. The ecological and environmental issues raised by the fluorinated frictional dielectric in the device cannot be disregarded when TENG is used extensively (for example, in blue energy applications). Thus, developing advanced tribo-dielectric material made of degradable and fluorine-free substances is imperative, which is significant to promote the safe and eco-friendly development of TENGs.

In conclusion, advanced dielectric materials have promoted the development of high-performance TENGs as promising energy harvesters and self-powered sensors. The roadmap in **Figure 29** clarifies the core points regarding the design, fabrication, and application of advanced tribo-dielectric materials. The tradeoff between triboelectric properties and functional adaptability should be considered for the comprehensive performance. We believe emerging advanced tribo-dielectric materials are expected to play a significant role in driving diverse applications of TENG in the future.

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## Conflict of Interest

The authors declare no conflict of interest.

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dielectric materials, energy harvesting, self-powered sensing, triboelectric materials, triboelectric nanogenerator

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